56. 12-Bit A/D Converter (S12ADFa)

In this section, “PCLK” is used to refer to PCLKB.

56.1 Overview

This MCU incorporates two units of 12-bit successive approximation A/D converters. Unit 0 for high-speed conversion can select analog input of up to 8 channels. Unit 1 for middle-speed conversion can select analog input of up to 21 channels, temperature sensor output, and internal reference voltage.

The 12-bit A/D converter converts analog input of up to 8 selected channels (unit 0) or up to 21 channels (unit 1), temperature sensor output, or internal reference voltage into a 12-bit digital value through successive approximation.

The A/D converter has three operating modes: single scan mode in which the analog inputs of eight arbitrarily selected channels (unit 0) and 21 channels (unit 1) are converted in ascending channel order; continuous scan mode in which the analog inputs of eight arbitrarily selected channels (unit 0) and 21 channels (unit 1) are continuously converted in ascending channel order; and group scan mode in which eight arbitrarily selected channels (unit 0) and 21 channels (unit 1) are arbitrarily divided into two groups (groups A and B) or three groups (groups A, B, and C) and converted in ascending channel order in each group.

In group scan mode, either two groups (groups A and B) or three groups (groups A, B, and C) is selected.

The conditions for scanning start of each group (A and B or A, B, and C) (synchronous trigger) can be independently selected, thus allowing A/D conversion of each group to be started independently.

During group priority operation, in addition to the above-mentioned operation, a trigger to start scanning for the priority group is accepted during scan for the low-priority group, and scan for the priority group is started after scan for the low-priority group was discontinued. The priority order is group A > group B > group C. Accordingly, as priority operation, when a trigger to start scanning for group B is accepted during scan for group C, group C scan is discontinued, and scan for group B is started. Likewise, when a trigger to start scanning for group A is accepted during scan for group C, group C scan is discontinued and scan for group A is started. In the same way, when a trigger to start scanning for group A is accepted during scan for group B, group B scan is discontinued and scan for group A is started.

The discontinued scan operation can be restarted after the scanning of the priority group is completed.

In double trigger mode, one analog input channel arbitrarily selected is converted in single scan mode or group scan mode (group A), and the resulting data of A/D conversion started by the first and second synchronous triggers are stored into different registers (duplication of A/D conversion data).

Self-diagnosis is executed once at the beginning of each scan, and one of the three voltages internally generated in the 12-bit A/D converter is converted.

The temperature sensor output and internal reference voltage can be selected with analog input of the channels, and A/D conversion is performed in the order of the analog input of the channels, temperature sensor output, and internal reference voltage. A/D conversion of the extended analog input is independently performed.

The MCU incorporates the comparison function (with windows A and B). In addition, the value of A/D conversion and the reference value of the low side can be compared by a comparator.

Table 56.1 lists the specifications of the 12-bit A/D converter and Table 56.2 lists the functions of the 12-bit A/D converter. Figure 56.1 and Figure 56.2 show block diagrams of the 12-bit A/D converter.

Table 56.1 Specifications of 12-Bit A/D Converter (1/2)

Item	Description
Number of units	Two units (S12AD and S12AD1)
Input channels	Eight channels for S12AD and 21 channels for S12AD1 + 1 extension
Extended analog function	Temperature sensor output, internal reference voltage
A/D conversion method	Successive approximation method
Resolution	12 bits
Conversion time	0.48 μs per channel (12-bit conversion mode) 0.45 μs per channel (10-bit conversion mode) 0.42 μs per channel (8-bit conversion mode) (A/D conversion clock: when ADCLK operates at 60 MHz)
A/D conversion clock	Peripheral module clock PCLK*1 and A/D conversion clock ADCLK*1 can be set so that the frequency ratio should be one of the following. PCLK to ADCLK frequency ratio = 1:1, 2:1, 4:1, 8:1 ADCLK is set using the clock generation circuit.
Data registers	• 29 registers for analog input (eight for S12AD and 21 for S12AD1), 1 for A/D-converted data duplication in double trigger mode per unit, and 2 for A/D-converted data duplication during extended operation in double trigger mode per unit. • One register for temperature sensor (S12AD1) • One register for internal reference (S12AD1) • One register for self-diagnosis per unit • The results of A/D conversion are stored in 12-bit A/D data registers. • 8-, 10-, and 12-bit accuracy output for the results of A/D conversion • The value obtained by adding up A/D-converted results is stored as a value in the number of bit for conversion accuracy + 2 bits/4 bits*2 in the A/D data registers in A/D-converted value addition mode. • Double trigger mode (selectable in single scan and group scan modes): The first piece of A/D-converted analog-input data on one selected channel is stored in the data register for the channel, and the second piece is stored in the duplication register. • Extended operation in double trigger mode (available for specific triggers): A/D-converted analog-input data on one selected channel is stored in the duplication register that is prepared for each type of trigger.
Operating modes	Operating modes can be set independently for two units. • Single scan mode: A/D conversion is performed only once on the analog inputs arbitrarily selected. A/D conversion is performed only once on the temperature sensor output (S12AD1). A/D conversion is performed only once on the internal reference voltage (S12AD1). A/D conversion is performed only once on the extended analog input (S12AD1). • Continuous scan mode: A/D conversion is performed repeatedly on the analog input, temperature sensor output (S12AD1), and internal reference voltage (S12AD1) of the arbitrarily selected channel. A/D conversion is performed repeatedly on the extended analog input (S12AD1). • Group scan mode: Two (groups A and B) or three (groups A, B, and C) can be selected as the number of the groups to be used. Only the combination of groups A and B can be selected when the number of the groups is two. Analog inputs, temperature sensor output (S12AD1), and internal reference voltage (S12AD1) that are arbitrarily selected are divided into two groups (group A and B) or three groups (group A, B, and C), and A/D conversion of the analog input selected on a group basis is performed only once. The conditions for scanning start of groups A, B, and C (synchronous trigger) can be independently selected, thus allowing A/D conversion of each group to be started independently. • Group scan mode (when group priority control selected): If a priority-group trigger is input during scanning of the low-priority group, scan of the low-priority group is stopped and scan of the priority group is started. The priority order is group A (highest) > group B > group C (lowest). Whether or not to restart scanning of the low-priority group after processing for the high-priority group completes, is selectable. Rescan can also be set to start either from the beginning of the selected channel or the channel on which A/D conversion is not completed.

Table 56.1 Specifications of 12-Bit A/D Converter (2/2)

Item	Description
Conditions for A/D conversion start	• Software trigger • Synchronous trigger Trigger by the multi-function timer pulse unit (MTU), 8-bit timer (TMR), 16-bit timer pulse unit (TPU), or event link controller (ELC). • Asynchronous trigger A/D conversion can be triggered by the external trigger ADTRG0# (S12AD) or ADTRG1# (S12AD1) pin (independently for two units).
Functions	• Channel-dedicated sample-and-hold function (three channels for S12AD only) • Variable sampling state count (settable for each channel) • Self-diagnosis of 12-bit A/D converter • Selectable A/D-converted value addition mode or average mode • Analog input disconnection detection assist function (discharge function/precharge function) • Double trigger mode (duplication of A/D conversion data) • 12-/10-/8-bit conversion switching • Automatic clear function of A/D data registers • Extended analog input • Comparison function (windows A and B)
Interrupt sources	• In the modes except double trigger mode and group scan mode, a scan end interrupt request (S12ADI or S12ADI1) can be generated on completion of single scan (independently for two units). • In double trigger mode, a scan end interrupt request (S12ADI or S12ADI1) can be generated on completion of double scan (independently for two units). • In group scan mode, a scan end interrupt request (S12ADI or S12ADI1) can be generated on completion of group A scan, whereas a scan end interrupt request (S12GBADI or S12GBADI1) for group B can be generated on completion of group B scan, and a group C scan end interrupt request (S12GCADI or S12GCADI1) can be generated on completion of group C scan. • When double trigger mode is selected in group scan mode, an A/D scan end interrupt request (S12ADI or S12ADI1) can be generated on completion of double scan of group A, and the corresponding scan end interrupt request (S12GBADI/S12GCADI or S12GBADI1/S12GCADI1) can be generated on completion of group B and group C scan. • A compare interrupt request (S12CMPAI, S12CMPAI1, S12CMPBI, or S12CMPBI1) can be generated upon a match with the comparison condition for the digital compare function. • The S12ADI/S12ADI1, S12GBADI/S12GBADI1, and S12GCADI/S12GCADI1 interrupts can activate the DMA controller (DMAC) and data transfer controller (DTC).
Event link	• An ELC event is generated upon completion of all scans • Able to start scanning by a trigger from the ELC
Low power consumption function	• Module stop state can be set.*3, *4

Note 1. The frequency of PCLK, the peripheral module clock, becomes that set in the SCKCR.PCKB[3:0] bits, and that of ADCLK, the A/D conversion clock, becomes the frequency set in the SCKCR.PCKC[3:0] bits for unit 0 (S12AD), and that set in the SCKCR.PCKD[3:0] bits for unit 1 (S12AD1).

Note 2. The number of extended bits during addition differs depending on the addition count.
2-bit extension: 1-time to 4-time conversion (add zero to three times)
4-bit extension: 16-time conversion (add 15 times)

Note 3. See section 11, Low Power Consumption for details.

Note 4. Wait for 1 μs or longer to start A/D conversion after release from the module stop state.

Table 56.2 Functions of 12-Bit A/D Converter

Item			Pin Name, Abbreviation Unit 0 (S12AD)	Unit 1 (S12AD1)
Analog input channels			AN000 to AN007	AN100 to AN120, internal reference voltage, temperature sensor output, extended input
Conditions for A/D conversion start	Software	Software trigger	Enabled	
	Asynchronous trigger	Trigger input pin	ADTRG0#	ADTRG1#
	Synchronous trigger	Compare match/input capture from MTU0.TGRA	TRGA0N	
		Compare match/input capture from MTU1.TGRA	TRGA1N	
		Compare match/input capture from MTU2.TGRA	TRGA2N	
		Compare match/input capture from MTU3.TGRA	TRGA3N	
		Compare match/input capture from MTU4.TGRA or underflow (trough) of MTU4.TCNT in complementary PWM mode	TRGA4N	
		Compare match/input capture from MTU6.TGRA	TRGA6N	
		Compare match/input capture from MTU7.TGRA or underflow (trough) of MTU7.TCNT in complementary PWM mode	TRGA7N	
		Compare match from MTU0.TGRE	TRG0N	
		Compare match between MTU4.TADCORA and MTU4.TCNT	TRG4AN	
		Compare match between MTU4.TADCORB and MTU4.TCNT	TRG4BN	
		Compare match between MTU4.TADCORA and MTU4.TCNT, or compare match between MTU4.TADCORB and MTU4.TCNT	TRG4AN or TRG4BN	
		Compare match between MTU4.TADCORA and MTU4.TCNT, and compare match between MTU4.TADCORB and MTU4.TCNT (when interrupt skipping function 2 is used)	TRG4ABN	
		Compare match between MTU7.TADCORA and MTU7.TCNT	TRG7AN	
		Compare match between MTU7.TADCORB and MTU7.TCNT	TRG7BN	
		Compare match between MTU7.TADCORA and MTU7.TCNT, or compare match between MTU7.TADCORB and MTU7.TCNT	TRG7AN or TRG7BN	
		Compare match between MTU7.TADCORA and MTU7.TCNT, and compare match between MTU7.TADCORB and MTU7.TCNT (when interrupt skipping function 2 is used)	TRG7ABN	
		Compare match between TMR0.TCORA and TMR0.TCNT	TMTRG0AN_0	
		Compare match between TMR2.TCORA and TMR2.TCNT	TMTRG0AN_1	
		Compare match/input capture from TPU0.TGRA or compare match/input capture from TPU1.TGRA or compare match/input capture from TPU2.TGRA or compare match/input capture from TPU3.TGRA or compare match/input capture from TPU4.TGRA	TPTRGAN	
		Compare match/input capture from TPU0.TGRA0	TPTRG0AN	
		ELC trigger	ELCTRG00N, ELCTRG01N, ELCTRG00N or ELCTRG01N	ELCTRG10N, ELCTRG11N, ELCTRG10N or ELCTRG11N
Channel-dedicated sample-and-hold function		Target channels	AN000 to AN002	—
Interrupts			S12ADI, S12GBADI, S12GCADI, S12CMPAI, S12CMPBI interrupt	S12ADI1, S12GBADI1, S12GCADI1, S12CMPAI1, S12CMPBI1 interrupt
Setting of module stop function*1, *2			MSTPCRA.MSTPA17 bit	MSTPCRA.MSTPA16 bit

Note: When setting an A/D conversion start trigger to ADTRG0# or ADTRG1#, set the pin mode control bit in the port mode register for the corresponding pin to 1 (peripheral functions), and set the pin function select bit in the pin function control register to ADTRG0# or ADTRG1#. See section 22, I/O Ports for details.

Note 1. See section 11, Low Power Consumption for details.

Note 2. Wait for 1 µs or longer to start A/D conversion after release from the module stop state.

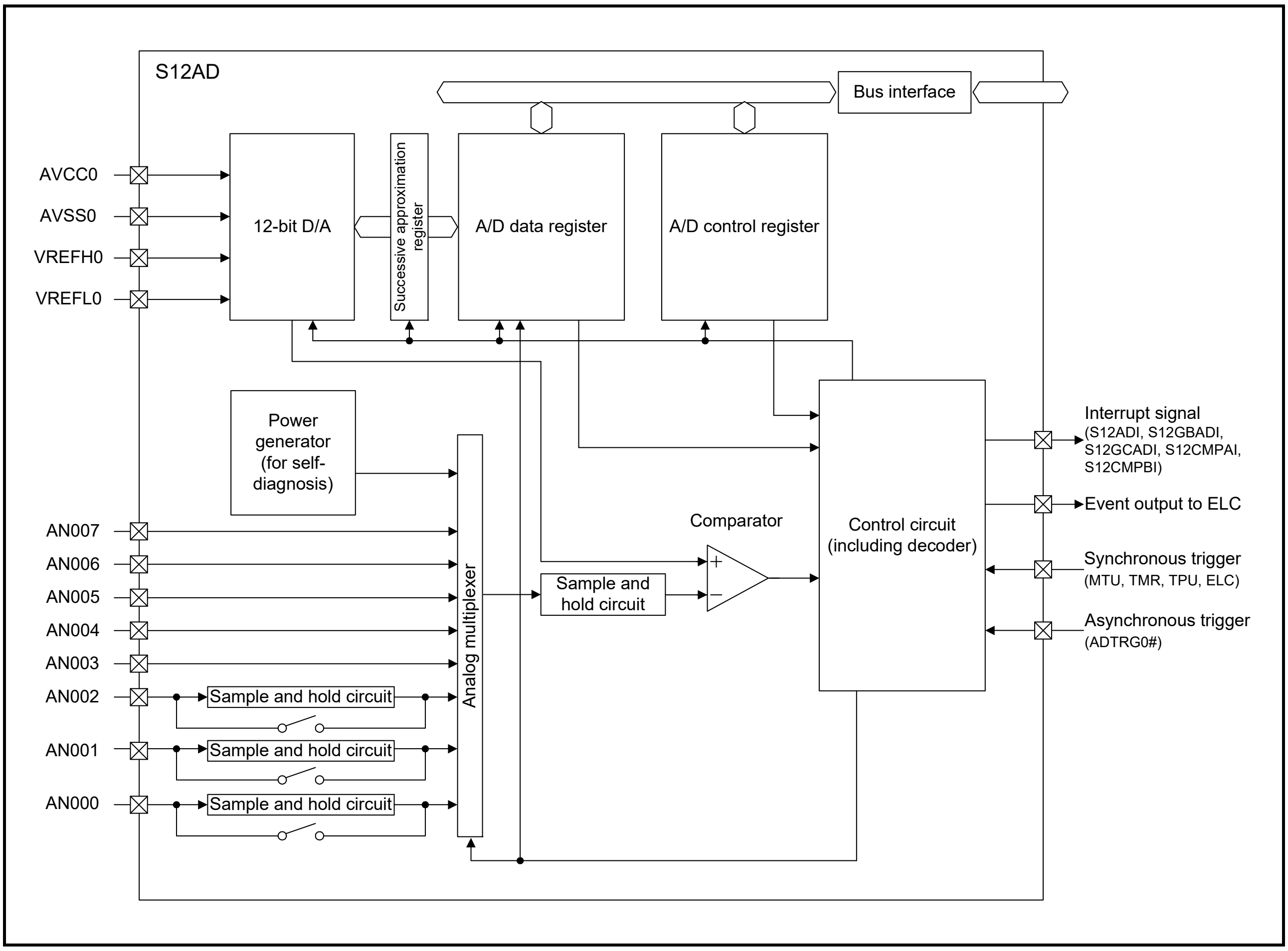


Figure 56.1 Block Diagram of 12-Bit A/D Converter (Unit 0)

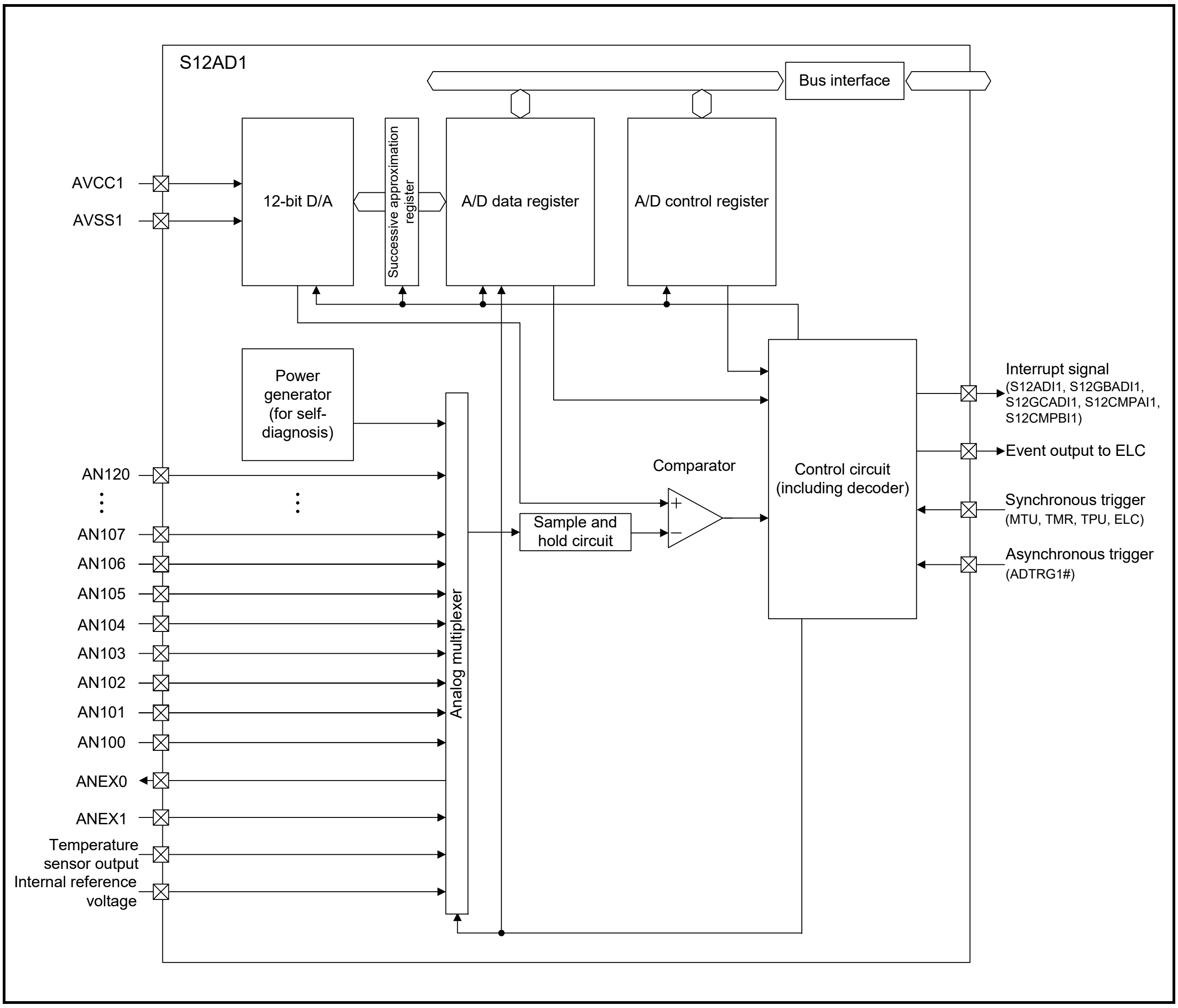


Figure 56.2 Block Diagram of 12-Bit A/D Converter (Unit 1)

Table 56.3 lists the input/output pins of the 12-bit A/D converter.

The 12-bit A/D converter consists of two units, unit 0 (S12AD) and unit 1 (S12AD1). These units can be operated independently. The input channels of S12AD and S12AD1 can be divided into three groups for operation.

Table 56.3 Input/Output Pins of 12-Bit A/D Converter

Unit	Pin Name	I/O	Function
Unit 0 (S12AD)	AVCC0	—	Analog block power supply pin
	AVSS0	—	Analog block ground pin
	VREFH0	Input	Reference power supply pin
	VREFL0	Input	Reference power supply ground pin
	AN000 to AN007	Input	Analog input pins
	ADTRG0#	Input	External trigger input pin for starting A/D conversion
Unit 1 (S12AD1)	AVCC1	—	Multiplexed analog block power supply and reference power supply pin functions
	AVSS1	—	Multiplexed analog block ground and reference power supply ground pin functions
	AN100 to AN120	Input	Analog input pins
	ANEX0	Output	Extended analog output pin
	ANEX1	Input	Extended analog input pin
	ADTRG1#	Input	External trigger input pin for starting A/D conversion

56.2 Register Descriptions

56.2.1 A/D Data Registers y (ADDRy) (y = 0 to 20), A/D Data Duplication Register (ADDBLDR), A/D Data Duplication Register A (ADDBLDRA), A/D Data Duplication Register B (ADDBLDRB), A/D Temperature Sensor Data Register (ADTSDR), A/D Internal Reference Voltage Data Register (ADOCDR)

Address(es): S12AD.ADDR0 0008 9020h, S12AD.ADDR1 0008 9022h, S12AD.ADDR2 0008 9024h,
S12AD.ADDR3 0008 9026h, S12AD.ADDR4 0008 9028h, S12AD.ADDR5 0008 902Ah,
S12AD.ADDR6 0008 902Ch, S12AD.ADDR7 0008 902Eh, S12AD.ADDBLDR 0008 9018h,
S12AD.ADDBLDRA 0008 9084h, S12AD.ADDBLDRB 0008 9086h,
S12AD1.ADDR0 0008 9120h, S12AD1.ADDR1 0008 9122h, S12AD1.ADDR2 0008 9124h,
S12AD1.ADDR3 0008 9126h, S12AD1.ADDR4 0008 9128h, S12AD1.ADDR5 0008 912Ah,
S12AD1.ADDR6 0008 912Ch, S12AD1.ADDR7 0008 912Eh, S12AD1.ADDR8 0008 9130h,
S12AD1.ADDR9 0008 9132h, S12AD1.ADDR10 0008 9134h, S12AD1.ADDR11 0008 9136h,
S12AD1.ADDR12 0008 9138h, S12AD1.ADDR13 0008 913Ah, S12AD1.ADDR14 0008 913Ch,
S12AD1.ADDR15 0008 913Eh, S12AD1.ADDR16 0008 9140h, S12AD1.ADDR17 0008 9142h,
S12AD1.ADDR18 0008 9144h, S12AD1.ADDR19 0008 9146h, S12AD1.ADDR20 0008 9148h,
S12AD1.ADDBLDR 0008 9118h, S12AD1.ADDBLDRA 0008 9184h, S12AD1.ADDBLDRB 0008 9186h,
S12AD1.ADTSDR 0008 911Ah, S12AD1.ADOCDR 0008 911Ch

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The ADDRy registers (y = 0 to 7 for S12AD; y = 0 to 20 for S12AD1) are 16-bit read-only registers which store the A/D conversion results.
The ADDBLDR register is a 16-bit read-only register used in double trigger mode. The ADDBLDR register stores the results of A/D conversion when the conversion is started by the second trigger.
The ADDBLDRA and ADDBLDRB registers are 16-bit read-only registers that store the A/D conversion results in response to the respective triggers during extended operation in double trigger mode.
The ADTSDR register is a 16-bit read-only register that stores the A/D converted value of the temperature sensor output.
The ADOCDR register is a 16-bit read-only register that stores the A/D conversion results of the internal reference voltage.
The format of each register differs depending on the conditions below.

- Settings of the A/D data register format select bit (ADCER.ADRFMT) (flush-right or flush-left)
- Settings of the A/D data register bit resolution setting bits (ADCER.ADPRC[1:0]) (12-, 10-, or 8-bit)
- Settings of the addition count select bits (ADADC.ADC[2:0]) (2-, 3-, 4-, or 16-time conversion)
- Settings of the average mode enable bit (ADADC.AVEE) (add or average)

The data formats for each given condition are shown below.

(1) When A/D-Converted Value Addition/Average Mode is Not Selected

- Flush-right format with setting of 12-bit resolution
  The A/D-converted value is stored in bits 11 to 0. Bits 15 to 12 are read as 0.
- Flush-right format with setting of 10-bit resolution
  The A/D-converted value is stored in bits 9 to 0. Bits 15 to 10 are read as 0.
- Flush-right format with setting of 8-bit resolution
  The A/D-converted value is stored in bits 7 to 0. Bits 15 to 8 are read as 0.
- Flush-left format with setting of 12-bit resolution
  The A/D-converted value is stored in bits 15 to 4. Bits 3 to 0 are read as 0.
- Flush-left format with setting of 10-bit resolution

The A/D-converted value is stored in bits 15 to 6. Bits 5 to 0 are read as 0.

- Flush-left format with setting of 8-bit resolution
  The A/D-converted value is stored in bits 15 to 8. Bits 7 to 0 are read as 0.

(2) When A/D-Converted Average Mode is Selected

- Flush-right format with setting of 12-bit resolution
  The mean value of the A/D-converted results of the same channel is stored in bits 11 to 0.
  Bits 15 to 12 are read as 0.
- Flush-right format with setting of 10-bit resolution
  The mean value of the A/D-converted results of the same channel is stored in bits 9 to 0.
  Bits 15 to 10 are read as 0.
- Flush-right format with setting of 8-bit resolution
  The mean value of the A/D-converted results of the same channel is stored in bits 7 to 0.
  Bits 15 to 8 are read as 0.
- Flush-left format with setting of 12-bit resolution
  The mean value of the A/D-converted results of the same channel is stored in bits 15 to 4.
  Bits 3 to 0 are read as 0.
- Flush-left format with setting of 10-bit resolution
  The mean value of the A/D-converted results of the same channel is stored in bits 15 to 6.
  Bits 5 to 0 are read as 0.
- Flush-left format with setting of 8-bit resolution
  The mean value of the A/D-converted results of the same channel is stored in bits 15 to 8.
  Bits 7 to 0 are read as 0.

A/D-converted value average mode can be set only when 2- or 4-time conversion is selected in A/D-converted value addition mode.

(3) When A/D-Converted Value Addition Mode is Selected

- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The value added by the A/D-converted value of the same channel is stored in bits 13 to 0.
  Bits 15 and 14 are read as 0.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The value added by the A/D-converted value of the same channel is stored in bits 11 to 0.
  Bits 15 to 12 are read as 0.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The value added by the A/D-converted value of the same channel is stored in bits 9 to 0.
  Bits 15 to 10 are read as 0.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The value added by the A/D-converted value of the same channel is stored in bits 15 to 0.
  When the 16-time conversion is selected, setting with 10- or 8-bit resolution is prohibited.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The value added by the A/D-converted value of the same channel is stored in bits 15 to 2.
  Bits 1 and 0 are read as 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The value added by the A/D-converted value of the same channel is stored in bits 15 to 4.

Bits 3 to 0 are read as 0.

- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The value added by the A/D-converted value of the same channel is stored in bits 15 to 6.
  Bits 5 to 0 are read as 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The value added by the A/D-converted value of the same channel is stored in bits 15 to 0.
  When the 16-time conversion is selected, setting with 10- or 8-bit resolution is prohibited.

When A/D-converted addition mode is selected, the value added by the A/D-converted value of the same channel is indicated. The number of A/D conversions can be set to 1, 2, 3, 4, or 16 times. If A/D-converted addition mode is selected, when the conversion count is set to 1 to 4 times, the value added by the A/D conversion result is retained in the A/D data register as 2-bit extended data of the conversion accuracy bits; when the conversion count is set to 16 times, the value added by the A/D conversion result is retained in the A/D data register as 4-bit extended data of the conversion accuracy bits. Even if A/D-converted value addition mode is selected, the value is stored in the A/D data register according to the settings of the A/D data register format select bits. When the 16-time conversion is selected, setting with 10- or 8-bit resolution is prohibited.

56.2.2 A/D Self-Diagnosis Data Register (ADRD)

Address(es): S12AD.ADRD 0008 901Eh, S12AD1.ADRD 0008 911Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADRD is a 16-bit read-only register that stores the A/D conversion results based on the 12-bit A/D converter's self-diagnosis. In addition to the A/D-converted value, the self-diagnosis status is included in. In the ADRD register, the different formats are used depending on the conditions below.

- Settings of the A/D data register format select bit (ADCER.ADRFMT) (flush-right or flush-left)
- Settings of the A/D data register bit resolution setting bits (ADCER.ADPRC[1:0]) (12-, 10-, or 8-bit)

The A/D-converted value addition mode and A/D-converted value average mode cannot be applied to the A/D self-diagnosis function. For details of self-diagnosis, see section 56.2.13, A/D Control Extended Register (ADCER).

The data formats for each given condition are shown below.

- Flush-right format with setting of 12-bit resolution
  The A/D-converted value is stored in bits 11 to 0. The self-diagnosis status is stored in bits 15 and 14.
  Bits 13 and 12 are read as 0.
- Flush-right format with setting of 10-bit resolution
  The A/D-converted value is stored in bits 9 to 0. The self-diagnosis status is stored in bits 15 and 14.
  Bits 13 to 10 are read as 0.
- Flush-right format with setting of 8-bit resolution
  The A/D-converted value is stored in bits 7 to 0. The self-diagnosis status is stored in bits 15 and 14.
  Bits 13 to 8 are read as 0.
- Flush-left format with setting of 12-bit resolution
  The A/D-converted value is stored in bits 15 to 4. The self-diagnosis status is stored in bits 1 and 0.
  Bits 3 and 2 are read as 0.
- Flush-left format with setting of 10-bit resolution
  The A/D-converted value is stored in bits 15 to 6. The self-diagnosis status is stored in bits 1 and 0.
  Bits 5 to 2 are read as 0.
- Flush-left format with setting of 8-bit resolution
  The A/D-converted value is stored in bits 15 to 8. The self-diagnosis status is stored in bits 1 and 0.
  Bits 7 to 2 are read as 0.

Table 56.4 Self-Diagnosis Status Description

Bits 15 and 14 for flush-right format setting Bits 1 and 0 for flush-left format setting	Self-diagnosis status
00b	Self-diagnosis has never been executed since power-on.
01b	Self-diagnosis using the voltage of 0 V has been executed.
10b	Self-diagnosis using the voltage of reference power supply × 1/2 has been executed.
11b	Self-diagnosis using the voltage of reference power supply has been executed.

Note: For details of self-diagnosis, see section 56.2.13, A/D Control Extended Register (ADCER).

56.2.3 A/D Control Register (ADCSR)

Address(es): S12AD.ADCSR 0008 9000h, S12AD1.ADCSR 0008 9100h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	ADST	ADCS[1:0]		ADIE	—	—	TRGE	EXTRG	DBLE	GBADIE	—	DBLANS[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	DBLANS[4:0]	Double Trigger Channel Select	These bits select one analog input channel for double triggered operation. The setting is only effective while double trigger mode is selected.	R/W
b5	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6	GBADIE	Group B Scan End Interrupt Enable	0: Disables interrupt generation upon group B scan completion. 1: Enables interrupt generation upon group B scan completion.	R/W
b7	DBLE	Double Trigger Mode Select	0: Deselects double trigger mode. 1: Selects double trigger mode.	R/W
b8	EXTRG	Trigger Select*1	0: A/D conversion is started by synchronous trigger. 1: A/D conversion is started by asynchronous trigger.	R/W
b9	TRGE	Trigger Start Enable	0: Disables A/D conversion to be started by synchronous or asynchronous trigger. 1: Enables A/D conversion to be started by synchronous or asynchronous trigger.	R/W
b11, b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	ADIE	Scan End Interrupt Enable	0: Disables interrupt generation upon scan completion. 1: Enables interrupt generation upon scan completion.	R/W
b14, b13	ADCS[1:0]	Scan Mode Select	b14 b13 0 0: Single scan mode 0 1: Group scan mode 1 0: Continuous scan mode 1 1: Setting prohibited	R/W
b15	ADST	A/D Conversion Start	0: Stops A/D conversion process. 1: Starts A/D conversion process.	R/W

Note 1. Starting A/D conversion using an external pin (asynchronous trigger)
Set the ADCSR.TRGE and EXTRG bits to 1 while a high-level signal is input to the external pin (ADTRG0# or ADTRG1#). Then, if the ADTRG0# or ADTRG1# signal is changed to low, the falling edge is detected and the scan process is started. In this case, the pulse width of the low-level input must be at least 1.5 clock cycles of PCLK.
The relationship between each unit and the external pin (asynchronous trigger) is shown below.

Unit	External pin (asynchronous trigger)
S12AD	ADTRG0#
S12AD1	ADTRG1#

ADCSR sets double trigger mode, A/D conversion start trigger; enables/disables scan end interrupt; selects the scan mode; and starts or stops A/D conversion.

DBLANS[4:0] Bits (Double Trigger Channel Select)

The DBLANS[4:0] bits select one of the channels for A/D conversion data duplication in double trigger mode. The A/D conversion results of the analog input of the channel selected by the DBLANS[4:0] bits are stored into the A/D data register y when conversion is started by the first trigger, and into the A/D data duplication register when started by the second trigger. Table 56.5 shows selection of the channel for double triggered operation.
When double trigger mode is selected, the channels selected by the ADANSA0 and ADANSA1 registers are invalid, and the channel selected by the DBLANS[4:0] bits is subjected to A/D conversion instead.
When double trigger mode is selected in group scan mode, double trigger mode operation is performed for group A only

and not performed for group B or C. Also, in double trigger mode, the analog inputs of multiple channels, temperature sensor outputs, or internal reference voltage cannot be selected for group A, but can be selected for groups B and C.
The DBLANS[4:0] bits should be set while the ADST bit is 0. They should not be set simultaneously when 1 is written to the ADST bit.

Table 56.5 Relationship between DBLANS[4:0] Bits Settings and Double Trigger Enabled Channels

S12AD (Unit 0)		S12AD1 (Unit 1)					
DBLANS[4:0]	Duplication Channel	DBLANS[4:0]	Duplication Channel	DBLANS[4:0]	Duplication Channel	DBLANS[4:0]	Duplication Channel
00000b	AN000	00000b	AN100	01000b	AN108	10000b	AN116
00001b	AN001	00001b	AN101	01001b	AN109	10001b	AN117
00010b	AN002	00010b	AN102	01010b	AN110	10010b	AN118
00011b	AN003	00011b	AN103	01011b	AN111	10011b	AN119
00100b	AN004	00100b	AN104	01100b	AN112	10100b	AN120
00101b	AN005	00101b	AN105	01101b	AN113		
00110b	AN006	00110b	AN106	01110b	AN114		
00111b	AN007	00111b	AN107	01111b	AN115		

Note: Duplication cannot be selected for the A/D conversion data of self-diagnosis, temperature sensor output, and internal reference voltage.

GBADIE Bit (Group B Scan End Interrupt Enable)

The GBADIE bit enables or disables group B scan end interrupt in group scan mode.
A scan end interrupt for group B is provided individually for each unit. Table 56.6 shows the relationship between each unit and the scan end interrupt for group B.

Table 56.6 Relationship between Each Unit and Group B Scan End Interrupt

Unit	Group B Scan End Interrupt
S12AD	S12GBADI
S12AD1	S12GBADI1

DBLE Bit (Double Trigger Mode Select)

Double trigger mode has a function to store the resulting data of A/D conversion started by the first and second synchronous triggers into separate registers.
When double trigger mode is selected, the channels specified in the ADANSA0 and ADANSA1 registers are invalid and the channel selected by the DBLANS[4:0] bits is effective instead. Double trigger mode can be only operated by the synchronous trigger selected by the ADSTRGR.TRSA[5:0] bits. Do not generate an asynchronous or software trigger.
The A/D conversion results started by the first trigger are stored into the A/D data register y and those started by the second trigger are stored into the A/D data duplication register. In this case, if the ADIE bit is set to 1, the interrupt is generated not upon completion of the first conversion but upon completion of the second conversion.
In continuous scan mode, double trigger mode should not be selected. In addition, double trigger mode should not be used for self-diagnosis, or conversion of the temperature sensor output and internal reference voltage. When using double trigger mode in group scan mode, A/D conversion of the internal reference voltage should not be selected for group A.
The DBLE bit should be set after the ADST bit has been set to 0.

EXTRG Bit (Trigger Select)

The EXTRG bit selects the synchronous trigger or the asynchronous trigger as the trigger for starting A/D conversion.
In group scan mode, the setting of this bit is valid for the selected trigger of group A.
For groups B and C, A/D conversion is started by the selected synchronous trigger regardless of this bit setting.

TRGE Bit (Trigger Start Enable)

The TRGE bit enables or disables A/D conversion by the synchronous trigger and the asynchronous trigger.
This bit should be set to 1 in group scan mode.

ADIE Bit (Scan End Interrupt Enable)

The ADIE bit enables or disables the A/D scan end interrupt (S12ADI) in scans except for groups B and C scan in group scan mode.
With double trigger mode deselected, the A/D scan conversion end is generated after the first scan is completed if the ADIE bit is set to 1.
When the extended analog input is selected also, if the ADIE bit is set to 1 when A/D conversion is completed, an A/D scan conversion end interrupt is generated.
With double trigger mode selected, the A/D scan conversion end is generated after the second scan is completed if the ADIE bit is set to 1 as long as the scan is started by the synchronous trigger selected by the ADSTRGR.TRSA[5:0] bits.
When scan is started by a software trigger, even with double trigger mode selected, the A/D scan conversion end interrupt is generated if the ADIE bit is set to 1 when the scan is completed. The A/D scan conversion end interrupt is provided individually for each unit. Table 56.7 shows the relationship between each unit and the A/D scan conversion end interrupt.

Table 56.7 Relationship between Each Unit and A/D scan conversion end Interrupt

Unit	A/D scan conversion end interrupt
S12AD	S12ADI
S12AD1	S12ADI1

ADCS[1:0] Bits (Scan Mode Select)

The ADCS[1:0] bits select the scan mode.
In single scan mode, A/D conversion is performed for the analog inputs selected with the ADANSA0 and ADANSA1 registers in the ascending order of the channel number, and when one cycle of A/D conversion is completed for all the selected channels, the scan conversion is stopped.*1
In continuous scan mode, while the ADCSR.ADST bit is 1, A/D conversion is performed for the analog inputs selected with the ADANSA0 and ADANSA1 registers in the ascending order of the channel number, and when one cycle of A/D conversion is completed for all the selected channels, A/D conversion is repeated from the first channel. If the ADCSR.ADST bit is set to 0 during continuous scan, A/D conversion is stopped even if scanning is in progress.*1
In group scan mode, A/D conversion is performed for the analog inputs (group A) selected with the ADANSA0 and ADANSA1 registers in the ascending order of the channel number after scanning is started by the synchronous trigger selected by the ADSTRGR.TRSA[5:0] bits, and when one cycle of A/D conversion is completed for all the selected channels, A/D conversion is stopped.*1 A/D conversion is also performed for the analog inputs (group B or C) selected with the ADANSB0 and ADANSB1 registers and the ADANSC0 and ADANSC1 registers in the ascending order of the channel number after A/D conversion is started by the synchronous trigger selected by the ADSTRGR.TRSB[5:0] bits and ADGCTRGR.TRSC[5:0] bits, and when A/D conversion is completed for all the selected channels, A/D conversion is stopped.*1
When selecting group scan mode, different channels and triggers should be selected for groups A, B, and C.
When using two groups while group scan mode is set, use groups A and B (ADGCTRGR.GRCE bit = 0). When using three groups, use groups A, B, and C (ADGCTRGR.GRCE bit = 1).
When selecting the extended analog input, select single scan mode or sequence scan mode.
The ADCS[1:0] bits should be set while the ADST bit is 0. They should not be set simultaneously when 1 is written to the ADST bit.

Note 1. When the temperature sensor output or internal reference voltage is selected, after A/D conversion of the analog input, A/D conversion is performed on the temperature sensor output and internal reference voltage in this order.

Table 56.8 Selection of Scan Mode, Double Trigger Mode, and Targets for A/D Conversion

Scan Mode Setup	Double Trigger Mode Setting	Targets for A/D Conversion					
		Self-diagnosis	Analogue Input (Including Group A)	Analogue Input (Group B and Group C)	Temperature Sensor Output	Internal Reference Voltage	Extended Analog Input
Single scan	DBLE = 0	✓	✓	×	✓	✓	✓
	DBLE = 1	×	✓ (1 channel only)	×	×	×	×
Sequence scan	DBLE = 0	✓	✓	×	✓	✓	✓
	DBLE = 1	×	×	×	×	×	×
Group scan	DBLE = 0	✓	✓	✓	✓	✓	×
	DBLE = 1	×	✓ (1 channel only)	✓	✓ (Group B and Group C)	✓ (Group B and Group C)	×

✓: Selectable, ×: Not selectable

Note: When selecting the extended analog input, deselect other targets for A/D conversion.

ADST Bit (A/D Conversion Start)

The ADST bit starts or stops A/D conversion process.

Before the ADST bit is set to 1, set the A/D conversion clock, the conversion mode, and conversion target analog input.

[Setting conditions]

- 1 is written by software.
- The synchronous trigger selected by the ADSTRGR.TRSA[5:0] bits is detected with ADCSR.EXTRG and ADCSR.TRGE bits being set to 0 and 1, respectively.
- The synchronous trigger selected by the ADSTRGR.TRSB[5:0] bits is detected with the ADCSR.TRGE bit being set to 1 in group scan mode.
- The asynchronous trigger is detected with the ADCSR.TRGE and ADCSR.EXTRG bits being set to 1 and the ADSTRGR.TRSA[5:0] bits being set to 000000b.
- With group priority control operation mode enabled (ADCSR.ADCS[1:0] bits = 01b and ADGSPCR.PGS bit = 1), a group B or C trigger is detected and A/D conversion of group B or C is started.
- With group priority control operation mode enabled (ADCSR.ADCS[1:0] bits = 01b and ADGSPCR.PGS bit = 1), the ADGSPCR.GBRSCN bit is set to 1 and A/D conversion of group B or C is restarted.
- With group priority control operation mode enabled (ADCSR.ADCS[1:0] bits = 01b and ADGSPCR.PGS bit = 1), the ADGSPCR.GBRP bit is set to 1 and A/D conversion of the lowest-priority group is started.

[Clearing conditions]

- 0 is written by software.
- The A/D conversion of all the selected channels, temperature sensor output, or the internal reference voltage (for S12AD1 only) is completed in single scan mode.
- A/D conversion of extended analog inputs is completed in single scan mode.
- Group A scan is completed in group scan mode.
- Group B scan is completed in group scan mode.
- Group C scan is completed in group scan mode.
- With group priority control operation mode enabled (ADCSR.ADCS[1:0] bits = 01b and ADGSPCR.PGS bit = 1), the ADGSPCR.GBRSCN bit is set to 1 and the scanning of the low-priority group started by a trigger is stopped.

Note: When group priority control operation mode has been enabled (ADCSR.ADCS[1:0] bits = 01b and ADGSPCR.PGS bit = 1), do not set the ADST bit to 1.

Note: When group priority control operation mode has been enabled (ADCSR.ADCS[1:0] bits = 01b and

ADGSPCR.PGS bit = 1) and ADGSPCR.GBRP = 1, do not set the ADST bit to 0. When forcibly terminating A/D conversion, follow the procedure for clearing the ADST bit.

Note: When the single-scan continuous function is used (ADGSPCR.GBRP bit = 1) with group priority control operation mode enabled (ADCSR.ADCS[1:0] bits = 01b and ADGSPCR.PGS bit = 1), the ADST bit remains as 1.

56.2.4 A/D Channel Select Register A0 (ADANSA0)

(1) S12AD.ADANSA0

Address(es): 0008 9004h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	ANSA0 07	ANSA0 06	ANSA0 05	ANSA0 04	ANSA0 03	ANSA0 02	ANSA0 01	ANSA0 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSA000	A/D Conversion Channel Select	0: AN000 to AN007 are not subjected to conversion. 1: AN000 to AN007 are subjected to conversion.	R/W
b1	ANSA001			R/W
b2	ANSA002			R/W
b3	ANSA003			R/W
b4	ANSA004			R/W
b5	ANSA005			R/W
b6	ANSA006			R/W
b7	ANSA007			R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

S12AD.ADANSA0 selects analog input channels for A/D conversion from among AN000 to AN007. In group scan mode, this register selects group A channels.

ANSA0n Bit (n = 00 to 07) (A/D Conversion Channel Select)

The ANSA0n bit select analog input channels for A/D conversion from among AN000 to AN007. The channels to be selected and the number of channels can be arbitrarily set. The ANSA000 bit corresponds to AN000 and the ANSA007 bit corresponds to AN007.

When double trigger mode is selected, the channel selected by S12AD.ADCSR.DBLANS[4:0] bits is selected in group A, and the ANSA0n bit setting is invalid.

The ANSA0n bit should be set while the S12AD.ADCSR.ADST bit is 0.

(2) S12AD1.ADANSA0

Address(es): 0008 9104h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	ANSA015	ANSA014	ANSA013	ANSA012	ANSA011	ANSA010	ANSA009	ANSA008	ANSA007	ANSA006	ANSA005	ANSA004	ANSA003	ANSA002	ANSA001	ANSA000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSA000	A/D Conversion Channel Select	0: AN100 to AN115 are not subjected to conversion. 1: AN100 to AN115 are subjected to conversion.	R/W
b1	ANSA001			R/W
b2	ANSA002			R/W
b3	ANSA003			R/W
b4	ANSA004			R/W
b5	ANSA005			R/W
b6	ANSA006			R/W
b7	ANSA007			R/W
b8	ANSA008			R/W
b9	ANSA009			R/W
b10	ANSA010			R/W
b11	ANSA011			R/W
b12	ANSA012			R/W
b13	ANSA013			R/W
b14	ANSA014			R/W
b15	ANSA015			R/W

S12AD1.ADANSA0 selects analog input channels for A/D conversion from among AN100 to AN115. In group scan mode, this register selects group A channels.

ANSA0n Bit (n = 00 to 15) (A/D Conversion Channel Select)

The ANSA0n bit select analog input channels for A/D conversion from among AN100 to AN115. The channels to be selected and the number of channels can be arbitrarily set. The ANSA000 bit corresponds to AN100 and the ANSA015 bit corresponds to AN115.

When double trigger mode is selected, the channel selected by the S12AD1.ADCSR.DBLANS[4:0] bits is selected in group A, and the ANSA0n bit setting is invalid.

The ANSA0n bit should be set while the S12AD1.ADCSR.ADST bit is 0.

56.2.5 A/D Channel Select Register A1 (ADANSA1)

(1) S12AD1.ADANSA1

Address(es): 0008 9106h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	ANSA1 04	ANSA1 03	ANSA1 02	ANSA1 01	ANSA1 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSA100	A/D Conversion Channel Select	0: AN116 to AN120 are not subjected to conversion. 1: AN116 to AN120 are subjected to conversion.	R/W
b1	ANSA101			R/W
b2	ANSA102			R/W
b3	ANSA103			R/W
b4	ANSA104			R/W
b15 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

S12AD1.ADANSA1 selects an analog input channel AN116 to AN120 for A/D conversion. In group scan mode, group A channels are to be selected.

ANSA1n Bit (n = 00 to 04) (A/D Conversion Channel Select)

The ANSA1n bit selects analog input channel AN116 to AN120 for A/D conversion. Channels to be selected and the number of channel can arbitrarily be specified. The ANSA100 bit corresponds to AN116 and the ANSA104 bit corresponds to AN120.

When double trigger mode is selected, the channel selected by the S12AD1.ADCSR.DBLANS[4:0] bits is selected in group A, and the ANSA1n bit setting is invalid.

The ANSA1n bit should be set while the S12AD1.ADCSR.ADST bit is 0.

56.2.6 A/D Channel Select Register B0 (ADANSB0)

(1) S12AD.ADANSB0

Address(es): 0008 9014h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	ANSB0 07	ANSB0 06	ANSB0 05	ANSB0 04	ANSB0 03	ANSB0 02	ANSB0 01	ANSB0 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSB000	A/D Conversion Channel Select	0: AN000 to AN007 are not subjected to conversion. 1: AN000 to AN007 are subjected to conversion.	R/W
b1	ANSB001			R/W
b2	ANSB002			R/W
b3	ANSB003			R/W
b4	ANSB004			R/W
b5	ANSB005			R/W
b6	ANSB006			R/W
b7	ANSB007			R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

S12AD.ADANSB0 selects analog input channels for A/D conversion from among AN000 to AN007 in group B when group scan mode is selected. The S12AD.ADANSB0 register is not used in any scan mode other than group scan mode.

ANSB0n Bit (n = 00 to 07) (A/D Conversion Channel Select)

The ANSB0n bit select analog input channels for A/D conversion from among AN000 to AN007 in group B when group scan mode is selected. The S12AD.ADANSB0 register is used for group scan mode only; not used for any other modes. The channels specified in group A (the channels corresponding to group A, selected with the S12AD.ADANSA0 and S12AD.ADANSA1 registers and the S12AD.ADCSR.DBLANS[4:0] bits in double trigger mode) should be excluded as the channels to be selected and the number of channels to be set.
The ANSB000 bit corresponds to AN000 and the ANSB007 bit corresponds to AN007.
The ANSB0n bit should be set while the S12AD.ADCSR.ADST bit is 0.

(2) S12AD1.ADANSB0

Address(es): 0008 9114h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	ANSB0 15	ANSB0 14	ANSB0 13	ANSB0 12	ANSB0 11	ANSB0 10	ANSB0 09	ANSB0 08	ANSB0 07	ANSB0 06	ANSB0 05	ANSB0 04	ANSB0 03	ANSB0 02	ANSB0 01	ANSB0 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSB000	A/D Conversion Channel Select	0: AN100 to AN115 are not subjected to conversion. 1: AN100 to AN115 are subjected to conversion.	R/W
b1	ANSB001			R/W
b2	ANSB002			R/W
b3	ANSB003			R/W
b4	ANSB004			R/W
b5	ANSB005			R/W
b6	ANSB006			R/W
b7	ANSB007			R/W
b8	ANSB008			R/W
b9	ANSB009			R/W
b10	ANSB010			R/W
b11	ANSB011			R/W
b12	ANSB012			R/W
b13	ANSB013			R/W
b14	ANSB014			R/W
b15	ANSB015			R/W

S12AD1.ADANSB0 selects analog input channels for A/D conversion from among AN100 to AN115 in group B when group scan mode is selected. The S12AD1.ADANSB0 register is not used in any scan mode other than group scan mode.

ANSB0n Bit (n = 00 to 15) (A/D Conversion Channel Select)

The ANSB0n bit select analog input channels for A/D conversion from among AN100 to AN115 in group B when group scan mode is selected. The S12AD1.ADANSB0 register is used for group scan mode only; not used for any other modes. The channels specified in group A (the channels corresponding to group A, selected with the S12AD1.ADANSA0 and S12AD1.ADANSA1 registers and the ADCSR.DBLANS[4:0] bits in double trigger mode) should be excluded as the channels to be selected and the number of channels to be set.
The ANSB000 bit corresponds to AN100 and the ANSB015 bit corresponds to AN115.
The ANSB0n bit should be set while the S12AD1.ADCSR.ADST bit is 0.

56.2.7 A/D Channel Select Register B1 (ADANSB1)

(1) S12AD1.ADANSB1

Address(es): 0008 9116h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	ANSB104	ANSB103	ANSB102	ANSB101	ANSB100
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSB100	A/D Conversion Channel Select	0: AN116 to AN120 are not subjected to conversion. 1: AN116 to AN120 are subjected to conversion.	R/W
b1	ANSB101			R/W
b2	ANSB102			R/W
b3	ANSB103			R/W
b4	ANSB104			R/W
b15 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

S12AD1.ADANSB1 selects analog input channels AN116 to AN120 for A/D conversion in group B when group scan mode is selected. The S12AD1.ADANSB1 register is not used in any scan mode other than group scan mode.

ANSB1n Bit (n = 00 to 04) (A/D Conversion Channel Select)

The ANSB1n bit select analog input channels AN116 to AN120 for A/D conversion in group B when group scan mode is selected. The S12AD1.ADANSB1 register is used for group scan mode only; not used for any other modes. The channels specified in group A (the channels corresponding to group A, selected with the S12AD1.ADANSA0, S12AD1.ADANSA1 registers and the S12AD1.ADCSR.DBLANS[4:0] bits in double trigger mode) should be excluded as the channels to be selected and the number of channels to be set.

The ANSB100 bit corresponds to AN116 and the ANSB104 bit corresponds to AN120.

The ANSB1n bit should be set while the S12AD1.ADCSR.ADST bit is 0.

56.2.8 A/D Channel Select Register C0 (ADANSC0)

(1) S12AD.ADANSC0

Address(es): 0008 90D4h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	ANSC007	ANSC006	ANSC005	ANSC004	ANSC003	ANSC002	ANSC001	ANSC000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSC000	A/D Conversion Channel Select	0: AN000 to AN007 are not subjected to conversion. 1: AN000 to AN007 are subjected to conversion.	R/W
b1	ANSC001			R/W
b2	ANSC002			R/W
b3	ANSC003			R/W
b4	ANSC004			R/W
b5	ANSC005			R/W
b6	ANSC006			R/W
b7	ANSC007			R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

S12AD.ADANSC0 selects analog input channels for A/D conversion from among AN000 to AN007 in group C when group scan mode is selected. The S12AD.ADANSC0 register is not used in any scan mode other than group scan mode.

ANSC0n Bit (n = 00 to 07) (A/D Conversion Channel Select)

The ANSC0n bit select analog input channels for A/D conversion from among AN000 to AN007 in group C when group scan mode is selected. The S12AD.ADANSC0 register is used for group scan mode only; not used for any other modes.
The channels specified in group A (the channels corresponding to group A, selected with the S12AD.ADANSA0 and S12AD.ADANSA1 registers and the S12AD.ADCSR.DBLANS[4:0] bits in double trigger mode) should be excluded as the channels to be selected and the number of channels to be set.
The ANSC000 bit corresponds to AN000 and the ANSC007 bit corresponds to AN007.
The ANSC0n bit should be set while the S12AD.ADCSR.ADST bit is 0.

(2) S12AD1.ADANSC0

Address(es): 0008 91D4h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	ANSC015	ANSC014	ANSC013	ANSC012	ANSC011	ANSC010	ANSC009	ANSC008	ANSC007	ANSC006	ANSC005	ANSC004	ANSC003	ANSC002	ANSC001	ANSC000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSC000	A/D Conversion Channel Select	0: AN100 to AN115 are not subjected to conversion. 1: AN100 to AN115 are subjected to conversion.	R/W
b1	ANSC001			R/W
b2	ANSC002			R/W
b3	ANSC003			R/W
b4	ANSC004			R/W
b5	ANSC005			R/W
b6	ANSC006			R/W
b7	ANSC007			R/W
b8	ANSC008			R/W
b9	ANSC009			R/W
b10	ANSC010			R/W
b11	ANSC011			R/W
b12	ANSC012			R/W
b13	ANSC013			R/W
b14	ANSC014			R/W
b15	ANSC015			R/W

S12AD1.ADANSC0 selects analog input channels for A/D conversion from among AN100 to AN115 in group C when group scan mode is selected. The S12AD1.ADANSC0 register is not used in any scan mode other than group scan mode.

ANSC0n Bit (n = 00 to 15) (A/D Conversion Channel Select)

The ANSC0n bit select analog input channels for A/D conversion from among AN100 to AN115 in group C when group scan mode is selected. The S12AD1.ADANSC0 register is used for group scan mode only; not used for any other modes.
The channels specified in group A (the channels corresponding to group A, selected with the S12AD1.ADANSA0 and S12AD1.ADANSA1 registers and the S12AD1.ADCSR.DBLANS[4:0] bits in double trigger mode) should be excluded as the channels to be selected and the number of channels to be set.
The ANSC000 bit corresponds to AN100 and the ANSC015 bit corresponds to AN115.
The ANSC0n bit should be set while the S12AD1.ADCSR.ADST bit is 0.

56.2.9 A/D Channel Select Register C1 (ADANSC1)

(1) S12AD1.ADANSC1

Address(es): 0008 91D6h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	ANSC1 04	ANSC1 03	ANSC1 02	ANSC1 01	ANSC1 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ANSC100	A/D Conversion Channel Select	0: AN116 to AN120 are not subjected to conversion. 1: AN116 to AN120 are subjected to conversion.	R/W
b1	ANSC101			R/W
b2	ANSC102			R/W
b3	ANSC103			R/W
b4	ANSC104			R/W
b15 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

S12AD1.ADANSC1 selects analog input channels AN116 to AN120 for A/D conversion in group C when group scan mode is selected. The S12AD1.ADANSC1 register is not used in any scan mode other than group scan mode.

ANSC1n Bit (n = 00 to 04) (A/D Conversion Channel Select)

The ANSC1n bit selects analog input channels AN116 to AN120 for A/D conversion in group C when group scan mode is selected. The S12AD1.ADANSC1 register is used for group scan mode only; not used for any other modes.

The channels specified in group A (the channels corresponding to group A, selected with the S12AD1.ADANSA0 and S12AD1.ADANSA1 registers and the S12AD1.ADCSR.DBLANS[4:0] bits in double trigger mode) should be excluded as the channels to be selected and the number of channels to be set.

The ANSC100 bit corresponds to AN116 and the ANSC104 bit corresponds to AN120

The ANSC1n bit should be set while the S12AD1.ADCSR.ADST bit is 0.

56.2.10 A/D-Converted Value Addition/Average Function Channel Select Register 0 (ADADS0)

(1) S12AD.ADADS0

Address(es): 0008 9008h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	ADS007	ADS006	ADS005	ADS004	ADS003	ADS002	ADS001	ADS000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ADS000	A/D-Converted Value Addition/ Average Channel Select	0: A/D-converted value addition/average mode for AN000 to AN007 is not selected. 1: A/D-converted value addition/average mode for AN000 to AN007 is selected.	R/W
b1	ADS001			R/W
b2	ADS002			R/W
b3	ADS003			R/W
b4	ADS004			R/W
b5	ADS005			R/W
b6	ADS006			R/W
b7	ADS007			R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

S12AD.ADADS0 selects channels AN000 to AN007 on which A/D conversion is performed successively 2, 3, 4, or 16 times and then converted values are added (integrated) or averaged.

ADS0n Bit (n = 00 to 07) (A/D-Converted Value Addition/Average Channel Select)

When the ADS0n bit of the number that is the same as that of A/D-converted channel selected by the S12AD.ADANSA0.ANSA0n bit (n = 00 to 07) or S12AD.ADCSR.DBLANS[4:0] bits and S12AD.ADANSB0.ANSB0n bit is set to 1, A/D conversion of analog input of the selected channels is performed successively 2, 3, 4, or 16 times that is set with the S12AD.ADADC.ADC[2:0] bits.

When the S12AD.ADADC.AVEE bit is 0, the value obtained by addition (integration) is stored in the A/D data register.

When the S12AD.ADADC.AVEE bit is 1, the mean value of the results obtained by addition (integration) is stored in the A/D data register.

As for the channel on which the A/D conversion is performed and addition/average mode is not selected, a normal one-time conversion is executed and the conversion result is stored to the A/D data register.

The ADS0n bit should be set while the S12AD.ADCSR.ADST bit is 0.

(2) S12AD1.ADADS0

Address(es): 0008 9108h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	ADS015	ADS014	ADS013	ADS012	ADS011	ADS010	ADS009	ADS008	ADS007	ADS006	ADS005	ADS004	ADS003	ADS002	ADS001	ADS000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ADS000	A/D-Converted Value Addition/ Average Channel Select	0: A/D-converted value addition/average mode for AN100 to AN115 is not selected. 1: A/D-converted value addition/average mode for AN100 to AN115 is selected.	R/W
b1	ADS001			R/W
b2	ADS002			R/W
b3	ADS003			R/W
b4	ADS004			R/W
b5	ADS005			R/W
b6	ADS006			R/W
b7	ADS007			R/W
b8	ADS008			R/W
b9	ADS009			R/W
b10	ADS010			R/W
b11	ADS011			R/W
b12	ADS012			R/W
b13	ADS013			R/W
b14	ADS014			R/W
b15	ADS015			R/W

S12AD1.ADADS0 selects channels AN100 to AN115 on which A/D conversion is performed successively 2, 3, 4, or 16 times and then converted values are added (integrated) or averaged.

ADS0n Bit (n = 00 to 15) (A/D-Converted Value Addition/Average Channel Select)

When the ADS0n bit of the number that is the same as that of A/D-converted channel selected by the S12AD1.ADANSA0.ANSA0n bit (n = 00 to 15) or S12AD1.ADCSR.DBLANS[4:0] bits and S12AD1.ADANSB0.ANSB0n bit and S12AD1.ADANSC0.ANSC0n bit is set to 1, A/D conversion of analog input of the selected channels is performed successively 2, 3, 4, or 16 times that is set with the S12AD1.ADADC.ADC[2:0] bits. When the S12AD1.ADADC.AVEE bit is 0, the value obtained by addition (integration) is stored in the A/D data register. When the S12AD1.ADADC.AVEE bit is 1, the mean value of the results obtained by addition (integration) is stored in the A/D data register.

As for the channel on which the A/D conversion is performed and addition/average mode is not selected, a normal one-time conversion is executed and the conversion result is stored to the A/D data register.

The ADS0n bit should be set while the S12AD1.ADCSR.ADST bit is 0.

56.2.11 A/D-Converted Value Addition/Average Function Channel Select Register 1 (ADADS1)

(1) S12AD1.ADADS1

Address(es): 0008 910Ah

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	ADS104	ADS103	ADS102	ADS101	ADS100
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ADS100	A/D-Converted Value Addition/ Average Channel Select	0: A/D-converted value addition/average mode for AN116 to AN120 is not selected. 1: A/D-converted value addition/average mode for AN116 to AN120 is selected.	R/W
b1	ADS101			R/W
b2	ADS102			R/W
b3	ADS103			R/W
b4	ADS104			R/W
b15 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

S12AD1.ADADS1 selects channels AN116 to AN120 on which A/D conversion is performed successively 2, 3, 4, or 16 times and then converted values are added (integrated) or averaged.

ADS1n Bit (n = 00 to 04) (A/D-Converted Value Addition/Average Channel Select)

When the ADS1n bit of the number that is the same as that of A/D-converted channel selected by the S12AD1.ADANSA1.ANSA1n bit (n = 00 to 04) or S12AD1.ADCSR.DBLANS[4:0] bits and S12AD1.ADANSB1.ANSB1n bit is set to 1, A/D conversion of analog input of the selected channels is performed successively 2, 3, 4, or 16 times that is set with the S12AD1.ADADC.ADC[2:0] bits. When the S12AD1.ADADC.AVEE bit is 0, the value obtained by addition (integration) is stored in the A/D data register. When the S12AD1.ADADC.AVEE bit is 1, the mean value of the results obtained by addition (integration) is stored in the A/D data register. As for the channel on which the A/D conversion is performed and addition/average mode is not selected, a normal one-time conversion is executed and the conversion result is stored to the A/D data register.
The ADS1n bit should be set while the S12AD1.ADCSR.ADST bit is 0.

Figure 56.3 shows a scanning operation sequence in which both the S12AD.ADADS0.ADS002 and S12AD.ADADS0.ADS006 bits are set to 1.
It is assumed that addition mode is selected (S12AD.ADADC.AVEE = 0) in continuous scan mode (S12AD.ADCSR.ADCS[1:0] = 10b), the addition count is set to three times (S12AD.ADADC.ADC[2:0] = 011b), and channels AN000 to AN007 are selected (S12AD.ADANSA0.ANSA0n = FFh). The conversion process begins with AN000. The AN002 conversion is performed successively four times (add three times), and the added (integrated) value is stored in A/D data register 2. After that the AN003 conversion is started. The AN006 conversion is performed successively four times and the added (integrated) value is stored in A/D data register 6. After conversion of AN007, the conversion operation is once again performed in the same sequence from AN000.

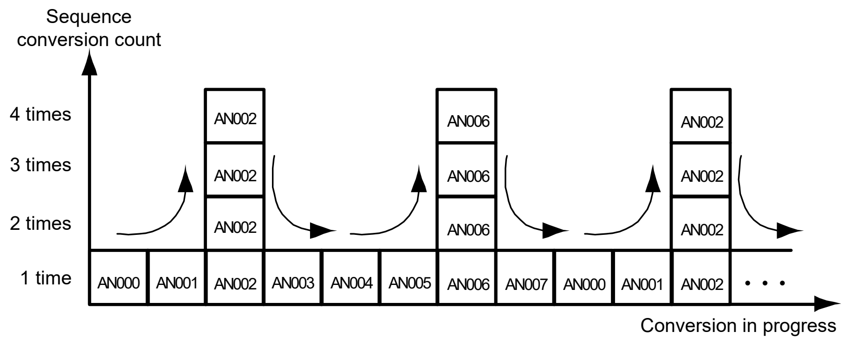


Figure 56.3 Scan Conversion Sequence with S12AD.ADADC.ADC[2:0] = 011b, S12AD.ADADC.AVEE = 0, S12AD.ADADS0.ADS002 = 1, and S12AD.ADADS0.ADS006 = 1

56.2.12 A/D-Converted Value Addition/Average Count Select Register (ADADC)

Address(es): S12AD.ADADC 0008 900Ch, S12AD1.ADADC 0008 910Ch

	b7	b6	b5	b4	b3	b2	b1	b0
	AVEE	—	—	—	—	ADC[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	ADC[2:0]	Addition Count Select	b2 b0 0 0 0: 1-time conversion (no addition; same as normal conversion) 0 0 1: 2-time conversion (addition once) 0 1 0: 3-time conversion (addition twice)*1 0 1 1: 4-time conversion (addition three times) 1 0 1: 16-time conversion (addition 15 times)*1 Settings other than above are prohibited.	R/W
b6 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	AVEE	Average Mode Enable	0: Addition mode is selected. 1: Average mode is selected.	R/W

Note 1. The AVEE bit is enabled only when 2-time or 4-time conversion is selected. When average mode is selected (ADADC.AVEE bit = 1), do not set 3-time conversion (ADADC.ADC[2:0] = 010b) nor 16-time conversion (ADADC.ADC[2:0] = 101b).

ADADC sets the addition count for A/D conversion of the channel, temperature sensor output, and internal reference voltage for which A/D-converted value addition/average mode is selected, and selects either addition or average mode.

ADC[2:0] Bits (Addition Count Select)

The ADC[2:0] bits set the addition count common to the channels for which A/D conversion and A/D-converted value addition/average mode is selected, including the channels selected in double trigger mode (by ADCSR.DBLANS[4:0] bits), and to A/D conversion of temperature sensor output or internal reference voltage.

When average mode is selected by setting the ADADC.AVEE bit to 1, do not set the addition count to one time (ADADC.ADC[2:0] = 000b), three times (ADADC.ADC[2:0] = 010b), or 16 times (ADADC.ADC[2:0] = 101b).

The ADC[2:0] bits should be set while the ADCSR.ADST bit is 0.

AVEE Bit (Average Mode Enable)

The AVEE bit selects addition or average mode for A/D conversion of the channel for which A/D conversion and A/D-converted value addition/average mode is selected, including the channels selected in double trigger mode (by ADCSR.DBLANS[4:0] bits), temperature sensor output, and internal reference voltage.

When average mode is selected by setting the ADADC.AVEE bit to 1, do not set the addition count to one time (ADADC.ADC[2:0] = 000b), three times (ADADC.ADC[2:0] = 010b), or 16 times (ADADC.ADC[2:0] = 101b). The mean value of 1-time, 3-time, and 16-time conversion cannot be obtained.

The AVEE bit should be set while the ADCSR.ADST bit is 0.

56.2.13 A/D Control Extended Register (ADCER)

Address(es): S12AD.ADCER 0008 900Eh, S12AD1.ADCER 0008 910Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	ADRFMT	—	—	—	DIAGM	DIAGLD	DIAGVAL[1:0]		—	—	ACE	—	—	ADPRC[1:0]		—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b2, b1	ADPRC[1:0]	A/D Conversion Resolution Setting	b2 b1 0 0: Perform A/D conversion with setting for 12-bit resolution 0 1: Perform A/D conversion with setting for 10-bit resolution 1 0: Perform A/D conversion with setting for 8-bit resolution 1 1: Setting prohibited	R/W
b4, b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b5	ACE	A/D Data Register Automatic Clearing Enable	0: Disables automatic clearing. 1: Enables automatic clearing.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b9, b8	DIAGVAL[1:0]	Self-Diagnosis Conversion Voltage Select	b9 b8 0 0: Setting prohibited in self-diagnosis voltage fixed mode 0 1: Uses the voltage of 0 V for self-diagnosis. 1 0: Uses the voltage of reference power supply × 1/2 for self-diagnosis. 1 1: Uses the voltage of reference power supply for self-diagnosis.	R/W
b10	DIAGLD	Self-Diagnosis Mode Select	0: Rotation mode for self-diagnosis voltage 1: Fixed mode for self-diagnosis voltage	R/W
b11	DIAGM	Self-Diagnosis Enable	0: Disables self-diagnosis of 12-bit A/D converter. 1: Enables self-diagnosis of 12-bit A/D converter.	R/W
b14 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15	ADRFMT	A/D Data Register Format Select	0: Flush-right is selected for the A/D data register format. 1: Flush-left is selected for the A/D data register format.	R/W

ADCER sets self-diagnosis mode, format of A/D data registers y (ADDRy), and automatic clearing of A/D data registers.

ADPRC[1:0] Bits (A/D Conversion Resolution Setting)

The ADPRC[1:0] bits select whether A/D conversion is performed with setting for 8-, 10-, or 12-bit resolution.
When the resolution of the A/D conversion is changed, the bit width of the valid data to be stored in the result register and time for A/D conversion are also changed. For details, refer to section 56.3.7, Analog Input Sampling Time and Scan Conversion Time.
The ADPRC[1:0] bits should be set while the ADCSR.ADST bit is 0.

ACE Bit (A/D Data Register Automatic Clearing Enable)

The ACE bit enables or disables automatic clearing (all 0) of ADDRy, ADRD, ADDBLDR, ADDBLDRA, ADDBLDRB, ADTSDR, or ADOCDR after any of these registers have been read by the CPU and DTC. Automatic clearing of the A/D data register is enabled to detect a failure which has not been updated in the A/D data register.

DIAGVAL[1:0] Bits (Self-Diagnosis Conversion Voltage Select)

The DIAGVAL[1:0] bits select the voltage value used in self-diagnosis voltage fixed mode. For details, refer to the descriptions of the ADCER.DIAGLD bit.
Self-diagnosis should not be executed by setting the ADCER.DIAGLD bit to 1 when the ADCER.DIAGVAL[1:0] bits are set to 00b.

DIAGLD Bit (Self-Diagnosis Mode Select)

The DIAGLD bit selects whether the three voltage values are rotated or the fixed voltage is used in self-diagnosis.
Setting this bit (ADCER.DIAGLD) to 0 allows conversion of the voltages in rotation mode where 0, the reference power supply × 1/2, and the reference power supply are converted in this order. When self-diagnosis rotation mode is selected after a reset, self-diagnosis is performed from 0 V. When self-diagnosis voltage fixed mode is selected, the fixed voltage specified by the ADCER.DIAGVAL[1:0] bits is converted. In self-diagnosis voltage rotation mode, the self-diagnosis voltage value does not return to 0 when scan conversion is completed. When scan conversion is restarted, therefore, rotation starts at the voltage value following the previous value. If fixed mode is switched to rotation mode, rotation starts at the fixed voltage value.
The DIAGLD bit should be set while the ADCSR.ADST bit is 0.

DIAGM Bit (Self-Diagnosis Enable)

The DIAGM bit enables or disables self-diagnosis.
Self-diagnosis is used to detect a failure of the 12-bit A/D converter. Specifically, one of the internally generated voltage values 0, the reference power supply × 1/2, and the reference power supply is converted. When conversion is completed, information on the converted voltage and the conversion result is stored into the self-diagnosis data register (ADRD). ADRD can then be read out by software to determine whether the conversion result falls within the normal range (normal) or not (abnormal). Self-diagnosis is executed once at the beginning of each scan, and one of the three voltages is converted. When self-diagnosis is selected in group scan mode, self-diagnosis is separately executed in groups A, B, and C.
The DIAGM bit should be set while the ADCSR.ADST bit is 0.

ADRFMT Bit (A/D Data Register Format Select)

The ADRFMT bit specifies flush-right or flush-left for the data to be stored in ADDRy, ADDBLDR, ADDBLDRA, ADDBLDRB, ADTSDR, ADOCDR, ADRD, ADCMPDR0, ADCMPDR1, ADWINLLB, or ADWINULB.
The ADRFMT bit should be set while the ADCSR.ADST bit is 0.

For details on the format of each data register, see section 56.2.1, A/D Data Registers y (ADDRy) (y = 0 to 20), A/D Data Duplication Register (ADDBLDR), A/D Data Duplication Register A (ADDBLDRA), A/D Data Duplication Register B (ADDBLDRB), A/D Temperature Sensor Data Register (ADTSDR), A/D Internal Reference Voltage Data Register (ADOCDR), section 56.2.2, A/D Self-Diagnosis Data Register (ADRD), section 56.2.30, A/D Comparison Function Window A Lower Level Setting Register (ADCMPDR0), section 56.2.31, A/D Comparison Function Window A Upper Level Setting Register (ADCMPDR1), section 56.2.37, A/D Comparison Function Window B Lower Level Setting Register (ADWINLLB), and section 56.2.38, A/D Comparison Function Window B Upper Level Setting Register (ADWINULB).

56.2.14 A/D Conversion Start Trigger Select Register (ADSTRGR)

Address(es): S12AD.ADSTRGR 0008 9010h, S12AD1.ADSTRGR 0008 9110h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	TRSA[5:0]						—	—	TRSB[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	TRSB[5:0]	A/D Conversion Start Trigger Select for Group B	Select the A/D conversion start trigger for group B in group scan mode.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13 to b8	TRSA[5:0]	A/D Conversion Start Trigger Select	Select the A/D conversion start trigger in single scan mode and continuous scan mode. In group scan mode, the A/D conversion start trigger for group A is selected.	R/W
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

ADSTRGR selects the A/D conversion start trigger.

TRSB[5:0] Bits (A/D Conversion Start Trigger Select for Group B)

The TRSB[5:0] bits select the trigger to start scanning of the analog input selected in group B. The TRSB[5:0] bits require to be set only in group scan mode and are not used in any other scan mode. For the scan conversion start trigger for group B, setting a software trigger or an asynchronous trigger is prohibited. Therefore, the TRSB[5:0] bits should be set to the value other than 000000b and the ADCSR.TRGE bit should be set to 1 in group scan mode.
When two groups are selected (ADGCTRGR.GRCE = 0) during group priority operation in group scan mode, setting the ADGSPCR.GBRP bit to 1 allows group B to continuously operate in single scan mode. When setting the ADGSPCR.GBRP bit to 1, set the TRSB[5:0] bits to 3Fh. Note that the issuance period of trigger for A/D conversion must be more than or equal to the actual scan conversion time (t_{SCAN}). If the issuance period is less than t_{SCAN}, A/D conversion by the trigger may have no effect.
When the trigger from the module (MTU) operated in 120 MHz is selected as an A/D conversion start trigger, a delay of the period for synchronization processing occurs. See section 56.3.7, Analog Input Sampling Time and Scan Conversion Time for details.
Table 56.9 lists the A/D conversion startup sources selected by the TRSB[5:0] bits.

TRSA[5:0] Bits (A/D Conversion Start Trigger Select)

The TRSA[5:0] bits select the trigger to start A/D conversion in single scan mode and continuous scan mode. In group scan mode, the trigger to start scanning of the analog input selected in group A is selected. When scanning is executed in group scan mode or double trigger mode, set the ADCSR.TRGE bit to 1.

- When using the A/D conversion startup source of a synchronous trigger, set the ADCSR.TRGE bit to 1 and set the ADCSR.EXTRG bit to 0.
- When using the asynchronous trigger, set the ADCSR.TRGE bit to 1 and set the ADCSR.EXTRG bit to 1.
- Software trigger (ADCSR.ADST) is enabled regardless of the settings of the ADCSR.TRGE bit, the ADCSR.EXTRG bit, and the TRSA[5:0] bits.

Note that the issuance period of trigger for A/D conversion must be more than or equal to the actual scan conversion time (t_{SCAN}). If the issuance period is less than t_{SCAN}, A/D conversion by a trigger may have no effect. When the trigger from the module (MTU) operated in 120 MHz is selected as an A/D conversion start trigger, a delay of the period for synchronization processing occurs. See section 56.3.7, Analog Input Sampling Time and Scan Conversion Time for details.

Table 56.10 lists the selection of A/D conversion start sources selected by the TRSA[5:0] bits.

Table 56.9 Selection of A/D Activation Sources by the TRSB[5:0] Bits

Module	Source	Remarks	TRSB[5]	TRSB[4]	TRSB[3]	TRSB[2]	TRSB[1]	TRSB[0]
Trigger source deselection state			1	1	1	1	1	1
MTU	TRGA0N	Compare match/input capture from MTU0.TGRA	0	0	0	0	0	1
	TRGA1N	Compare match/input capture from MTU1.TGRA	0	0	0	0	1	0
	TRGA2N	Compare match/input capture from MTU2.TGRA	0	0	0	0	1	1
	TRGA3N	Compare match/input capture from MTU3.TGRA	0	0	0	1	0	0
	TRGA4N	Compare match/input capture from MTU4.TGRA or underflow (trough) of MTU4.TCNT in complementary PWM mode	0	0	0	1	0	1
	TRGA6N	Compare match/input capture from MTU6.TGRA	0	0	0	1	1	0
	TRGA7N	Compare match/input capture from MTU7.TGRA or underflow (trough) of MTU7.TCNT in complementary PWM mode	0	0	0	1	1	1
	TRG0N	Compare match from MTU0.TGRE	0	0	1	0	0	0
	TRG4AN	Compare match between MTU4.TADCORA and MTU4.TCNT	0	0	1	0	0	1
	TRG4BN	Compare match between MTU4.TADCORB and MTU4.TCNT	0	0	1	0	1	0
	TRG4AN or TRG4BN	Compare match between MTU4.TADCORA and MTU4.TCNT, or compare match between MTU4.TADCORB and MTU4.TCNT	0	0	1	0	1	1
	TRG4ABN	Compare match between MTU4.TADCORA and MTU4.TCNT, and compare match between MTU4.TADCORB and MTU4.TCNT (when interrupt skipping function 2 is used)	0	0	1	1	0	0
	TRG7AN	Compare match between MTU7.TADCORA and MTU7.TCNT	0	0	1	1	0	1
	TRG7BN	Compare match between MTU7.TADCORB and MTU7.TCNT	0	0	1	1	1	0
	TRG7AN or TRG7BN	Compare match between MTU7.TADCORA and MTU7.TCNT, or compare match between MTU7.TADCORB and MTU7.TCNT	0	0	1	1	1	1
	TRG7ABN	Compare match between MTU7.TADCORA and MTU7.TCNT, and compare match between MTU7.TADCORB and MTU7.TCNT (when interrupt skipping function 2 is used)	0	1	0	0	0	0
TMR	TMTRG0AN_0	Compare match between TMR0.TCORA and TMR0.TCNT	0	1	1	1	0	1
	TMTRG0AN_1	Compare match between TMR2.TCORA and TMR2.TCNT	0	1	1	1	1	0
TPU	TPTRGAN	Compare match/input capture from TPUn.TGRAn (n = 0 to 5)	0	1	1	1	1	1
	TPTRG0AN	Compare match/input capture from TPU0.TGRA0	1	0	0	0	0	0
ELC	ELCTRG00N, ELCTRG10N	A/D start source 0 from the ELC	0	1	0	0	0	1
	ELCTRG01N, ELCTRG11N	A/D start source 1 from the ELC	0	1	0	0	1	0
	ELCTRG00N or ELCTRG01N, ELCTRG10N or ELCTRG11N	A/D start source 0 from the ELC or A/D start source 1 from the ELC	0	1	1	0	0	1

Table 56.10 Selection of A/D Activation Sources by the TRSA[5:0] Bits

Module	Source	Remarks	TRSA[5]	TRSA[4]	TRSA[3]	TRSA[2]	TRSA[1]	TRSA[0]
Trigger source deselection state			1	1	1	1	1	1
MTU	TRGA0N	Compare match/input capture from MTU0.TGRA	0	0	0	0	0	1
	TRGA1N	Compare match/input capture from MTU1.TGRA	0	0	0	0	1	0
	TRGA2N	Compare match/input capture from MTU2.TGRA	0	0	0	0	1	1
	TRGA3N	Compare match/input capture from MTU3.TGRA	0	0	0	1	0	0
	TRGA4N	Compare match/input capture from MTU4.TGRA or underflow (trough) of MTU4.TCNT in complementary PWM mode	0	0	0	1	0	1
	TRGA6N	Compare match/input capture from MTU6.TGRA	0	0	0	1	1	0
	TRGA7N	Compare match/input capture from MTU7.TGRA or underflow (trough) of MTU7.TCNT in complementary PWM mode	0	0	0	1	1	1
	TRG0N	Compare match from MTU0.TGRE	0	0	1	0	0	0
	TRG4AN	Compare match between MTU4.TADCORA and MTU4.TCNT	0	0	1	0	0	1
	TRG4BN	Compare match between MTU4.TADCORB and MTU4.TCNT	0	0	1	0	1	0
	TRG4AN or TRG4BN	Compare match between MTU4.TADCORA and MTU4.TCNT, or compare match between MTU4.TADCORB and MTU4.TCNT	0	0	1	0	1	1
	TRG4ABN	Compare match between MTU4.TADCORA and MTU4.TCNT, and compare match between MTU4.TADCORB and MTU4.TCNT (when interrupt skipping function 2 is used)	0	0	1	1	0	0
	TRG7AN	Compare match between MTU7.TADCORA and MTU7.TCNT	0	0	1	1	0	1
	TRG7BN	Compare match between MTU7.TADCORB and MTU7.TCNT	0	0	1	1	1	0
	TRG7AN or TRG7BN	Compare match between MTU7.TADCORA and MTU7.TCNT, or compare match between MTU7.TADCORB and MTU7.TCNT	0	0	1	1	1	1
	TRG7ABN	Compare match between MTU7.TADCORA and MTU7.TCNT, and compare match between MTU7.TADCORB and MTU7.TCNT (when interrupt skipping function 2 is used)	0	1	0	0	0	0
TMR	TMTRG0AN_0	Compare match between TMR0.TCORA and TMR0.TCNT	0	1	1	1	0	1
	TMTRG0AN_1	Compare match between TMR2.TCORA and TMR2.TCNT	0	1	1	1	1	0
TPU	TPTRGAN	Compare match/input capture from TPUn.TGRAn (n = 0 to 5)	0	1	1	1	1	1
	TPTRG0AN	Compare match/input capture from TPU0.TGRA0	1	0	0	0	0	0
ELC	ELCTRG00N/ ELCTRG10N	A/D start source 0 from the ELC	0	1	0	0	0	1
	ELCTRG01N/ ELCTRG11N	A/D start source 1 from the ELC	0	1	0	0	1	0
	ELCTRG00N or ELCTRG01N, ELCTRG10N or ELCTRG11N	A/D start source 0 from the ELC or A/D start source 1 from the ELC	0	1	1	0	0	1

56.2.15 A/D Conversion Extended Input Control Register (ADEXICR)

Address(es): S12AD1.ADEXICR 0008 9112h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	EXOEN	EXSEL[1:0]		—	OCSB	TSSB	OCSA	TSSA	—	—	—	—	—	—	OCSAD	TSSAD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	TSSAD	Temperature Sensor Output A/D-Converted Value Addition/Average Mode Select	0: Temperature sensor output A/D-converted value addition/average mode is not selected. 1: Temperature sensor output A/D-converted value addition/average mode is selected.	R/W
b1	OCSAD	Internal Reference Voltage A/D-Converted Value Addition/Average Mode Select	0: Internal reference voltage A/D-converted value addition/average mode is not selected. 1: Internal reference voltage A/D-converted value addition/average mode is selected.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	TSSA	Temperature Sensor Output A/D Conversion Select	0: A/D conversion of temperature sensor output is not performed. 1: A/D conversion of temperature sensor output is performed.	R/W
b9	OCSA	Internal Reference Voltage A/D Conversion Select	0: A/D conversion of internal reference voltage is not performed. 1: A/D conversion of internal reference voltage is performed.	R/W
b10	TSSB	Temperature Sensor Output A/D Conversion Select	0: A/D conversion of temperature sensor output is not performed. 1: A/D conversion of temperature sensor output is performed.	R/W
b11	OCSB	Internal Reference Voltage A/D Conversion Select	0: A/D conversion of internal reference voltage is not performed. 1: A/D conversion of internal reference voltage is performed.	R/W
b12	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b14, b13	EXSEL[1:0]	Extended Analog Input Select	b14 b13 0 0: Analog input channels (ANn) 0 1: ANEX1 1 0: Setting prohibited 1 1: Setting prohibited	R/W
b15	EXOEN	Extended Analog Output Control	0: Output disabled 1: Output enabled	R/W

ADEXICR specifies the settings of A/D conversion of the temperature sensor output, internal reference voltage, or extended analog input.

TSSAD Bit (Temperature Sensor Output A/D-Converted Value Addition/Average Mode Select)

When A/D conversion of the temperature sensor output is selected and the TSSAD bit is set to 1, A/D conversion of the temperature sensor output is performed in sequence for the count set in the ADADC.ADC[2:0] bits (2, 3, 4, or 16 times). The added (accumulated) total value is returned to the A/D temperature sensor data register (ADTSDR) when the ADADC.AVEE bit is 0 and the average is returned to ADTSDR when the ADADC.AVEE bit is 1. The TSSAD bit should be set while the ADCSR.ADST bit is 0.

OCSAD Bit (Internal Reference Voltage A/D-Converted Value Addition/Average Mode Select)

When the OCSAD bit is set to 1, A/D conversion of the internal reference voltage is selected and performed successively 2, 3, 4, or 16 times that is set with the ADADC.ADC[2:0] bits. When the ADADC.AVEE bit is 0, the value obtained by addition (integration) is stored in the A/D internal reference voltage data register (ADOCDR). When the ADADC.AVEE bit is 1, the mean value is stored in ADOCDR.

The OCSAD bit should be set while the ADCSR.ADST bit is 0.

TSSA Bit (Temperature Sensor Output A/D Conversion Select)

This bit selects A/D conversion of the temperature sensor output for group A in single scan mode, sequence scan mode, or group scan mode. When A/D conversion of the temperature sensor output is performed, set the ADCSR.DBLE bit to 0.

The TSSA bit should be set while the ADCSR.ADST bit is 0.

OCSA Bit (Internal Reference Voltage A/D Conversion Select)

This bit selects A/D conversion of the internal reference voltage for group A in single scan mode, sequence scan mode, or group scan mode. When A/D conversion of the internal reference voltage is performed, set the ADCSR.DBLE bit to 0. The OCSA bit should be set while the ADCSR.ADST bit is 0. Wait for at least 400 ns after setting the OCSA bit to 1 to start A/D conversion.

TSSB Bit (Temperature Sensor Output A/D Conversion Select)

This bit selects A/D conversion of the temperature sensor output for group B in group scan mode.

This bit should be set while the ADCSR.ADST bit is 0. When the TSSA bit is 1, do not set the TSSB bit to 1.

OCSB Bit (Internal Reference Voltage A/D Conversion Select)

This bit selects A/D conversion of the internal reference voltage for group B in group scan mode.

This bit should be set while the ADCSR.ADST bit is 0. When the OCSA bit is 1, do not set the OCSB bit to 1. Wait for at least 400 ns after setting the OCSB bit to 1 to start A/D conversion.

EXSEL[1:0] Bits (Extended Analog Input Select)

These bits can select extended analog input ANEX1 besides analog input channels (ANn).

When ANEX1 is selected, ANEX0 should be input to ANEX1 through an external operational amplifier. In addition, only analog input channels AN100 to AN107 are selectable. Do not select channels AN108 to AN120. For details, refer to section 56.3.5.1, Usage of ANEX1.

EXOEN Bit (Extended Analog Output Control)

This bit controls the extended analog output (ANEX0). When the output is enabled, the multiplexed value of AN100 to AN107 among the analog input channels of unit 1. Do not enable output while the EXSEL[1:0] bits are 00b.

56.2.16 A/D Group C Extended Input Control Register (ADGCEXCR)

Address(es): S12AD1.ADGCEXCR 0008 91D8h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	OCSC	TSSC
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	TSSC	Group C Temperature Sensor Output A/D Conversion Select	0: Temperature sensor output is not A/D converted 1: Temperature sensor output is A/D converted	R/W
b1	OCSC	Group C Internal Reference Voltage A/D Conversion Select	0: Internal reference voltage is not A/D converted 1: Internal reference voltage is A/D converted	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The ADGCEXCR register specifies extended input for group C.

TSSC Bit (Group C Temperature Sensor Output A/D Conversion Select)

This bit selects A/D conversion of the temperature sensor output for group C in group scan mode.
The TSSC bit should be set while the ADCSR.ADST bit is 0. The TSSC bit should not be set to 1 while the TSSA or TSSB bit is 1.

OCSC Bit (Group C Internal Reference Voltage A/D Conversion Select)

This bit selects A/D conversion of the internal reference voltage for group C in group scan mode.
The OCSC bit should be set while the ADCSR.ADST bit is 0. The OCSC bit should not be set while the OCSA or OCSB bit is 1. Wait for at least 400 ns after setting the OCSC bit to 1 to start A/D conversion.

56.2.17 A/D Group C Trigger Select Register (ADGCTRGR)

Address(es): S12AD.ADGCTRGR 0008 90D9h, S12AD1.ADGCTRGR 0008 91D9h

	b7	b6	b5	b4	b3	b2	b1	b0
	GRCE	GCADIE	TRSC[5:0]					
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	TRSC[5:0]	Group C A/D Conversion Start Trigger Select	Select the A/D conversion start trigger for group C in group scan mode.	R/W
b6	GCADIE	Group C Scan End Interrupt Enable	0: Disables interrupt generation after completion of group C scan 1: Enables interrupt generation after completion of group C scan	R/W
b7	GRCE	Group C A/D Conversion Operation Enable	Enables A/D conversion operation for group C. 0: Group C is not used 1: Group C is used	R/W

ADGCTRGR enables operation for group C and selects the A/D conversion start trigger. For details on group priority operation, see Table 56.16 and Table 56.17.

TRSC[5:0] Bits (Group C A/D Conversion Start Trigger Select)

These bits select the trigger to start scanning of the analog input selected in group C. These bits are used for group scan mode only; not used for any other modes. Software trigger or asynchronous trigger cannot be set as the scan conversion trigger for group C. When using group C in group scan mode, set the TRSC[5:0] bits to a value other than 000000b, set the ADCSR.TRGE bit to 1, and set the GRCE bit to 1.

When group C is used during group priority control in group scan mode and the ADGSPCR.GBRP bit is set to 1, group C can be continuously operated in single scan mode. When continuously operating group C in single scan mode, set the TRSC[5:0] bits to 3Fh and disable trigger selection.

Note that the issuance period of trigger for A/D conversion must be more than or equal to the actual scan conversion time (t_{SCAN}). If the issuance period is less than tSCAN, A/D conversion by a trigger may have no effect.

When the trigger from the module (MTU) operated in 120 MHz is selected as an A/D conversion start trigger, a delay of the period for synchronization processing occurs. See section 56.3.7, Analog Input Sampling Time and Scan Conversion Time for details.

Table 56.11 lists the selection of A/D conversion start sources selected by the TRSA[5:0] bits for group C.

Table 56.11 Selection of A/D Activation Sources by the TRSC[5:0] Bits

Module	Source	Remarks	TRSC[5]	TRSC[4]	TRSC[3]	TRSC[2]	TRSC[1]	TRSC[0]
Trigger source deselection state			1	1	1	1	1	1
MTU	TRGA0N	Compare match/input capture from MTU0.TGRA	0	0	0	0	0	1
	TRGA1N	Compare match/input capture from MTU1.TGRA	0	0	0	0	1	0
	TRGA2N	Compare match/input capture from MTU2.TGRA	0	0	0	0	1	1
	TRGA3N	Compare match/input capture from MTU3.TGRA	0	0	0	1	0	0
	TRGA4N	Compare match/input capture from MTU4.TGRA or underflow (trough) of MTU4.TCNT in complementary PWM mode	0	0	0	1	0	1
	TRGA6N	Compare match/input capture from MTU6.TGRA	0	0	0	1	1	0
	TRGA7N	Compare match/input capture from MTU7.TGRA or underflow (trough) of MTU7.TCNT in complementary PWM mode	0	0	0	1	1	1
	TRG0N	Compare match from MTU0.TGRE	0	0	1	0	0	0
	TRG4AN	Compare match between MTU4.TADCORA and MTU4.TCNT	0	0	1	0	0	1
	TRG4BN	Compare match between MTU4.TADCORB and MTU4.TCNT	0	0	1	0	1	0
	TRG4AN or TRG4BN	Compare match between MTU4.TADCORA and MTU4.TCNT, or compare match between MTU4.TADCORB and MTU4.TCNT	0	0	1	0	1	1
	TRG4ABN	Compare match between MTU4.TADCORA and MTU4.TCNT, and compare match between MTU4.TADCORB and MTU4.TCNT (when interrupt skipping function 2 is used)	0	0	1	1	0	0
	TRG7AN	Compare match between MTU7.TADCORA and MTU7.TCNT	0	0	1	1	0	1
	TRG7BN	Compare match between MTU7.TADCORB and MTU7.TCNT	0	0	1	1	1	0
	TRG7AN or TRG7BN	Compare match between MTU7.TADCORA and MTU7.TCNT, or compare match between MTU7.TADCORB and MTU7.TCNT	0	0	1	1	1	1
	TRG7ABN	Compare match between MTU7.TADCORA and MTU7.TCNT, and compare match between MTU7.TADCORB and MTU7.TCNT (when interrupt skipping function 2 is used)	0	1	0	0	0	0
TMR	TMTRG0AN_0	Compare match between TMR0.TCORA and TMR0.TCNT	0	1	1	1	0	1
	TMTRG0AN_1	Compare match between TMR2.TCORA and TMR2.TCNT	0	1	1	1	1	0
TPU	TPTRGAN	Compare match/input capture from TPUn.TGRAn (n = 0 to 5)	0	1	1	1	1	1
	TPTRG0AN	Compare match/input capture from TPU0.TGRA0	1	0	0	0	0	0
ELC	ELCTRG00N/ ELCTRG10N	A/D start source 0 from the ELC	0	1	0	0	0	1
	ELCTRG01N/ ELCTRG11N	A/D start source 1 from the ELC	0	1	0	0	1	0
	ELCTRG00N or ELCTRG01N, ELCTRG10N or ELCTRG11N	A/D start source 0 from the ELC or A/D start source 1 from the ELC	0	1	1	0	0	1

GCADIE Bit (Group C Scan End Interrupt Enable)

This bit enables or disables scan end interrupt generation for group C. A scan end interrupt for group C is provided individually for each unit. Table 56.12 shows the relationship between each unit and the scan end interrupt for group C.

Table 56.12 Relationship between Each Unit and Group C Scan End Interrupt

Unit	Group C Scan End Interrupt
S12AD	S12GCADI
S12AD1	S12GCADI1

GRCE Bit (Group C A/D Conversion Operation Enable)

When using group C in group scan mode, set the GRCE bit to 1.
When the GRCE bit is 0, trigger input for group C is disabled.
During group priority operation (the ADGSPCR.PGS bit is 1) with group C used, when the ADGSPCR.GBRP bit is set to 1, single scan for group C is continuously operated. When the GRCE bit is set to 1, single scan for group B is not

continuously operated.
The GRCE bit should be set while the ADCSR.ADST bit is 0.

56.2.18 A/D Sampling State Register n (ADSSTRn) (n = 0 to 15, L, T, O)

Address(es): S12AD.ADSSTR0 0008 90E0h, S12AD.ADSSTR1 0008 90E1h, S12AD.ADSSTR2 0008 90E2h, S12AD.ADSSTR3 0008 90E3h, S12AD.ADSSTR4 0008 90E4h, S12AD.ADSSTR5 0008 90E5h, S12AD.ADSSTR6 0008 90E6h, S12AD.ADSSTR7 0008 90E7h, S12AD1.ADSSTRL 0008 91DDh, S12AD1.ADSSTRT 0008 91DEh, S12AD1.ADSSTRO 0008 91DFh, S12AD1.ADSSTR0 0008 91E0h, S12AD1.ADSSTR1 0008 91E1h, S12AD1.ADSSTR2 0008 91E2h, S12AD1.ADSSTR3 0008 91E3h, S12AD1.ADSSTR4 0008 91E4h, S12AD1.ADSSTR5 0008 91E5h, S12AD1.ADSSTR6 0008 91E6h, S12AD1.ADSSTR7 0008 91E7h, S12AD1.ADSSTR8 0008 91E8h, S12AD1.ADSSTR9 0008 91E9h, S12AD1.ADSSTR10 0008 91EAh, S12AD1.ADSSTR11 0008 91EBh, S12AD1.ADSSTR12 0008 91ECh, S12AD1.ADSSTR13 0008 91EDh, S12AD1.ADSSTR14 0008 91EEh, S12AD1.ADSSTR15 0008 91EFh

	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	1	0	1	1

The ADSSTRn register sets the sampling time for analog input.
If one state is one ADCLK (A/D conversion clock) cycle and the ADCLK clock is 60 MHz, one state is 16.7 ns. The initial value is 11 states. If the impedance of analog input signal source is too high to secure sufficient sampling time or if the ADCLK clock is slow, the sampling time can be adjusted. The ADSSTRn register should be set while the ADCSR.ADST bit is 0. The lower-limit value for sampling time differs depending on the PCLK to ADCLK frequency ratio.
Set a value that is 5 states or more when PCLK to ADCLK frequency ratio = 1:1, 2:1, 4:1, or 8:1.

Table 56.13 shows the relationship between the A/D sampling state register and the relevant channels.
For details, refer to section 56.3.7, Analog Input Sampling Time and Scan Conversion Time.

Table 56.13 Relationship between A/D Sampling State Register and Relevant Channels

Unit	Register Name	Relevant Channels
S12AD	ADSSTR0 register	AN000, Self-Diagnosis
	ADSSTR1 register	AN001
	ADSSTR2 register	AN002
	ADSSTR3 register	AN003
	ADSSTR4 register	AN004
	ADSSTR5 register	AN005
	ADSSTR6 register	AN006
	ADSSTR7 register	AN007
S12AD1	ADSSTR0 register	AN100, Self-Diagnosis
	ADSSTR1 register	AN101
	ADSSTR2 register	AN102
	ADSSTR3 register	AN103
	ADSSTR4 register	AN104
	ADSSTR5 register	AN105
	ADSSTR6 register	AN106
	ADSSTR7 register	AN107
	ADSSTR8 register	AN108
	ADSSTR9 register	AN109
	ADSSTR10 register	AN110
	ADSSTR11 register	AN111
	ADSSTR12 register	AN112
	ADSSTR13 register	AN113
	ADSSTR14 register	AN114
	ADSSTR15 register	AN115
	ADSSTRL register	AN116 to AN120
	ADSSTRT register	Temperature sensor output
	ADSSTRO register	Internal reference voltage

Note 1. When performing A/D conversion of the temperature sensor output or internal reference voltage, the sampling time should be 5 µs or longer.

56.2.19 A/D Sample-and-Hold Circuit Control Register (ADSHCR)

Address(es): S12AD.ADSHCR 0008 9066h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	SHANS[2:0]			SSTSH[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	SSTSH[7:0]	Channel-Dedicated Sample-and-Hold Circuit Sampling Time Setting	Set the sampling time (4 to 255 states).	R/W
b10 to b8	SHANS[2:0]	Channel-Dedicated Sample-and-Hold Circuit Bypass Select	Select whether to use or not use (bypass) AN000 to AN002 channel-dedicated sample-and-hold circuits. 0: Bypass the channel-dedicated sample-and-hold circuits. 1: Use the channel-dedicated sample-and-hold circuits.	R/W
b15 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

ADSHCR sets the parameters related to channel-dedicated sample-and-hold circuits.

SSTSH[7:0] Bits (Channel-Dedicated Sample-and-Hold Circuit Sampling Time Setting)

These bits set the sampling time for the channel-dedicated sample-and-hold circuits when the ADSHMSR.SHMD bit is 0. If one state is one ADCLK (A/D conversion clock) cycle and the ADCLK clock is 60 MHz, one state is 16.7 ns. The initial value is 24 states. If the impedance of analog input signal source is too high to secure sufficient sampling time or if the ADCLK clock is slow, the sampling time can be adjusted. The SSTSH[7:0] bits should be set while the ADCSR.ADST bit is 0. The sampling time must be set to a value that is 4 states or more and is 255 or less. Also, the sampling time should be 0.4 µs or longer.

For example, when ADCLK is set to 60 MHz, the lower-limit of the sampling state setting value is 24 states.

SHANS[2:0] Bits (Channel-Dedicated Sample-and-Hold Circuit Bypass Select)

These bits select whether to use or not use (bypass) AN000 to AN002 channel-dedicated sample-and-hold circuits. The SHANS[0] bit selects AN000, SHANS[1] bit selects AN001, and SHANS[2] bit selects AN002. The SHANS[2:0] bits should be set while the ADCSR.ADST bit is 0.

If any channel from among AN000 to AN002 is selected for group B or C while operation is in group scan mode under group priority control, make the setting to bypass the channel-dedicated sample-and-hold circuit.

56.2.20 A/D Sample-and-Hold Operating Mode Select Register (ADSHMSR)

Address(es): S12AD.ADSHMSR 0008 907Ch

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	SHMD
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SHMD	Channel-Dedicated Sample-and-Hold Circuit Operating Mode Setting	0: Constant sampling of the channel-dedicated sample-and-hold circuits is disabled. 1: Constant sampling of the channel-dedicated sample-and-hold circuits is enabled.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register enables or disables constant sampling of the sample-and-hold circuit dedicated for channels.

SHMD Bit (Channel-Dedicated Sample-and-Hold Circuit Operating Mode Setting)

When the SHMD bit is set to 1, constant sampling of the sample-and-hold circuit for the channels selected by the ADSHCR.SHANS[2:0] bits is enabled. The SHMD bit should be set while the ADCSR.ADST bit is 0.
When the constant sampling function is enabled, constant sampling is performed while the 12-bit A/D converter is in the wait state and holding is performed when the A/D conversion is in progress.

56.2.21 A/D Disconnection Detection Control Register (ADDISCR)

Address(es): S12AD.ADDISCR 0008 907Ah, S12AD1.ADDISCR 0008 917Ah

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	ADNDIS[4:0]				
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	ADNDIS[4:0]	A/D Disconnection Detection Assist Setting	b4 ADNDIS[4]: Discharge/precharge selected 0: Discharge 1: Precharge b3 to b0 ADNDIS[3:0]: Discharge/precharge period	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

ADDISCR sets the disconnection detection assist function.

ADNDIS[4:0] Bits (A/D Disconnection Detection Assist Setting)

These bits select either precharge or discharge and the period of precharge/discharge for the A/D disconnection detection assist function. Setting the ADNDIS[4] bit = 1 allows to select precharge and setting the ADNDIS[4] bit = 0 allows to select discharge. The period of precharge/discharge can be set with the ADNDIS[3:0] bits. When the ADNDIS[3:0] bits = 0000b, the disconnection detection assist function is not effective. Setting of the ADNDIS[3:0] bits to 0001b is prohibited. Except for the case of ADNDIS[3:0] = 0000b or 0001b, the specified value indicates the number of states for the period of precharge/discharge. The ADNDIS[4:0] bits should be set when the ADCSR.ADST bit is 0. When ADNDIS[3:0] are set to any value other than 0000b and the disconnection detection assist function is enabled, the channel-dedicated disconnection detection assist function is also enabled. Be sure to secure the wait time for the sample-and-hold circuit when using the channel-dedicated disconnection detection assist function.

The disconnection detection assist function cannot be used when the temperature sensor output or internal reference voltage is converted or when the self-diagnosis function is used.

56.2.22 A/D Group Scan Priority Control Register (ADGSPCR)

Address(es): S12AD.ADGSPCR 0008 9080h, S12AD1.ADGSPCR 0008 9180h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	GBRP	LGRRS	—	—	—	—	—	—	—	—	—	—	—	—	GBRSCN	PGS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PGS	Group Priority Control Setting*1	0: Operation is without group priority control 1: Operation is with group priority control	R/W
b1	GBRSCN	Low-Priority Group Restart Setting*2	(Enabled only when PGS = 1. Reserved when PGS = 0.) 0: Scanning for the group is not restarted after having been discontinued due to group priority control. 1: Scanning for the group is restarted after having been discontinued due to group priority control.	R/W
b13 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b14	LGRRS	Restart Channel Select	(Enabled when PGS = 1 and GBRSCN = 1. Reserved when PGS = 0 or GBRSCN = 0.) 0: Scanning is restarted from the scan start channel. 1: Scanning is restarted from the channel on which A/D conversion is not completed.	R/W
b15	GBRP	Single Scan Continuous Start*3	(Enabled only when PGS = 1. Reserved when PGS = 0.) 0: Single scan is not continuously activated. 1: Single scan for the lowest-priority group is continuously activated.	R/W

Note 1. When the PGS bit is to be set to 1, the ADCSR.ADCS[1:0] bits must be set to 01b (group scan mode). If the bits are set to any other values, proper operation is not guaranteed.
Note 2. When the GBRSCN bit is to be set to 1, the frequency ratio of peripheral module clock PCLK to A/D conversion clock ADCLK should be set to 1:1.
Note 3. When the GBRP bit has been set to 1, single scan is performed continuously for the lowest-priority group regardless of the setting of the GBRSCN bit.

ADGSPCR is used to discontinue scanning of the low-priority group and make settings for priority control of scanning for the priority group in group scan mode.
For the settings on group priority operation, see Table 56.16 and Table 56.17.

PGS Bit (Group Priority Control Setting)

This bit sets the priority of operation in group scan mode. Set this bit to 1 when giving priority to operation on the group.
When the PGS bit is to be set to 1, the ADCSR.ADCS[1:0] bits must be set to 01b (group scan mode).
During group priority operation, a trigger to start scanning for the priority group is accepted during scan for the low-priority group, and scan for the priority group is started after scan for the low-priority group was discontinued. The priority order is group A > group B > group C.
When a trigger to start scanning for group B is accepted during scan for group C, group C scan is discontinued, and scan for group B is started. When a trigger to start scanning for group A is accepted during scan for group C, group C scan is discontinued, and scan for group A is started.
Likewise, when a trigger to start scanning for group A is accepted during scan for group B, group B scan is discontinued and scan for group A is started.
When setting the PGS bit to 0, clearing should be performed by software according to section 56.6.2, Notes on Stopping A/D Conversion. When setting the PGS bit to 1, follow the procedure described in section 56.3.4.3, Operation under Group Priority Control.

GBRSCN Bit (Low-Priority Group Restart Setting)

This bit controls the restarting of scan operation during group priority control.
If a scan operation on the low-priority group has been stopped by a priority group trigger input with the GBRSCN bit set to 1, the scan operation is restarted after the scanning of the priority group is completed. Also, if a low-priority trigger is input during scan for the priority group, the scan operation on the low-priority group is restarted after the scan for the priority group is completed.
If the GBRSCN bit has been set to 0, triggers that are input during A/D conversion are ignored. Also, the ADCSR.ADST bit must be 0 when the GBRSCN bit is to be set.
The setting of the GBRSCN bit is enabled when the PGS bit is set to 1.

LGRRS Bit (Restart Channel Select)

This bit sets the channel on which scan is restarted during group priority control. The setting of the LGRRS bit is enabled when the PGS and GBRSCN bits are set to 1.
If a scan operation on the low-priority group has been stopped due to group priority operation with the LGRRS bit set to 0, the scan operation is restarted from the start channel after the scan for the priority group is completed.
If a scan operation on the low-priority group has been stopped due to group priority operation with the LGRRS bit set to 1, the scan operation is restarted*1 on the channel on which A/D conversion is not competed after the scan for the priority group is completed.
The LGRRS bit should be set while the ADCSR.ADST bit is 0.

Note 1. If A/D conversion on the addition set channel is not completed for the set number of times when scanning is stopped, A/D conversion on the channel is restarted for the set number of times when scanning is restarted.

GBRP Bit (Single Scan Continuous Start)

This bit is set when the lowest-priority group is continuously operated in single scan mode while group priority operation is set. The lowest-priority group is group C when groups A, B, and C are used; group B when groups A and B are used.
Setting the GBRP bit to 1 starts a single scan on the lowest-priority group. On completion of the scan, another single scan on the lowest-priority group is automatically started.
If scanning has been stopped due to group priority operation, single scan on the lowest-priority group is automatically restarted on completion of the A/D conversion on the priority group.
Disable the trigger input for the lowest-priority group before setting the GBRP bit to 1. When the GBRP bit is set to 1, only the lowest-priority group is scanned again even if the GBRSCN bit is 0.
The ADCSR.ADST bit must be 0 when the GBRP bit is to be set.
The setting of the GBRP bit is enabled when the PGS bit is 1.

56.2.23 A/D Comparison Function Control Register (ADCMPCR)

Address(es): S12AD.ADCMPCR 0008 9090h, S12AD1.ADCMPCR 0008 9190h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	CMPAIE	WCMPE	CMPBIE	—	CMPAE	—	CMPBE	—	—	—	—	—	—	—	CMPAB[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	CMPAB[1:0]	Window A/B Complex Conditions Setting	b1 b0 0 0: Window A comparison condition matched OR window B comparison condition matched 0 1: Window A comparison condition matched EXOR window B comparison condition matched 1 0: Window A comparison condition matched AND window B comparison condition matched 1 1: Setting is prohibited	R/W
b8 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b9	CMPBE	Comparison Window B Enable	0: Comparison window B disabled 1: Comparison window B enabled	R/W
b10	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b11	CMPAE	Comparison Window A Enable	0: Comparison window A disabled 1: Comparison window A enabled	R/W
b12	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b13	CMPBIE	Comparison Window B Interrupt Enable	0: Comparison interrupt by a match with the comparison condition (window B) is disabled 1: Comparison interrupt by a match with the comparison condition (window B) is enabled	R/W
b14	WCMPE	Window Function Setting	0: The window function is disabled Window A or B operates as a comparator for comparing between one value on the low side and the A/D converted value 1: The window function is enabled Window A or B operates as a window comparator for comparing between two values on the high and low sides and the A/D converted value	R/W
b15	CMPAIE	Comparison Window A Interrupt Enable	0: Comparison interrupt by a match with the comparison condition (window A) is disabled 1: Comparison interrupt by a match with the comparison condition (window A) is enabled	R/W

The ADCMPCR register is used to set the settings for the comparison window function (windows A and B).

CMPAB[1:0] Bits (Window A/B Complex Conditions Setting)

The CMPAB[1:0] bits are enabled when single scan mode and window A/B are both enabled (CMPAE bit = 1 and CMPBE bit = 1). These bits specify the monitor conditions of the ADWINMON.MONCOMB bits. The CMPAB[1:0] bits should be set when the ADCSR.ADST bit is 0.

CMPBE Bit (Comparison Window B Enable)

This bit is used to disable or enable comparison window B. The CMPBE bit should be set when the ADCSR.ADST bit is 0.

To set the following registers, set this bit to 0.

- A/D channel select registers A0/A1/B0/B1 (ADANSA0, ADANSA1, ADANSB0, and ADANSB1)
- Bits OCSB, TSSB, OCSA, and TSSA in the A/D conversion extended input control register (ADEXICR.OCSB, TSSB, OCSA, and TSSA)

- Bits OCSC and TSSC in the A/D group C extended input control register (ADGCEXCR.OCSC and TSSC)
- The CMPCHB[5:0] bits in the window B channel select register (ADCMPBNSR.CMPCHB[5:0])

CMPAE Bit (Comparison Window A Enable)

This bit is used to disable or enable comparison window A. The CMPAE bit should be set when the ADCSR.ADST bit is 0.

To set the following registers, set this bit to 0.

- A/D channel select registers A0/A1/B0/B1 (ADANSA0, ADANSA1, ADANSB0, and ADANSB1)
- Bits OCSB, TSSB, OCSA, and TSSA in the A/D conversion extended input control register (ADEXICR.OCSB, TSSB, OCSA, and TSSA)
- Bits OCSC and TSSC in the A/D group C extended input control register (ADGCEXCR.OCSC and TSSC)
- Window A channel select registers 0 and 1 (ADCMPANSR0 and ADCMPANSR1)
- Window A extended input select register (ADCMPANSER)

CMPBIE Bit (Comparison Window B Interrupt Enable)

Enables or disables comparison interrupts by a window B comparison condition match. A single comparison interrupt exists for each unit. Table 56.14 lists the relationship between each unit and its comparison interrupt.

WCMPE Bit (Window Function Setting)

This bit is used to disable or enable the window function. The WCMPE bit should be set when the ADCSR.ADST bit is 0.

CMPAIE Bit (Comparison Window A Interrupt Enable)

Enables or disables comparison interrupts by a window A comparison condition match. A single comparison interrupt exists for each unit. Table 56.14 lists the relationship between each unit and its comparison interrupt.

Table 56.14 Relationship Between Each Unit and its Comparison Interrupt Unit

Unit	Comparison Interrupt	
	When Window A Comparison Condition is Met	When Window B Comparison Condition is Met
S12AD	S12CMPAI	S12CMPBI
S12AD1	S12CMPAI1	S12CMPBI1

56.2.24 A/D Comparison Function Window A Channel Select Register 0 (ADCMPANSR0)

(1) S12AD.ADCMPANSR0

Address(es): 0008 9094h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	CMPCHA007	CMPCHA006	CMPCHA005	CMPCHA004	CMPCHA003	CMPCHA002	CMPCHA001	CMPCHA000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPCHA000	Comparison Window A Channel Select	0: AN000 to AN007 are excluded from the targets for comparison window A 1: AN000 to AN007 are included as the targets for comparison window A	R/W
b1	CMPCHA001			R/W
b2	CMPCHA002			R/W
b3	CMPCHA003			R/W
b4	CMPCHA004			R/W
b5	CMPCHA005			R/W
b6	CMPCHA006			R/W
b7	CMPCHA007			R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register selects analog inputs AN000 to AN007 of the channels that perform comparison with the conditions of comparison window A.

CMPCHA0n Bit (n = 00 to 07) (Comparison Window A Channel Select)

The comparison function is enabled when the CMPCHA0n bit with the same index number as the A/D conversion channel selected by the ADANSA0.ANSA0n bit (n = 00 to 07), ADANSB0.ANSB0n bit, and ADANSC0.ANSC0n bit is set to 1.

The CMPCHA0n bit should be set to 1 while the ADCSR.ADST bit is 0.

(2) S12AD1.ADCMPANSR0

Address(es): 0008 9194h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	CMPCHA015	CMPCHA014	CMPCHA013	CMPCHA012	CMPCHA011	CMPCHA010	CMPCHA009	CMPCHA008	CMPCHA007	CMPCHA006	CMPCHA005	CMPCHA004	CMPCHA003	CMPCHA002	CMPCHA001	CMPCHA000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPCHA000	Comparison Window A Channel Select	0: AN100 to AN115 are excluded from the targets for comparison window A 1: AN100 to AN115 are included as the targets for comparison window A	R/W
b1	CMPCHA001			R/W
b2	CMPCHA002			R/W
b3	CMPCHA003			R/W
b4	CMPCHA004			R/W
b5	CMPCHA005			R/W
b6	CMPCHA006			R/W
b7	CMPCHA007			R/W
b8	CMPCHA008			R/W
b9	CMPCHA009			R/W
b10	CMPCHA010			R/W
b11	CMPCHA011			R/W
b12	CMPCHA012			R/W
b13	CMPCHA013			R/W
b14	CMPCHA014			R/W
b15	CMPCHA015			R/W

This register selects analog inputs AN100 to AN115 of the channels that perform comparison with the conditions of comparison window A.

CMPCHA0n Bit (n = 00 to 15) (Comparison Window A Channel Select)

The comparison function is enabled when the CMPCHA0n bit with the same index number as the A/D conversion channel selected by the ADANSA0.ANSA0n bit (n = 00 to 15), ADANSB0.ANSB0n bit, and ADANSC0.ANSC0n bit is set to 1.

The CMPCHA0n bit should be set while the ADCSR.ADST bit is 0.

56.2.25 A/D Comparison Function Window A Channel Select Register 1 (ADCMPANSR1)

Address(es): S12AD1.ADCMPANSR1 0008 9196h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	CMPCHA104	CMPCHA103	CMPCHA102	CMPCHA101	CMPCHA100
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPCHA100	Comparison Window A Channel Select	0: AN116 to AN120 are excluded from the targets for comparison window A 1: AN116 to AN120 are included as the targets for comparison window A	R/W
b1	CMPCHA101			R/W
b2	CMPCHA102			R/W
b3	CMPCHA103			R/W
b4	CMPCHA104			R/W
b15 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register selects analog input channels AN116 to AN120 that compare with the conditions of comparison window A.

CMPCHA1n Bit (n = 00 to 04) (Comparison Window A Channel Select)

The comparison function is enabled when the CMPCHA1n bit with the same index number as the A/D conversion channel selected by the ADANSA0.ANSA0n bit (n = 00 to 04), ADANSB0.ANSB0n bit, and ADANSC0.ANSC0n bit is set to 1.

The CMPCHA1n bit should be set while the ADCSR.ADST bit is 0.

56.2.26 A/D Comparison Function Window A Extended Input Select Register (ADCMPANSER)

Address(es): S12AD1.ADCMPANSER 0008 9192h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	CMPSOC	CMPSTS
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPSTS	Temperature Sensor Output Comparison Select	0: Temperature sensor output is excluded from the targets for comparison window A 1: Temperature sensor output is included as the targets for comparison window A	R/W
b1	CMPSOC	Internal Reference Voltage Compare Select	0: Internal reference voltage is excluded from the targets for comparison window A 1: Internal reference voltage is included as the targets for comparison window A	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register selects whether the temperature sensor output or internal reference voltage is compared under the conditions of comparison window A.

CMPSTS Bit (Temperature Sensor Output Comparison Select)

The comparison window A is enabled when the CMPSTS bit is set to 1 while the ADEXICR.TSSA, ADEXICR.TSSB, or ADGCEXCR.TSSC bit is 1. The CMPSTS bit should be set while the ADCSR.ADST bit is 0.

CMPSOC Bit (Internal Reference Voltage Compare Select)

The comparison window A function is enabled when the ADEXICR.OCSA, ADEXICR.OCSB, or ADGCEXCR.OCSC bit is 1. The CMPSOC bit should be set while the ADCSR.ADST bit is 0.

56.2.27 A/D Comparison Function Window A Comparison Condition Setting Register 0 (ADCMPLR0)

(1) S12AD.ADCMPLR0

Address(es): 0008 9098h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	CMPLCHA007	CMPLCHA006	CMPLCHA005	CMPLCHA004	CMPLCHA003	CMPLCHA002	CMPLCHA001	CMPLCHA000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPLCHA000	Comparison Window A Comparison Condition Select	When the window function is disabled (the ADCMPCR.WCMPE bit is 0) 0: ADCMPDR0 register value > A/D converted value 1: ADCMPDR0 register value < A/D converted value When the window function is enabled (the ADCMPCR.WCMPE bit is 1) 0: A/D converted value < ADCMPDR0 register value or ADCMPDR1 register value < A/D converted value 1: ADCMPDR0 register value < A/D converted value < ADCMPDR1 register value	R/W
b1	CMPLCHA001			R/W
b2	CMPLCHA002			R/W
b3	CMPLCHA003			R/W
b4	CMPLCHA004			R/W
b5	CMPLCHA005			R/W
b6	CMPLCHA006			R/W
b7	CMPLCHA007			R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register specifies the conditions for comparison of the values in the S12AD.ADCMPDR0/ADCMPDR1 register and the value of A/D conversion. The S12AD.ADCMPLR0 register should be set while the S12AD.ADCSR.ADST bit is 0.

CMPLCHA0n Bit (n = 00 to 07) (Comparison Window A Comparison Condition Select)

This bit specifies the comparison conditions of the channels (AN000 to AN007) targeted for the window A comparison conditions. The condition can be specified for each analog input for comparison. The CMPLCHA000 bit corresponds to AN000 and the CMPLCHA007 bit corresponds to AN007.

(2) S12AD1.ADCMPLR0

Address(es): 0008 9198h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	CMPLCHA015	CMPLCHA014	CMPLCHA013	CMPLCHA012	CMPLCHA011	CMPLCHA010	CMPLCHA009	CMPLCHA008	CMPLCHA007	CMPLCHA006	CMPLCHA005	CMPLCHA004	CMPLCHA003	CMPLCHA002	CMPLCHA001	CMPLCHA000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPLCHA000	Comparison Window A Comparison Condition Select	When the window function is disabled (the ADCMPCR.WCMPE bit is 0) 0: ADCMPDR0 register value > A/D converted value 1: ADCMPDR0 register value < A/D converted value When the window function is enabled (the ADCMPCR.WCMPE bit is 1) 0: A/D converted value < ADCMPDR0 register value or ADCMPDR1 register value < A/D converted value 1: ADCMPDR0 register value < A/D converted value < ADCMPDR1 register value	R/W
b1	CMPLCHA001			R/W
b2	CMPLCHA002			R/W
b3	CMPLCHA003			R/W
b4	CMPLCHA004			R/W
b5	CMPLCHA005			R/W
b6	CMPLCHA006			R/W
b7	CMPLCHA007			R/W
b8	CMPLCHA008			R/W
b9	CMPLCHA009			R/W
b10	CMPLCHA010			R/W
b11	CMPLCHA011			R/W
b12	CMPLCHA012			R/W
b13	CMPLCHA013			R/W
b14	CMPLCHA014			R/W
b15	CMPLCHA015			R/W

This register specifies the conditions for comparing between the values of the S12AD1.ADCMPDR0/ADCMPDR1 registers and the result of A/D conversion. The S12AD1.ADCMPLR0 register should be set while the S12AD1.ADCSR.ADST bit is 0.

CMPLCHA0n Bit (n = 00 to 15) (Comparison Window A Comparison Condition Select)

This bit specifies the conditions for comparison with the target channels for window A (AN100 to AN115). The conditions can be specified for each analog input for comparison. The CMPLCHA000 bit corresponds to AN100 and the CMPLCHA015 bit corresponds to AN115.

When the result of comparison of each analog input matches with the pre-set condition, the S12AD1.ADCMPSR0.CMPSTCHA0n flag (n = 00 to 15) is set to 1 and a comparison interrupt (S12CMPAI1) is generated. The conditions for comparison are shown in Figure 56.4.

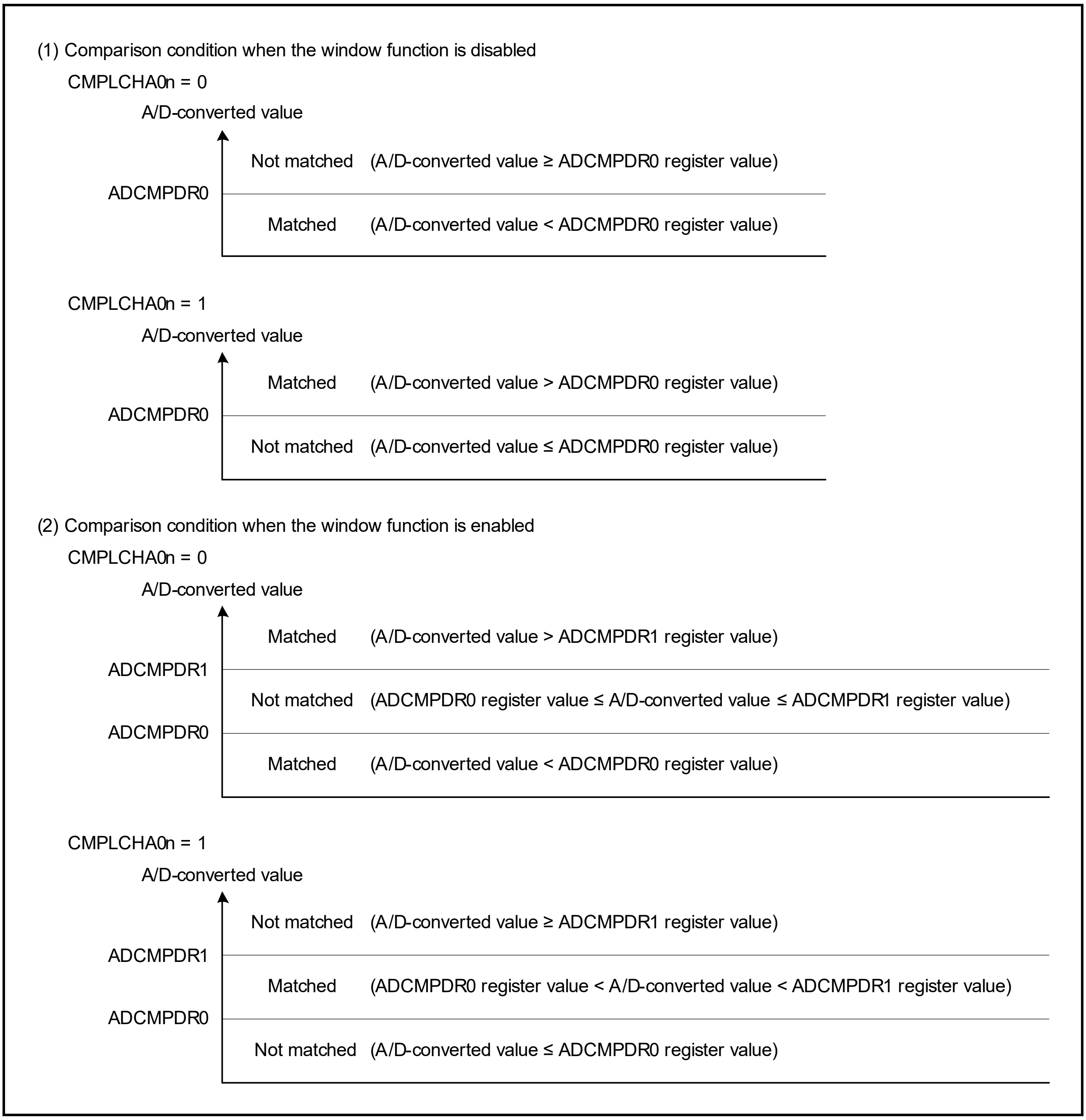


Figure 56.4 Details of Comparison Condition: Comparison Function Window A

56.2.28 A/D Comparison Function Window A Comparison Condition Setting Register 1 (ADCMPLR1)

Address(es): S12AD1.ADCMPLR1 0008 919Ah

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	CMPLCHA104	CMPLCHA103	CMPLCHA102	CMPLCHA101	CMPLCHA100
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPLCHA100	Comparison Window A Comparison Condition Select	When the window function is disabled (the ADCMPCR.WCMPE bit is 0) 0: ADCMPDR0 register value > A/D converted value 1: ADCMPDR0 register value < A/D converted value When the window function is enabled (the ADCMPCR.WCMPE bit is 1) 0: A/D converted value < ADCMPDR0 register value or ADCMPDR1 register value < A/D converted value 1: ADCMPDR0 register value < A/D converted value < ADCMPDR1 register value	R/W
b1	CMPLCHA101			R/W
b2	CMPLCHA102			R/W
b3	CMPLCHA103			R/W
b4	CMPLCHA104			R/W
b15 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register specifies the conditions for comparison of the value of the ADCMPDR0/ADCMPDR1 register and the result of A/D conversion. The ADCMPLR1 register should be set while the ADCSR.ADST bit is 0.

CMPLCHA1n Bit (n = 00 to 04) (Comparison Window A Comparison Condition Select)

This bit specifies the conditions for comparison with the target channels for window A (AN116 to AN120). The condition can be specified for each analog input for comparison. The CMPLCHA100 bit corresponds to AN116 and the CMPLCHA104 bit corresponds to AN120.

When the result of comparison of each analog input matches with the pre-set condition, the ADCMPSR1.CMPSTCHA1n flag (n = 00 to 04) is set to 1. A comparison interrupt (S12CMPAI1) is generated.

Figure 56.4 shows the conditions for comparison.

56.2.29 A/D Comparison Function Window A Extended Input Comparison Condition Setting Register (ADCMPLER)

Address(es): S12AD1.ADCMPLER 0008 9193h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	CMPLOC	CMPLTS
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPLTS	Comparison Window A Temperature Sensor Output Comparison Condition Select	When the window A function is disabled (the ADCMPCR.WCMPE bit is 0) 0: ADCMPDR0 register value > A/D converted value 1: ADCMPDR0 register value < A/D converted value When the window A function is enabled (the ADCMPCR.WCMPE bit is 1) 0: A/D converted value < ADCMPDR0 register value or A/D converted value > ADCMPDR1 register value 1: ADCMPDR0 register value < A/D converted value < ADCMPDR1 register value	R/W
b1	CMPLOC	Comparison Window A Internal Reference Voltage Comparison Condition Select	When the window A function is disabled (the ADCMPCR.WCMPE bit is 0) 0: ADCMPDR0 register value > A/D converted value 1: ADCMPDR0 register value < A/D converted value When the window A function is enabled (the ADCMPCR.WCMPE bit is 1) 0: A/D converted value < ADCMPDR0 register value or A/D converted value > ADCMPDR1 register value 1: ADCMPDR0 register value < A/D converted value < ADCMPDR1 register value	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register specifies the conditions for comparison of the value of the ADCMPDR0/ADCMPDR1 register and the result of A/D conversion. The ADCMPLER register should be set while the ADCSR.ADST bit is 0.

CMPLTS Bit (Comparison Window A Temperature Sensor Output Comparison Condition Select)

This bit selects the comparison conditions of temperature sensor output targeted for window A.
When the result of comparison of temperature sensor output matches with the pre-set condition, the ADCMPSER.CMPFTS flag is set to 1. A comparison interrupt (S12CMPAI1) is generated.
Figure 56.4 shows the conditions for comparison.

CMPLOC Bit (Comparison Window A Internal Reference Voltage Comparison Condition Select)

This bit selects the comparison conditions of the internal reference voltage targeted for window A.
When the result of comparison of the internal reference voltage matches with the pre-set condition, the ADCMPSER.CMPFOC flag is set to 1. A comparison interrupt (S12CMPAI1) is generated.
Figure 56.4 shows the conditions for comparison.

56.2.30 A/D Comparison Function Window A Lower Level Setting Register (ADCMPDR0)

Address(es): S12AD.ADCMPDR0 0008 909Ch, S12AD1.ADCMPDR0 0008 919Ch

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

This is a readable and writable register that specifies the reference data when the comparison window A function is used. The ADCMPDR0 register specifies the lower level of window A.
Writing to the ADCMPDR0 register is enabled even when A/D conversion is in progress. Rewriting the register value during the A/D conversion dynamically changes the reference data.
Satisfy the condition of [Upper limit level ≥ Lower limit level (ADCMPDR1 setting value ≥ ADCMPDR0 setting value)] when setting.

The data formats of the register for each given condition are shown below.

- Setting of the A/D data register format select bit (flush-right or -left format)
- Setting of the A/D conversion resolution setting bit (12, 10, or 8 bits)
- Setting of the A/D converted value addition/averaging function channel select register (when A/D converted value average mode is selected or not selected)
- Setting of the A/D converted value addition/average count select register (when addition/averaging mode is selected and addition count is selected)

Note: When the comparison values are set in a different format from that of the A/D data register y (ADDRy), correct result of comparison cannot be obtained.

(1) When A/D converted value addition/averaging mode is not selected

- Flush-right format with setting for 12-bit resolution
  The comparison level (lower) to be compared is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format with setting for 10-bit resolution
  The comparison level (lower) to be compared is set in bits 9 to 0. Write 0 to bits 15 to 10.
- Flush-right format with setting for 8-bit resolution
  The comparison level (lower) to be compared is set in bits 7 to 0. Write 0 to bits 15 to 8.
- Flush-left format with setting for 12-bit resolution
  The comparison level (lower) to be compared is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format with setting for 10-bit resolution
  The comparison level (lower) to be compared is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format with setting for 8-bit resolution
  The comparison level (lower) to be compared is set in bits 15 to 8. Write 0 to bits 7 to 0.

(2) When A/D converted value addition/averaging mode is selected

- Flush-right format with setting for 12-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format with setting for 10-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 9 to 0. Write 0 to bits 15 to 10.

- Flush-right format with setting for 8-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 7 to 0. Write 0 to bits 15 to 8.
- Flush-left format with setting for 12-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format with setting for 10-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format with setting for 8-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 15 to 8. Write 0 to bits 7 to 0.

A/D-converted value average mode can be set only when 2- or 4-time conversion is selected in A/D-converted value addition mode.

(3) When A/D converted value averaging mode is selected

- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 13 to 0. Write 0 to bits 15 and 14.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 11 to 0. Write 0 to bits 15 and 12.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 9 to 0. Write 0 to bits 15 and 10.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The comparison level (lower) to be compared with the value of A/D conversion of the same channel is set in bits 15 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The comparison level (lower) to be compared with the A/D converted value of the same channel is set in bits 15 to 2. Write 0 to bits 1 and 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The comparison level (lower) to be compared with the A/D converted value of the same channel is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The comparison level (lower) to be compared with the A/D converted value of the same channel is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The comparison level (lower) to be compared with the A/D converted value of the same channel is set in bits 15 to 0.

When A/D converted value addition mode is selected, the value for comparison with the result of cumulative A/D converted value of the same channel is specified. Conversion 1, 2, 3, 4, or 16 times is selectable. When A/D converted value addition mode is selected and if conversion 1 to 4 is selected, the resolution of the results of conversion will be extended by 2 bits and so set a value for comparison with the results in the ADCMPDR0 register. When conversion 16 times is selected, the resolution of the results of conversion will be extended by 4 bits and so set a value for comparison with the results in the ADCMPDR0 register.
Even when A/D converted value addition mode is selected, set the value for reference based on the setting of the A/D data register format select bit.

56.2.31 A/D Comparison Function Window A Upper Level Setting Register (ADCMPDR1)

Address(es): S12AD.ADCMPDR1 0008 909Eh, S12AD1.ADCMPDR1 0008 919Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

This is a readable and writable register that specifies the data for reference when the comparison window A function is used. The register specifies the upper level of window A.
A write operation to the ADCMPDR1 register is enabled even during A/D conversion. Rewriting the value of the register during A/D conversion dynamically changes the reference data.
Satisfy the condition of [upper limit level ≥ lower limit level (ADCMPDR1 setting value ≥ ADCMPDR0 setting value)] when setting.
The ADCMPDR1 register is not used when the window function is disabled.

The data formats of the register for each given condition are shown below.

- Setting of the A/D data register format select bit (flush-right or -left format)
- Setting of the A/D conversion resolution setting bit (12, 10, or 8 bits)
- Setting of the A/D converted value addition/averaging function channel select register (when A/D converted value averaging mode is selected or not selected)
- Setting of the A/D converted value addition/averaging time select register (when addition/averaging mode is select and addition count is selected)

Note: When the comparison values are set in a different format from that of the A/D data register y (ADDRy), correct result of comparison cannot be obtained.

(1) When A/D-converted value addition/averaging mode is not selected

- Flush-right format with the setting for 12-bit resolution
  The comparison level (upper) is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format with the setting for 10-bit resolution
  The comparison level (upper) is set in bits 9 to 0. Write 0 to bits 15 to 10.
- Flush-right format with the setting for 8-bit resolution
  The comparison level (upper) is set in bits 7 to 0. Write 0 to bits 15 to 8.
- Flush-left format with the setting for 12-bit resolution
  The comparison level (upper) is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format with the setting for 10-bit resolution
  The comparison level (upper) is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format with the setting for 8-bit resolution
  The comparison level (upper) is set in bits 15 to 8. Write 0 to bits 7 to 0.

(2) When A/D-converted averaging mode is selected

- Flush-right format with the setting for 12-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format with the setting for 10-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 9 to

0. Write 0 to bits 15 to 10.

- Flush-right format with the setting for 8-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 7 to 0. Write 0 to bits 15 to 8.
- Flush-left format with the setting for 12-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format with the setting for 10-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format with the setting for 8-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 8. Write 0 to bits 7 to 0.

A/D-converted value average mode can be set only when 2- or 4-time conversion is selected in A/D-converted value addition mode.

(3) When A/D converted value averaging mode is selected

- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The comparison level (upper) to be compared with the A/D converted value of the same channel is set bits 13 to 0. Write 0 to bits 15 to 14.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The comparison level (upper) to be compared with the A/D converted value of the same channel is set bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The comparison level (upper) to be compared with the A/D converted value of the same channel is set bits 9 to 0. Write 0 to bits 15 to 10.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The comparison level (upper) to be compared with the A/D converted value of the same channel is set bits 15 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The comparison level (upper) to be compared with the A/D converted value of the same channel is set in bits 15 to 2. Write 0 to bits 1 and 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The comparison level (upper) to be compared with the A/D converted value of the same channel is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The comparison level (upper) to be compared with the A/D converted value of the same channel is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The comparison level (upper) to be compared with the A/D converted value of the same channel is set in bits 15 to 0.

When A/D converted value addition mode is selected, the value for comparison with the result of cumulative A/D converted value of the same channel is specified. Conversion 1, 2, 3, 4, or 16 times is selectable. When A/D converted value addition mode is selected and if conversion 1 to 4 is selected, the resolution of the results of conversion will be extended by 2 bits and so set a value for comparison with the results in the ADCMPDR1 register. When conversion 16 times is selected, the resolution of the results of conversion will be extended by 4 bits and so set a value for comparison with the results in the ADCMPDR1 register.
Even when A/D converted value addition mode is selected, set the value for reference based on the setting of the A/D data register format select bit.

56.2.32 A/D Comparison Function Window A Channel Status Register 0 (ADCMPSR0)

(1) S12AD.ADCMPSR0

Address(es): 0008 90A0h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	CMPSTCHA007	CMPSTCHA006	CMPSTCHA005	CMPSTCHA004	CMPSTCHA003	CMPSTCHA002	CMPSTCHA001	CMPSTCHA000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPSTCHA000	Comparison Window A Flag	When window A operation is enabled (ADCMPCR.CMPAE = 1) These bits indicate the comparison result of the channels (AN000 to AN007) for the window A comparison condition 0: Comparison condition is not satisfied. 1: Comparison condition is satisfied.	R/W
b1	CMPSTCHA001			R/W
b2	CMPSTCHA002			R/W
b3	CMPSTCHA003			R/W
b4	CMPSTCHA004			R/W
b5	CMPSTCHA005			R/W
b6	CMPSTCHA006			R/W
b7	CMPSTCHA007			R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This is a register that stores the result of comparison by the comparison window A function.

CMPSTCHA0n Flag (n = 00 to 07) (Comparison Window A Flag)

These are status flags that indicate the result of comparison of the channels for the window A comparison conditions. When the conversion result matches with the comparison condition that is set in the ADCMPLR0.CMPLCHA0n bit (n = 00 to 07) upon completion of A/D conversion, these flags are set to 1. When the ADCMPCR.CMPIE bit is 1, a comparison interrupt request (S12CMPAI) is generated upon the setting of the flag.
The CMPSTCHA000 flag corresponds to AN000 and the CMPSTCHA007 flag corresponds to AN007.
1 cannot be written to the CMPSTCHA0n flag.
[Setting condition]

- When ADCMPCR.CMPAE = 1 and the condition set in the ADCMPLR0.CMPLCHA0n bit is satisfied.

[Clearing condition]

- When 0 is written after reading the state of 1.

(2) S12AD1.ADCMPSR0

Address(es): 0008 91A0h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	CMPST CHA015	CMPST CHA014	CMPST CHA013	CMPST CHA012	CMPST CHA011	CMPST CHA010	CMPST CHA009	CMPST CHA008	CMPST CHA007	CMPST CHA006	CMPST CHA005	CMPST CHA004	CMPST CHA003	CMPST CHA002	CMPST CHA001	CMPST CHA000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPSTCHA000	Comparison Window A Flag	When window A operation is enabled (ADCMPCR.CMPAE = 1) These bits indicate the comparison result of the channels (AN100 to AN115) for the window A comparison condition 0: The comparison condition is not satisfied. 1: The comparison condition is satisfied	R/W
b1	CMPSTCHA001			R/W
b2	CMPSTCHA002			R/W
b3	CMPSTCHA003			R/W
b4	CMPSTCHA004			R/W
b5	CMPSTCHA005			R/W
b6	CMPSTCHA006			R/W
b7	CMPSTCHA007			R/W
b8	CMPSTCHA008			R/W
b9	CMPSTCHA009			R/W
b10	CMPSTCHA010			R/W
b11	CMPSTCHA011			R/W
b12	CMPSTCHA012			R/W
b13	CMPSTCHA013			R/W
b14	CMPSTCHA014			R/W
b15	CMPSTCHA015			R/W

This is a register that stores the result of comparison of comparison window A.

CMPSTCHA0n Flag (n = 00 to 15) (Comparison Window A Flag)

These are status flags that indicate the result of comparison of the channels (AN100 to AN115) of the comparison condition for window A. When the conversion result matches with the comparison condition specified in the ADCMPLR0.CMPLCHA0n bit (n = 00 to 15) upon completion of A/D conversion, this flag is set to 1. When the ADCMPCR.CMPIE bit is 1, a comparison interrupt request (S12CMPAI1) is generated upon the setting of the flag. The CMPSTCHA000 flag corresponds to AN100 and the CMPSTCHA015 flag corresponds to AN115.
1 cannot be written to the CMPSTCHA0n flag.
[Setting condition]

- When ADCMPCR.CMPAE = 1, the condition set in the ADCMPLR0.CMPLCHA0n bit is satisfied.

[Clearing condition]

- When 0 is written after reading the state of 1.

56.2.33 A/D Comparison Function Window A Channel Status Register 1 (ADCMPSR1)

Address(es): S12AD1.ADCMPSR1 0008 91A2h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	CMPSTCHA104	CMPSTCHA103	CMPSTCHA102	CMPSTCHA101	CMPSTCHA100
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPSTCHA100	Comparison Window A Flag	When window A operation is enabled (ADCMPCR.CMPAE = 1) These bits indicate the comparison result of the channels (AN116 to AN120) for window A comparison condition 0: The comparison condition is not satisfied. 1: The comparison condition is satisfied.	R/W
b1	CMPSTCHA101			R/W
b2	CMPSTCHA102			R/W
b3	CMPSTCHA103			R/W
b4	CMPSTCHA104			R/W
b15 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This is a register that stores the result of the comparison of the comparison window A function.

CMPSTCHA1n Flag (n = 00 to 04) (Comparison Window A Flag)

These are status flags that indicate the result of comparison of the channels (AN116 to AN120) of the comparison condition for window A. When the conversion result matches with the comparison condition specified in the ADCMPLR1.CMPLCHA1n bit (n = 00 to 04) upon completion of A/D conversion, this flag is set to 1. When the ADCMPCR.CMPIE bit is 1, a comparison interrupt request (S12CMPAI1) is generated upon the setting of the flag. The CMPSTCHA100 flag corresponds to AN116 and the CMPSTCHA104 flag corresponds to AN120.
1 cannot be written to the CMPSTCHA1n flag.
[Setting condition]

- When ADCMPCR.CMPAE = 1, the condition set in the ADCMPLR1.CMPLCHA1n bit is satisfied.

[Clearing condition]

- When 0 is written after reading the state of 1.

56.2.34 A/D Comparison Function Window A Extended Input Channel Status Register (ADCMPSER)

Address(es): S12AD1.ADCMPSER 0008 91A4h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	CMPFOC	CMPFTS
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPFTS	Comparison Window A Temperature Sensor Output Comparison Flag	When window A operation is enabled (ADCMPCR.CMPAE = 1) This bit indicates the comparison result of the temperature sensor output 0: The comparison condition is not satisfied. 1: The comparison condition is satisfied.	R/W
b1	CMPFOC	Comparison Window A Internal Reference Voltage Comparison Flag	When window A operation is enabled (ADCMPCR.CMPAE = 1) This bit indicates the comparison result of the internal reference voltage 0: The comparison condition is not satisfied. 1: The comparison condition is satisfied.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This is a register that stores the result of the comparison of the comparison window A function.

CMPFTS Flag (Comparison Window A Temperature Sensor Output Comparison Flag)

This is a status flag that indicates the result of comparison of the temperature sensor output. When the conversion result matches with the comparison condition specified in the ADCMPLER.CMPLTS bit upon completion of A/D conversion, this flag is set to 1. When the ADCMPCR.CMPIE bit is 1, a comparison interrupt request (S12CMPAI1) is generated upon the setting of the flag.

1 cannot be written to the CMPFTS flag.

[Setting condition]

- When ADCMPCR.CMPAE = 1, the condition set in the ADCMPLER.CMPLTS bit is satisfied.

[Clearing condition]

- When 0 is written after reading the state of 1.

CMPFOC Flag (Comparison Window A Internal Reference Voltage Comparison Flag)

This is a status flag that indicates the result of comparison of the internal reference voltage. When the conversion result matches with the comparison condition specified in the ADCMPLER.CMPLOC bit upon completion of A/D conversion, this flag is set to 1. When the ADCMPCR.CMPIE bit is 1, a comparison interrupt request (S12CMPAI1) is generated upon the setting of the flag.

1 cannot be written to the CMPFOC flag.

[Setting condition]

- When ADCMPCR.CMPAE = 1, the condition set in the ADCMPLER.CMPLOC bit is satisfied.

[Clearing condition]

- When 0 is written after reading the state of 1.

56.2.35 A/D Comparison Function Window A/B Status Monitoring Register (ADWINMON)

Address(es): S12AD.ADWINMON 0008 908Ch, S12AD1.ADWINMON 0008 918Ch

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	MONCMPB	MONCMPA	—	—	—	MONCOMB
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MONCOMB	Combination Result Monitoring	When window A operation is enabled (ADCMPCR.CMPAE = 1) These bits indicate the comparison result of the internal reference voltage 0: The comparison condition is not satisfied. 1: The comparison condition is satisfied.	R
b3 to b1	—	Reserved	These bits are read as 0.	R
b4	MONCMPA	Comparison Result Monitor A	0: The comparison condition for window A is not satisfied. 1: The comparison condition for window A is satisfied.	R
b5	MONCMPB	Comparison Result Monitor B	0: The comparison condition for window B is not satisfied. 1: The comparison condition for window B is satisfied.	R
b7, b6	—	Reserved	These bits are read as 0.	R

This register can monitor the results of comparison and the combined result.

MONCOMB Bit (Combination Result Monitoring)

This is a dedicated bit for reading the combined result of comparison condition A and B that were set in the ADCMPCR.CMPAB[1:0] bits as a combined condition.

[Setting condition]

- When ADCMPCR.CMPAE = 1 and ADCMPCR.CMPBE = 1, the condition set in the ADCMPCR.CMPAB[1:0] bits is satisfied.

[Clearing condition]

- No match with the combined condition set in the ADCMPCR.CMPAB[1:0] bits
- When ADCMPCR.CMPAE = 0 or ADCMPCR.CMPBE = 0

MONCMPA Bit (Comparison Result Monitor A)

This is a dedicated bit for reading 1 when the condition set in the ADCMPLR0, ADCMPLR1, or ADCMPLER matches with the A/D converted value of the channel targeted for window A and for reading 0 when such condition is not satisfied.

[Setting condition]

- When ADCMPCR.CMPAE = 1, the condition set in the ADCMPLR0.CMPLCHA0n bit is satisfied.

[Clearing condition]

- When ADCMPCR.CMPAE = 1, the condition set in the ADCMPLR0.CMPLCHA0n bit is not satisfied.
- When ADCMPCR.CMPAE = 0, (ADCMPCR.CMPAE is automatically cleared to 0)

MONCMPB Bit (Comparison Result Monitor B)

This is a dedicated bit for reading 1 when the condition set in the ADCMPBNSR.CMPLB matches with the A/D converted value of the channel targeted for window B and for reading 0 when such condition is not satisfied.

[Setting condition]

- When ADCMPCR.CMPBE = 1, the condition set in the ADCMPBNSR.CMPLB bit is satisfied.

[Clearing condition]

- When ADCMPCR.CMPBE = 1, the condition set in the ADCMPBNSR.CMPLB bit is not satisfied.
- When ADCMPCR.CMPBE = 0, (ADCMPCR.CMPBE is automatically cleared to 0)

56.2.36 A/D Comparison Function Window B Channel Select Register (ADCMPBNSR)

(1) S12AD.ADCMPBNSR

Address(es): 0008 90A6h

	b7	b6	b5	b4	b3	b2	b1	b0
	CMPLB	—	CMPCHB[5:0]					
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	CMPCHB[5:0]	Comparison Window B Channel Select	Select the channel that compares with the condition of comparison window B b5 b0 0 0 0 0 0 0: AN000 0 0 0 0 0 1: AN001 0 0 0 0 1 0: AN002 : : 0 0 0 1 1 0: AN006 0 0 0 1 1 1: AN007 Settings other than above are prohibited.	R/W
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	CMPLB	Comparison Window B Comparison Condition Setting	When the window function is disabled (the ADCMPCR.WCMPE bit is 0) 0: ADWINLLB register value > A/D converted value 1: ADWINLLB register value < A/D converted value When the window function is enabled (the ADCMPCR.WCMPE bit is 1) 0: A/D converted value < ADWINLLB register value or ADWINULB register value < A/D converted value 1: ADWINLLB register value < A/D converted value < ADWINULB register value	R/W

This register is used to specify the comparison window B function.

CMPCHB[5:0] Bits (Comparison Window B Channel Select)

These bits select the channel to be compared with the comparison window B from AN000 to AN007.
When the number of the A/D conversion channel selected in the ADANSAy.ANSAyn bit (y = 0 to 1, n = 00 to 07) and ADANSBy.ANSByn bit, the comparison window B function is enabled.
The CMPCHB[5:0] bits should be set while the ADCSR.ADST bit is 0.

CMPLB Bit (Comparison Window B Comparison Condition Setting)

This bit specifies the condition for comparison of the channels for window B. When the comparison result of each analog input matches the pre-set condition, the ADCMPBSR.CMPSTB flag is set to 1, a comparison interrupt request (S12CMPBI) is generated. Figure 56.5 shows the conditions for comparison.

(2) S12AD1.ADCMPBNSR

Address(es): 0008 91A6h

	b7	b6	b5	b4	b3	b2	b1	b0
	CMPLB	—	CMPCHB[5:0]					
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	CMPCHB[5:0]	Comparison Window B Channel Select	Select the channel that compares with the condition of comparison window B b5 b0 0 0 0 0 0 0: AN100 0 0 0 0 0 1: AN101 0 0 0 0 1 0: AN102 : : 0 0 0 1 1 0: AN106 0 0 0 1 1 1: AN107 : : 0 1 0 0 0 0: AN116 0 1 0 0 0 1: AN117 0 1 0 0 1 0: AN118 0 1 0 0 1 1: AN119 0 1 0 1 0 0: AN120 1 0 0 0 0 0: Temperature sensor 1 0 0 0 0 1: Internal reference voltage Settings other than above are prohibited.	R/W
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	CMPLB	Comparison Window B Comparison Condition Setting	When window function is disabled (the ADCMPCR.WCMPE bit is 0) 0: ADWINLLB register value > A/D converted value 1: ADWINLLB register value < A/D converted value When window function is enabled (the ADCMPCR.WCMPE bit is 1) 0: A/D converted value < ADWINLLB register value or ADWINULB register value < A/D converted value 1: ADWINLLB register value < A/D converted value < ADWINULB register value	R/W

This register is used to specify the compare window B function.

CMPCHB[5:0] Bits (Comparison Window B Channel Select)

These bits select the channels to be compared under the comparison window B condition from AN100 to AN120, temperature sensor, internal reference voltage.
When the number of the A/D conversion channel selected in the ADANSAy.ANSAyn bit (y = 0 to 1, n = 00 to 15) and ADANSBy.ANSByn bit is specified, the comparison window B function is enabled.
The CMPCHB[5:0] bits should be set while the ADCSR.ADST bit is 0.

CMPLB Bit (Comparison Window B Comparison Condition Setting)

This bit specifies the condition for comparison of the channels for window B. When the comparison result of each analog input matches the pre-set condition, the ADCMPBSR.CMPSTB flag is set to 1, a comparison interrupt request (S12CMPBI1) is generated. Figure 56.5 shows the conditions for comparison.

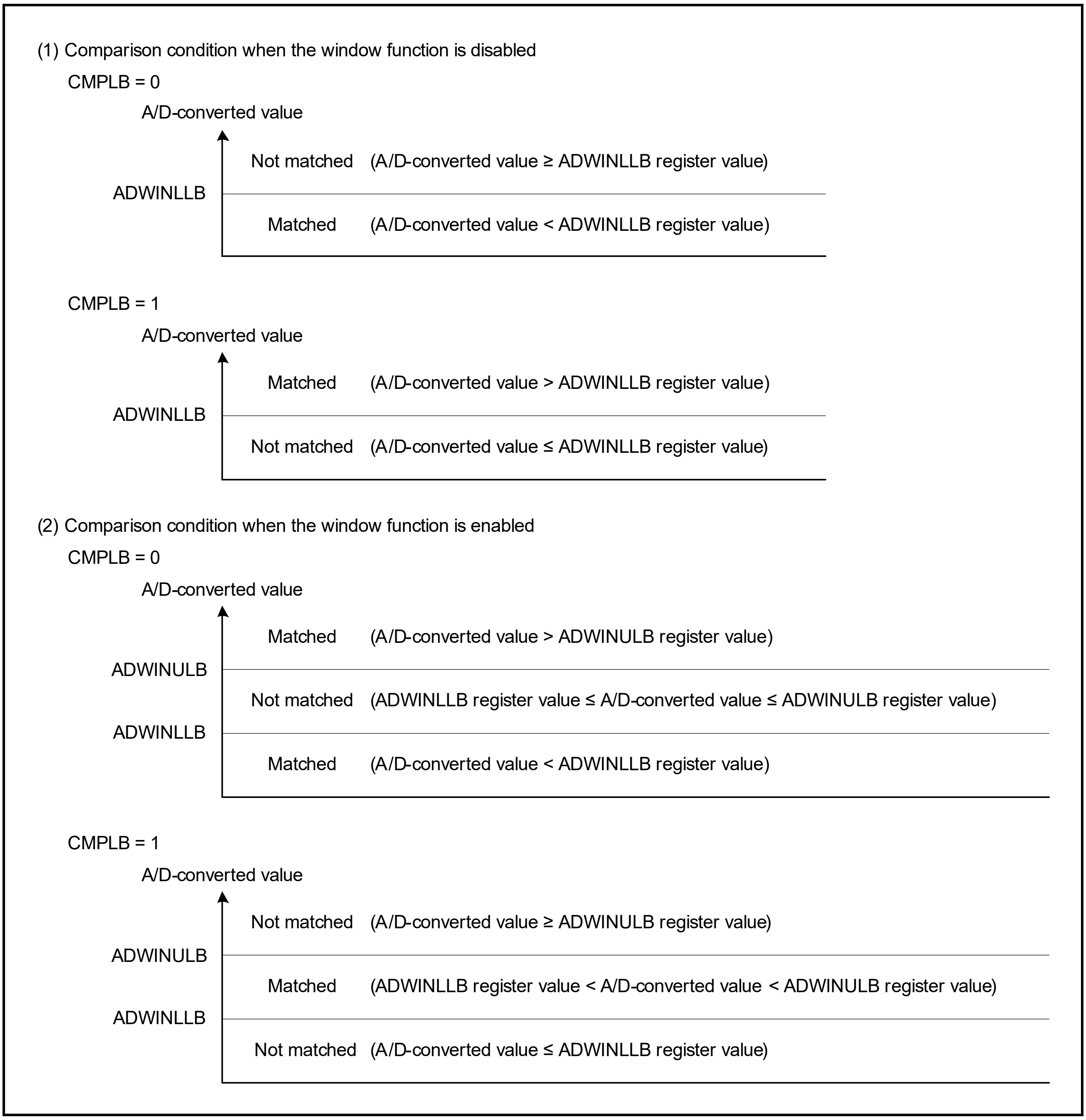


Figure 56.5 Details of Comparison Condition: Comparison Function Window B

56.2.37 A/D Comparison Function Window B Lower Level Setting Register (ADWINLLB)

Address(es): S12AD.ADWINLLB 0008 90A8h, S12AD1.ADWINLLB 0008 91A8h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

This is a readable and writable register that specifies reference data when comparison window B is in use. The register specifies the lower level of window B.
A write operation to the ADCMPDR1 register is enabled even during A/D conversion. Rewriting the value of the register during A/D conversion dynamically changes the reference data.
Satisfy the condition of [upper limit level ≥ lower limit level (ADWINULB setting value ≥ ADWINLLB setting value)] when setting.

The data formats of the register for each given condition are shown below.

- Setting of the A/D data register format select bit (flush-right or -left format)
- Setting of the A/D conversion resolution setting bit (12, 10, or 8 bits)
- Setting of the A/D converted value addition/averaging function channel select register (A/D converted value averaging mode: selected or not selected)
- Setting of the A/D converted value addition/averaging count select register (addition/averaging mode selected, addition count selected)

Note: When the comparison values are set in a different format from that of the A/D data register y (ADDRy), correct result of comparison cannot be obtained.

(1) When A/D-converted value addition/averaging mode is not selected

- Flush-right format with the setting for 12-bit resolution
  The comparison level (lower) is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format with the setting for 10-bit resolution
  The comparison level (lower) is set in bits 9 to 0. Write 0 to bits 15 to 10.
- Flush-right format with the setting for 8-bit resolution
  The comparison level (lower) is set in bits 7 to 0. Write 0 to bits 15 to 8.
- Flush-left format with the setting for 12-bit resolution
  The comparison level (lower) is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format with the setting for 10-bit resolution
  The comparison level (lower) is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format with the setting for 8-bit resolution
  The comparison level (lower) is set in bits 15 to 8. Write 0 to bits 7 to 0.

(2) When A/D-converted averaging mode is selected

- Flush-right format with the setting for 12-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format with the setting for 10-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 9 to 0. Write 0 to bits 15 to 10.

- Flush-right format with the setting for 8-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 7 to 0. Write 0 to bits 15 to 8.
- Flush-left format with the setting for 12-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format with the setting for 10-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format with the setting for 8-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 15 to 8. Write 0 to bits 7 to 0.

A/D-converted value average mode can be set only when 2- or 4-time conversion is selected in A/D-converted value addition mode.

(3) When A/D-converted addition mode is selected

- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 13 to 0. Write 0 to bits 15 and 14.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 9 to 0. Write 0 to bits 15 and 10.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 15 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 15 to 2. Write 0 to bits 1 and 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The comparison level (lower) to be compared with the A/D-converted results of the same channel is set in bits 15 to 0.

When A/D converted value addition mode is selected, the value for comparison with the result of cumulative A/D converted value of the same channel is specified. Conversion 1, 2, 3, 4, or 16 times is selectable. When A/D converted value addition mode is selected and if conversion 1 to 4 is selected, the resolution of the results of conversion will be extended by 2 bits and so set a value for comparison with the results in the ADWINLLB register. When conversion 16 times is selected, the resolution of the results of conversion will be extended by 4 bits and so set a value for comparison with the results in the ADWINLLB register.
Even when A/D converted value addition mode is selected, set the value for reference based on the setting of the A/D data register format select bit.

56.2.38 A/D Comparison Function Window B Upper Level Setting Register (ADWINULB)

Address(es): S12AD.ADWINULB 0008 90AAh, S12AD1.ADWINULB 0008 91AAh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

This is a readable and writable register that specifies the data for reference when the comparison window B function is used. The register specifies the upper level of window B.

A write operation to the register is enabled even during A/D conversion. Rewriting the value of the register during A/D conversion dynamically changes the reference data.

Satisfy the condition of [upper limit level ≥ lower limit level (ADWINULB setting value ≥ ADWINLLB setting value)] when setting.

The register is not used when the window function is disabled.

The format of the register differs depending on the conditions below.

- Setting of the A/D data register format select bit (flush-right or -left format)
- Setting of the A/D conversion resolution setting bit (12, 10, or 8 bits)
- Setting of the A/D-converted value addition/averaging function channel select register (when A/D converted value averaging mode is selected or not selected)
- Setting of the A/D converted value addition/averaging count select register (when addition/averaging mode is selected and addition count is selected)

Note: When the comparison values are set in a different format from that of the A/D data register y (ADDRy), correct result of comparison cannot be obtained.

(1) When A/D-converted value addition/averaging mode is not selected

- Flush-right format with the setting for 12-bit resolution
  The comparison level (upper) is set in bits 11 to 0. Write 0 to bits 15 to b12.
- Flush-right format with the setting for 10-bit resolution
  The comparison level (upper) is set in bits 9 to 0. Write 0 to bits 15 to b10.
- Flush-right format with the setting for 8-bit resolution
  The comparison level (upper) is set in bits 7 to 0. Write 0 to bits 15 to b8.
- Flush-left format with the setting for 12-bit resolution
  The comparison level (upper) is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format with the setting for 10-bit resolution
  The comparison level (upper) is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format with the setting for 8-bit resolution
  The comparison level (upper) is set in bits 15 to 8. Write 0 to bits 7 to 0.

(2) When A/D-converted averaging mode is selected

- Flush-right format with the setting for 12-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format with the setting for 10-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 9 to

0. Write 0 to bits 15 to 10.

- Flush-right format with the setting for 8-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 7 to 0. Write 0 to bits 15 to 8.
- Flush-left format with the setting for 12-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format with the setting for 10-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format with the setting for 8-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 8. Write 0 to bits 7 to 0.

A/D-converted value average mode can be set only when 2- or 4-time conversion is selected in A/D-converted value addition mode.

(3) When A/D-converted addition mode is selected

- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 13 to 0. Write 0 to bits 15 and 14.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 11 to 0. Write 0 to bits 15 to 12.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 9 to 0. Write 0 to bits 15 to 10.
- Flush-right format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 13 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 12-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 2. Write 0 to bits 1 and 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 10-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 4. Write 0 to bits 3 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 1 to 4 times) with setting of 8-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 6. Write 0 to bits 5 to 0.
- Flush-left format (in A/D-converted value addition mode and when number of conversions is selected for 16 times) with setting of 12-bit resolution
  The comparison level (upper) to be compared with the A/D-converted results of the same channel is set in bits 15 to 0.

When A/D converted value addition mode is selected, the value for comparison with the result of cumulative A/D converted value of the same channel is specified. Conversion 1, 2, 3, 4, or 16 times is selectable. When A/D converted value addition mode is selected and if conversion 1 to 4 is selected, the resolution of the results of conversion will be extended by 2 bits and so set a value for comparison with the results in the ADWINLLB register. When conversion 16 times is selected, the resolution of the results of conversion will be extended by 4 bits and so set a value for comparison with the results in the ADWINLLB register.
Even when A/D converted value addition mode is selected, set the value for reference based on the setting of the A/D data register format select bit.

56.2.39 A/D Comparison Function Window B Channel Status Register (ADCMPBSR)

(1) S12AD.ADCMPBSR

Address(es): 0008 90ACh

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	CMPSTB
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPSTB	Comparison Window B Flag	0: The comparison condition is not satisfied. 1: The comparison condition is satisfied.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is used to store results of comparison of the comparison window B function.

CMPSTB Flag (Comparison Window B Flag)

This is a status flag that indicates the result of comparison of the channels for window B comparison conditions (AN000 to AN007.
When the conversion result matches with the comparison condition that is set in the ADCMPBNSR.CMPCHB[5:0] bits upon completion of A/D conversion, this flag is set to 1. When the ADCMPCR.CMPIE bit is 1, a comparison interrupt request (S12CMPBI) is generated upon the setting of the flag.
1 cannot be written to the CMPSTB flag.
[Setting condition]

- When ADCMPCR.CMPBE = 1 and the condition set in the ADCMPBNSR.CMPLB bit is satisfied.

[Clearing condition]

- When 0 is written after reading the state of 1.

(2) S12AD1.ADCMPBSR

Address(es): 0008 91ACh

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	CMPSTB
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CMPSTB	Comparison Window B Flag	0: The comparison condition is not satisfied. 1: The comparison condition is satisfied.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This is a register that stores the result of comparison by comparison window B function.

CMPSTB Flag (Comparison Window B Flag)

This is a status flag that indicates the result of comparison of the channels for window B comparison conditions (AN100 to AN120, temperature sensor, internal reference voltage).

When the comparison condition that is set in the ADCMPBNSR.CMPCHB[5:0] bits is a match upon completion of A/D conversion, this flag is set to 1. When the ADCMPCR.CMPIE bit is 1, a comparison interrupt request (S12CMPBI1) is generated upon the setting of the flag.

1 cannot be written to the CMPSTB flag.

[Setting condition]

- When ADCMPCR.CMPBE = 1 and the condition set in the ADCMPBNSR.CMPLB bit is satisfied.

[Clearing condition]

- When 0 is written after reading the state of 1.

56.2.40 A/D Conversion Time Setting Register (ADSAM)

Address(es): S12AD.ADSAM 0008 906Eh, S12AD1.ADSAM 0008 916Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	SAM	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R
b5	SAM	Conversion Time Setting	0: Sets conversion time for high-speed. 1: Sets conversion time for middle-speed.	R/W
b15 to b6	—	Reserved	These bits are read as 0. The write value should be 0.	R

This register specifies conversion time for unit 0 (high-speed) and unit 1 (middle-speed).
Set the PSW.I bit to 0 (interrupt disabled) and the ADSAMPR.PRO[1:0] bits to 11 (writing enabled) before rewriting the register. Furthermore, disable writing to the register by setting the ADSAMPR.PRO[1:0] bits to 10 immediately after rewriting the register.

SAM Bit (Conversion Time Setting)

Set this bit to 0 for unit 0 which is a high-speed A/D converter. Setting this bit to 0 sets conversion time 13 states for 12-bit resolution, 11 states for 10-bit resolution, or 9 states for 8-bit resolution.
Set this bit to 1 for unit 1 which is a middle-speed A/D converter. Setting this bit to 1 sets conversion time 15 states for 12-bit resolution, 13 states for 10-bit resolution, or 11 states for 8-bit resolution.
Set the bit while the ADCSR.ADST bit is 0.
Note that the electrical characteristics cannot be satisfied when any setting other than the above is made to the respective units.

56.2.41 A/D Conversion Time Setting Protection Release Register (ADSAMPR)

Address(es): S12AD.ADSAMPR 0008 9063h, S12AD1.ADSAMPR 0008 9163h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	PRO[1:0]	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	PRO[1:0]	A/D Conversion Time Setting Register Protection	b1 b0 0 0: Writing to or reading from the A/D conversion time setting register is disabled (initial value. The value is read as 00b). 1 0: Writing to or reading from the A/D conversion time setting register is disabled (The value is read as 00b). 1 1: Writing to or reading from the A/D conversion time setting register is enabled (The value is read as 01b). Settings other than above are prohibited.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R

This register enables accesses to the A/D conversion time setting register (ADSAM).

Set the PRO[1:0] bits to 10b (writing disabled) after setting the A/D conversion time setting register (ADSAM).
The PRO[1:0] bits are read as 00b while writing to the bits is disabled, and 01b while writing to the bits is enabled.

56.3 Operation

56.3.1 Scanning Operation

In scanning, A/D conversion is performed sequentially on the analog inputs of the specified channels.
A scan conversion is performed in three operating modes: single scan mode, continuous scan mode, and group scan mode. In single scan mode, one or more specified channels are scanned once. In continuous scan mode, one or more specified channels are scanned repeatedly until the ADCSR.ADST bit is cleared to 0 from 1 by software. In group scan mode, the selected channels of groups A, B, and C are scanned once after starting to be scanned according to the respective synchronous trigger.
In single scan mode and continuous scan mode, A/D conversion is performed for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n. In group scan mode, A/D conversion is performed for ANn channels of groups A, B, and C selected by the ADANSA0 and ADANSA1 registers, ADANSB0 and ADANSB1 registers, and ADANSC0 and ADANSC1 registers, respectively, starting from the channel with the smallest number n.
When self-diagnosis is selected, it is executed once at the beginning of each scan and one of the three voltages internally generated in the 12-bit A/D converter is converted.
The temperature sensor output and internal reference voltage can be selected with analog input of the channels, and A/D conversion is performed in the order of the analog input of the channels, temperature sensor output, and internal reference voltage. When the extended analog is selected, perform A/D conversion in single scan mode or sequence scan mode.
Double trigger mode is to be used with single scan mode or group scan mode. With double trigger mode being enabled, A/D conversion data of a channel selected by the ADCSR.DBLANS[4:0] bits is duplicated only if the conversion is started by the synchronous trigger selected by the ADSTRGR.TRSA[5:0] bits. In group scan mode, the double trigger function can be used only for group A.
Extended double trigger mode indicates a state when the following synchronous trigger (two synchronous trigger sources enabled) is selected by the TRSA[5:0] bits in the A/D conversion start trigger select register (ADSTRGR) in double trigger mode.

- TRG4AN or TRG4BN (the ADSTRGR.TRSA[5:0] bits are set to 001011b)
- TRG7AN or TRG7BN (the ADSTRGR.TRSA[5:0] bits are set to 001111b)
- ELCTRG00N or ELCTRG01N, or ELCTRG10N or ELCTRG11N (Set the ADSTRGR.TRSA[5:0] bits to 011001b.)

In extended double trigger mode, in addition to normal operations in double trigger mode, A/D conversion data is stored in A/D data duplication register A (ADDBLDRA) or A/D data duplication register B (ADDBLDRB) depending on the trigger type. If two types of triggers have occurred simultaneously in this mode, A/D conversion data is not sorted by the trigger sources and is stored in data duplication register B (ADDBLDRB).
Note that if a new trigger is input during A/D conversion caused by another trigger, the new trigger is ignored.
When any of AN000 to AN002 channels is set as a channel-dedicated sample-and-hold circuit by the S12AD.ADSHCR.SHANS[2:0] bits, the target analog input is sampled and held before the first A/D conversion of each scan.

56.3.2 Single Scan Mode

56.3.2.1 Basic Operation (Without Channel-Dedicated Sample-and-Hold Circuits)

In basic operation of single scan mode, A/D conversion is performed once on the analog input of the specified channels as below.

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, A/D conversion is performed for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.

(2) Each time A/D conversion of a single channel is completed, the A/D conversion result is stored into the corresponding A/D data register (ADDRy).

(3) When A/D conversion of all the selected channels is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).

(4) The ADCSR.ADST bit remains 1 (A/D conversion start) during A/D conversion, and is automatically cleared to 0 when A/D conversion of all the selected channels is completed. Then the 12-bit A/D converter enters a wait state.

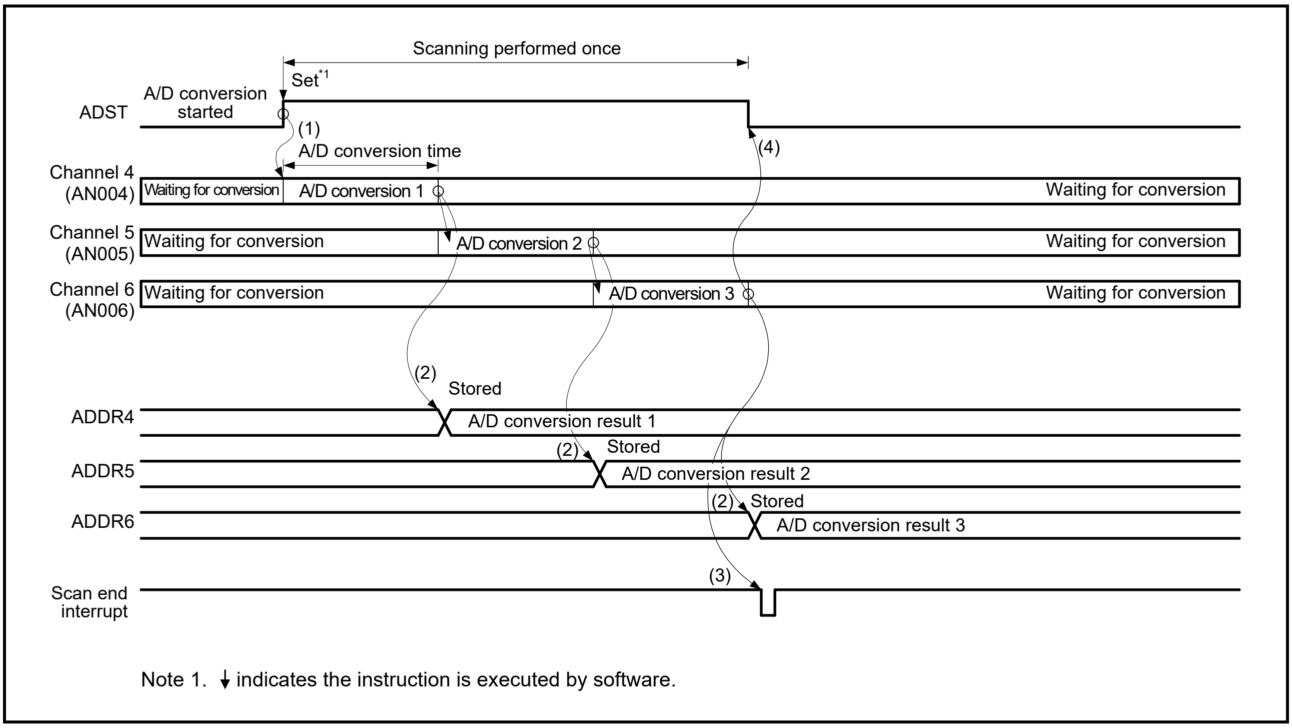


Figure 56.6 Example of Operation in Single Scan Mode (Basic Operation: AN004, AN005, AN006 Selected)

56.3.2.2 Basic Operation (With Channel-Dedicated Sample-and-Hold Circuits, With Constant Sampling Disabled)

When a channel-dedicated sample-and-hold circuit is used, sample-and-hold operations are performed first, and this is followed by A/D conversion once of the analog inputs on all selected channels. The ADSHCR.SHANS[2:0] bits are used to select the channels for which the channel-dedicated sample-and-hold circuits are to be used.

(1) Analog input sampling of all channels for which the channel-dedicated sample-and-hold circuits are to be used is started when the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input.
(2) After sample-and-hold operation, A/D conversion is performed for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.
(3) Each time A/D conversion of a single channel is completed, the result of A/D conversion is stored in the corresponding A/D data register (ADDRy).
(4) When A/D conversion of all the selected channels is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
(5) The ADCSR.ADST bit remains 1 (A/D conversion start) during A/D conversion, and is automatically cleared to 0 when A/D conversion of all the selected channels is completed. Then the 12-bit A/D converter enters a waiting state.

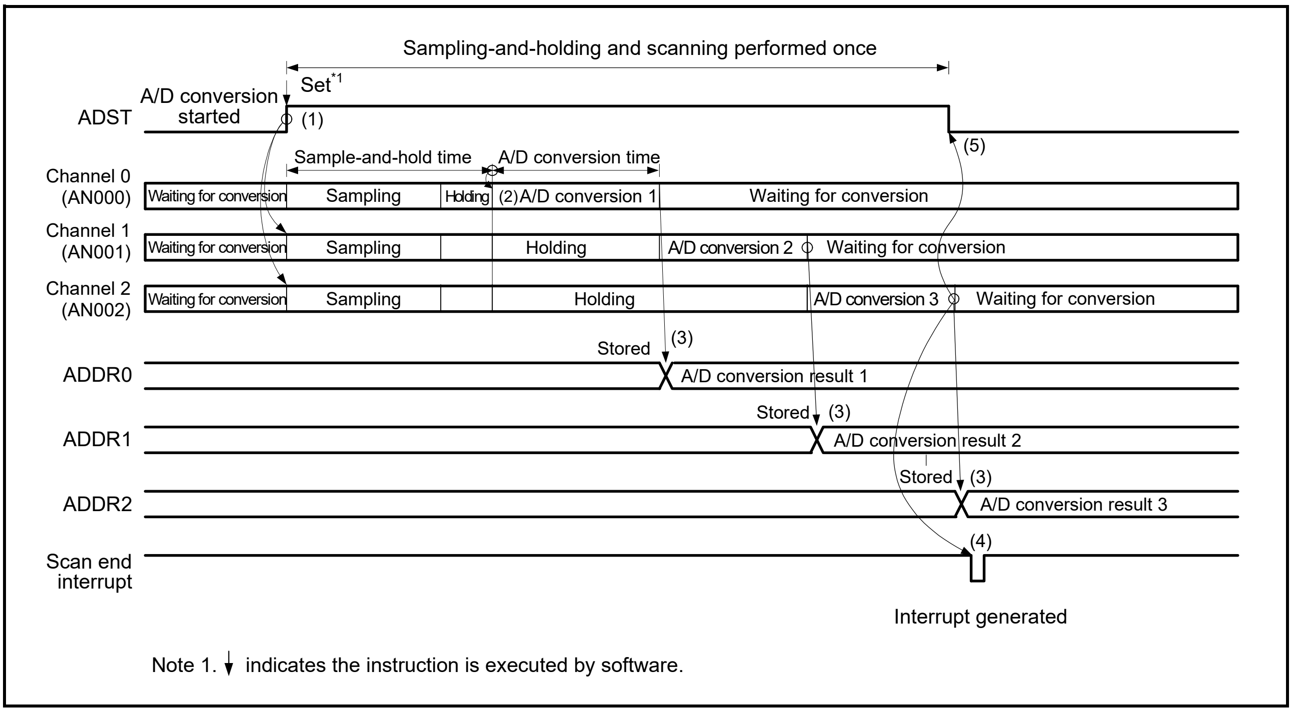


Figure 56.7 Example of Operation in Single Scan Mode (Channel-Dedicated Sample-and-Hold Circuits Used; AN000, AN001, AN002 Selected, With Constant Sampling Disabled)

56.3.2.3 Basic Operation (With Channel-Dedicated Sample-and-Hold Circuits, With Constant Sampling Enabled)

When constant sampling is enabled and the channel sample-and-hold circuit is used, a single A/D conversion is performed for analog input of all channels selected after sampling and holding is executed.
The channels for channel sample-and-hold circuit can be selected in the ADSHCR.SHANS[2:0] bits.

(1) When the ADSHMSR.SHMD bit is set to 1, the sample-and-hold circuit selected in the ADSHCR.SHANS[2:0] bits starts constant sampling.
(2) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, all channels that use the channel sample-and-holding circuit start to holding analog inputs. After setting the ADSHMSR.SHMD bit to 1 and 400 ns minimum (when allowable impedance of the source of the enable signal is 1 kΩ) are elapsed, set the ADCSR.ADST bit to 1.
(3) When the stabilizing time for the sample-and-hold circuit is elapsed, A/D conversion is performed for ANn channels selected in the ADANSA0 and ADANSA1 registers starting from the channel with the smallest number n.
(4) Each time A/D conversion of a single channels is completed, the A/D converted value is stored in the corresponding A/D data register (ADDRy), the sample-and-hold circuit resumes constant sampling.
(5) After completion of A/D conversion of all selected channels, if the ADCSR.ADIE bit is set to 1 (when interrupt generation after scanning is enabled), a scan end interrupt request is generated.
(6) The ADCSR.ADST bit holds 1 (start of A/D conversion) during A/D conversion and is automatically cleared upon completion of A/D conversion of all selected channels, and the 12-bit A/D converter enters into a wait state. When a single scan is performed in sequence, set the time of constant sampling of the sample-and-hold circuit to 400 ns or longer (when allowable impedance of the source of the enabling signal is 1 kΩ).
(7) When the ADSHMSR.SHMD bit is cleared to 0, the sample-and-hold circuit stops.

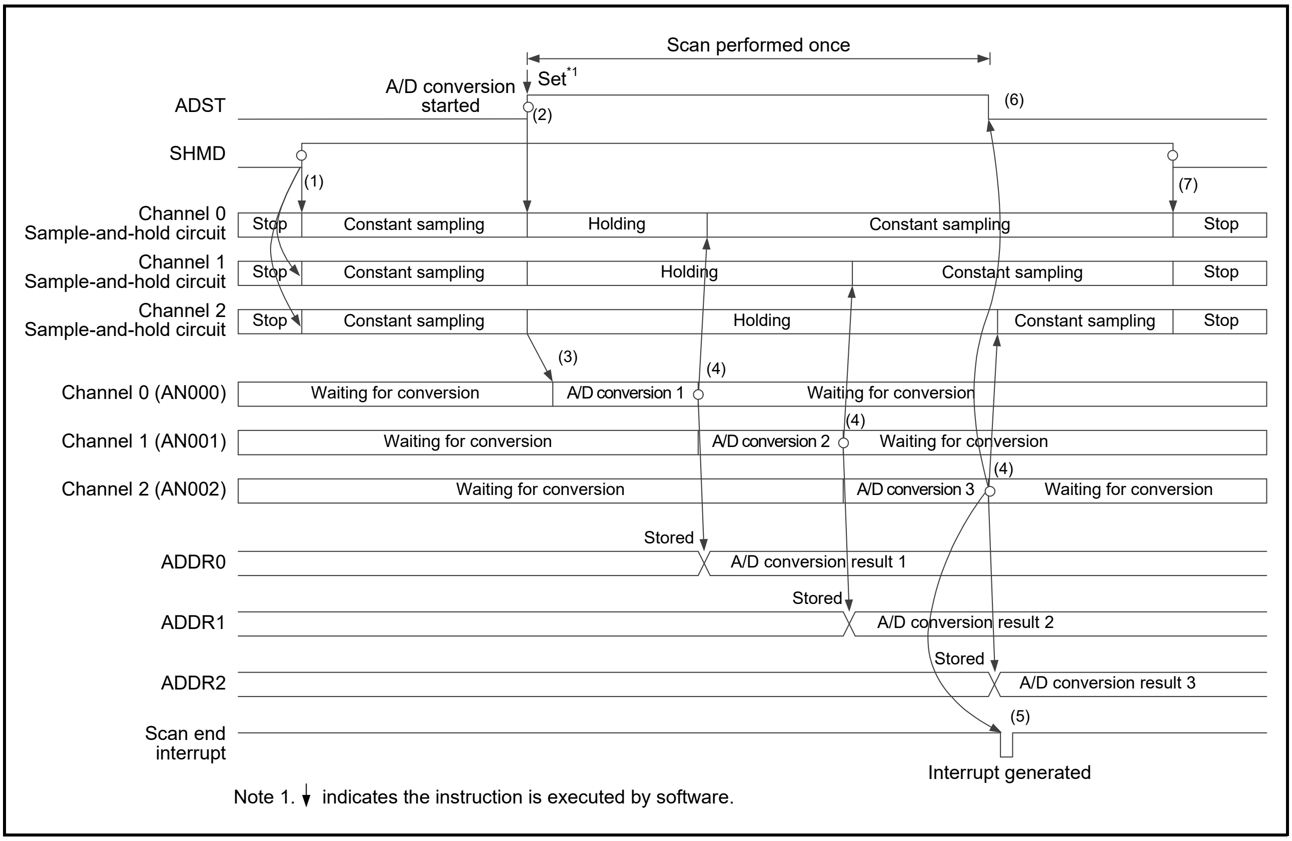


Figure 56.8 Example of Operation in Single Scan Mode (Channel-Dedicated Sample-and-Hold Circuits Used; AN000 to AN002 Selected, With Constant Sampling Enabled)

56.3.2.4 Channel Selection and Self-Diagnosis (Without Channel-Dedicated Sample-and-Hold Circuits)

When channels and self-diagnosis are selected, A/D conversion is performed once for the reference voltage supplied to the 12-bit A/D converter as below. After that, A/D conversion is performed only once on the analog input of the selected channels.

(1) A/D conversion for self-diagnosis is started when the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input.
(2) When A/D conversion for self-diagnosis is completed, A/D conversion result is stored into the A/D self-diagnosis data register (ADRD), and A/D conversion is performed for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.
(3) Each time A/D conversion of a single channel is completed, the A/D conversion result is stored into the corresponding A/D data register (ADDRy).
(4) When A/D conversion of all the selected channels is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
(5) The ADST bit remains 1 (A/D conversion start) during A/D conversion, and is automatically cleared to 0 when A/D conversion of all the selected channels is completed. Then the 12-bit A/D converter enters a wait state.

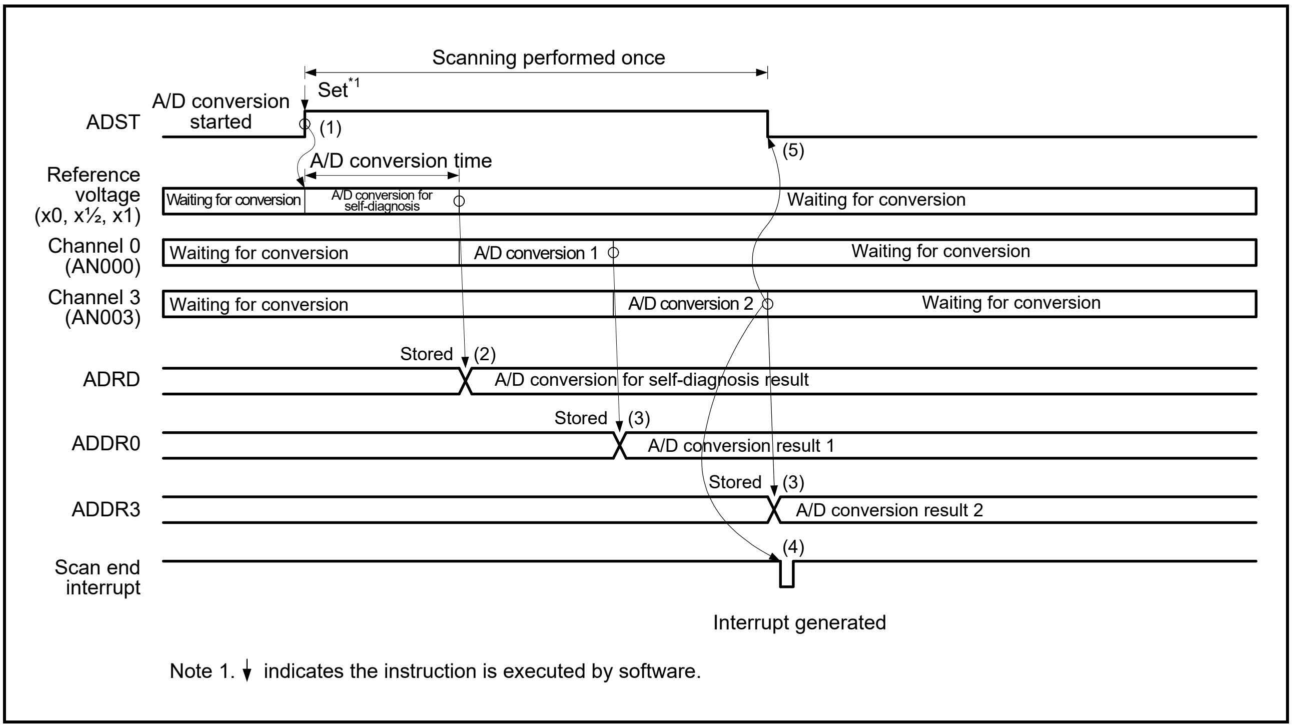


Figure 56.9 Example of Operation in Single Scan Mode (Basic Operation: AN000, AN003 Selected + Self-Diagnosis)

56.3.2.5 Channel Selection and Self-Diagnosis (With Channel-Dedicated Sample-and-Hold Circuits, With Constant Sampling Disabled)

When channels and self-diagnosis are selected and a channel-dedicated sample-and-hold circuit is used, sample-and-hold operations are performed first, and A/D conversion is performed once for the reference voltage supplied to the 12-bit A/D converter as below. After that, A/D conversion is performed only once on the analog input of the selected channels.

(1) Analog input sampling of all channels for which the channel-dedicated sample-and-hold circuits are to be used is started when the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input.
(2) A/D conversion for self-diagnosis is started after completion of sampling and holding.
(3) When A/D conversion for self-diagnosis is completed, A/D conversion result is stored into the A/D self-diagnosis data register (ADRD), and A/D conversion is performed for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.
(4) Each time A/D conversion of a single channel is completed, the A/D conversion result is stored into the corresponding A/D data register (ADDRy).
(5) When A/D conversion of all the selected channels is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt upon scanning completion enabled).
(6) The ADCSR.ADST bit remains 1 (A/D conversion start) during A/D conversion, and is automatically cleared to 0 when A/D conversion of all the selected channels is completed. Then the 12-bit A/D converter enters a wait state.

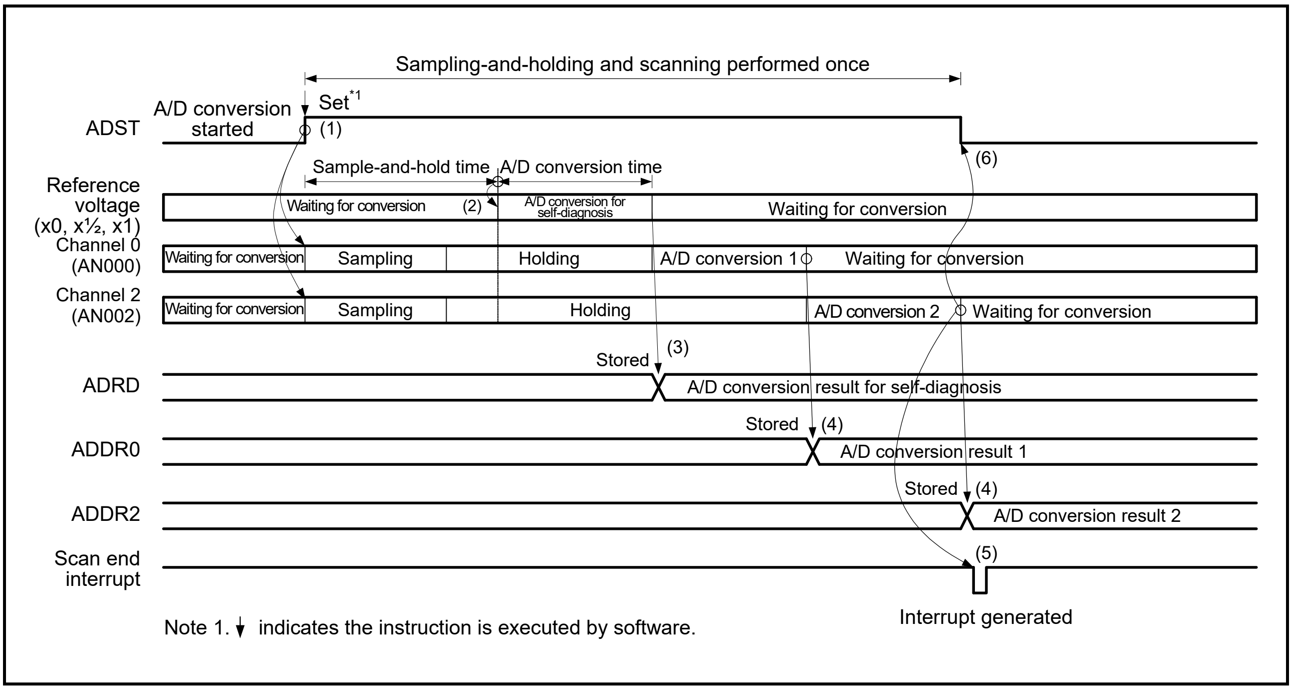


Figure 56.10 Example of Operation in Single Scan Mode (Channel-Dedicated Sample-and-Hold Circuits Used: AN000, AN002 Selected + Self-Diagnosis, with Constant Sampling Disabled)

56.3.2.6 Channel Selection and Self-Diagnosis (With Channel-Dedicated Sample-and-Hold Circuits, With Constant Sampling Enabled)

When self-diagnosis is selected with channels and a channel-dedicated sample-and-hold circuit is used with constant sampling enabled, sample-and-hold operations are performed first, and A/D conversion is performed on the reference voltage supplied to the 12-bit A/D converter as described below. After that, A/D conversion is performed only once on the analog input of the selected channels.

(1) When the ADSHMSR.SHMD bit is set to 1, the sample-and-hold circuit selected in the ADSHCR.SHANS[2:0] bits starts constant sampling.
(2) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, holding of analog input of all channels for which the channel-dedicated sample-and-hold circuits are to be used is started. Set the ADSHMSR.SHMD bit to 1 and wait for at least 400 ns (when impedance of the source of the enabling signal is 1 kΩ) to set the ADCSR.ADST bit to 1.
(3) When the stabilizing time for the sample-and-hold circuit is elapsed, A/D conversion by self-diagnosis is started.
(4) When A/D conversion by self-diagnosis is completed, the A/D converted values are stored in the self-diagnosis data register (ADRD) and A/D conversion is performed for ANn channels selected in the ADANSA0, ADANSA1 registers starting from the channel with the smallest number n.
(5) Each time A/D conversion of a single channel is completed, the A/D converted value is stored in the corresponding A/D data register (ADDRy), the sample-and-hold circuit resumes constant sampling.
(6) After completion of A/D conversion of all selected channels, if the ADCSR.ADIE bit is set to 1 (when interrupt generation after scanning is enabled), a scan end interrupt request is generated.
(7) The ADCSR.ADST bit holds 1 (A/D conversion start) during A/D conversion and is automatically cleared upon completion of A/D conversion of all selected channels, and the 12-bit A/D converter enters into a wait state. When a single scan is performed in sequence, set the time of constant sampling of the sample-and-hold circuit to at least 400 ns (when impedance of the source of the enabling signal is 1 kΩ).
(8) When the ADSHMSR.SHMD bit is cleared to 0, the sample-and-hold circuit stops.

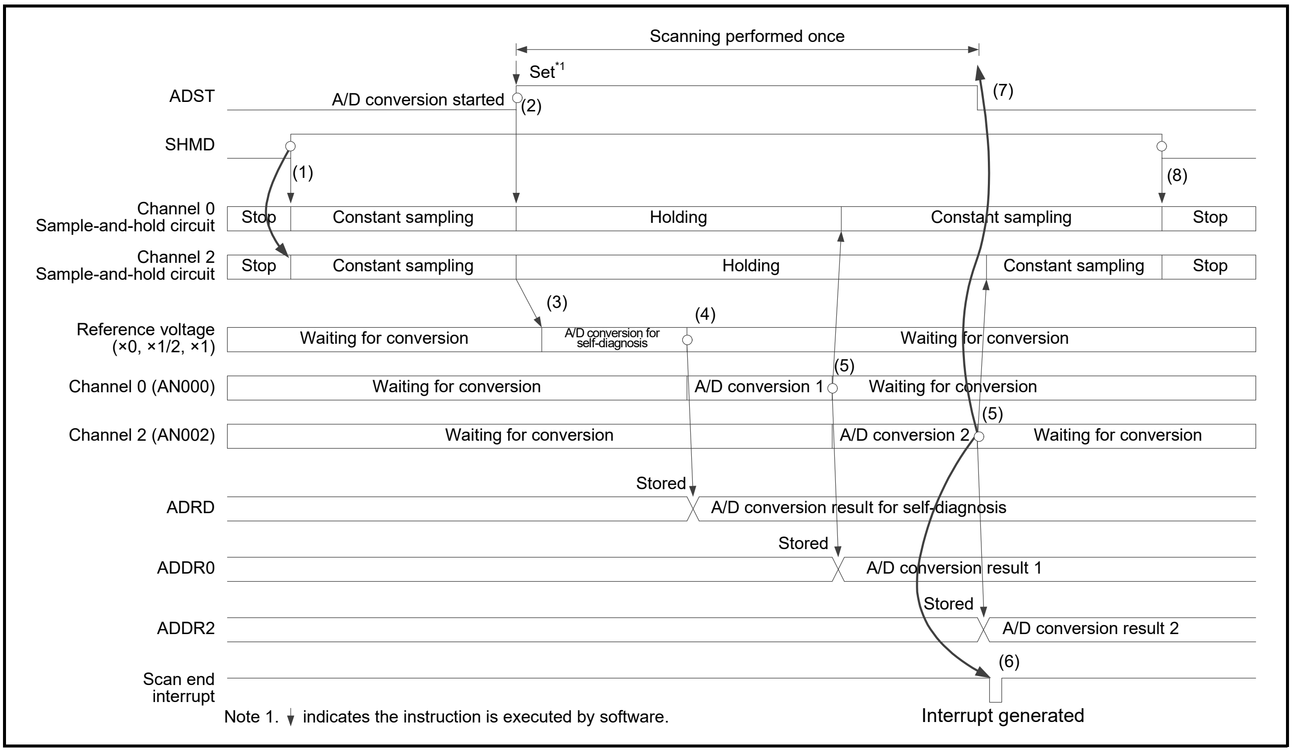


Figure 56.11 Example of Operation in Single Scan Mode (Channel-Dedicated Sample-and-Hold Circuits Used; AN000, AN002 Selected + Self-Diagnosis, With Constant Sampling Enabled)

56.3.2.7 A/D Conversion With Temperature Sensor Output/Internal Reference Voltage Selected

When temperature sensor output or internal reference voltage is selected with channels, A/D conversion of analog input of the channels is performed as described below. After that, A/D conversion is performed only once for the temperature sensor output or internal reference voltage. When temperature sensor output and internal reference voltage are both selected, A/D conversion is performed for the temperature sensor output first and then internal reference voltage. Deselecting channels and selecting only one of temperature sensor output and internal reference voltage is possible.

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, and A/D conversion is performed for ANn channels selected by the ADANSA0 or ADANSA1 register, starting from the channel with the smallest number n.
(2) Each time A/D conversion of a single channel is completed, the A/D converted value is stored in the corresponding A/D data register (ADDRy), and then A/D conversion for temperature sensor output is started.
(3) When A/D conversion of the temperature sensor output is completed, the A/D converted value is stored in the corresponding A/D temperature sensor data register (ADTSDR), and then A/D conversion for internal reference voltage is started.
(4) After completion of A/D conversion of the internal reference voltage, the A/D converted value is stored in the corresponding A/D internal reference voltage data register (ADOCDR), and if the ADCSR.ADIE bit is set to 1 (when the generation of an interrupt after scanning is enabled), a scan end interrupt request is generated.
(5) The ADCSR.ADST bit holds 1 (A/D conversion start) during A/D conversion, and the bit is automatically cleared upon completion of the A/D conversion. The 12-bit A/D converter enters into a wait state.

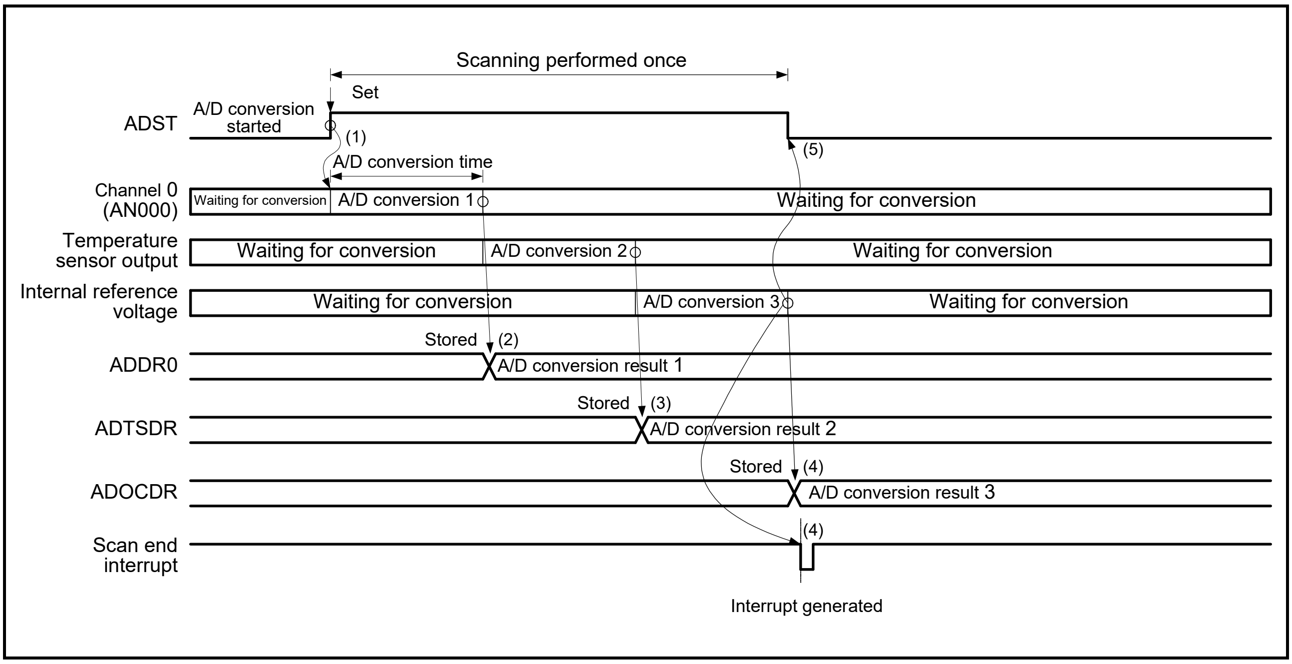


Figure 56.12 Example of Operation in Single Scan Mode (Basic Operation: AN000, Temperature Sensor Output, and Internal Reference Voltage Selected)

56.3.2.8 A/D Conversion in Double Trigger Mode

In single scan mode with double trigger mode, single scan operation started by synchronous trigger is performed twice as below.
Deselect self-diagnosis and set the temperature sensor A/D conversion select bit (ADEXICR.TSSA, ADEXICR.TSSB) and internal reference voltage A/D conversion select bit (ADEXICR.OCSA, ADEXICR.OCSB) to 0.
Duplication of A/D conversion data is enabled by setting the channel numbers to be duplicated to the ADCSR.DBLANS[4:0] bits and setting the ADCSR.DBLE bit to 1. When the ADCSR.DBLE bit is set to 1, channel selection using the ADANSA0 and ADANSA1 registers is invalid. In double trigger mode, synchronous triggers should be selected using the ADSTRGR.TRSA[5:0] bits, the ADCSR.EXTRG bit should be set to 0, and the ADCSR.TRGE bit should be set to 1. Software trigger should not be used.

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by synchronous trigger input, A/D conversion is started on the single channel selected by the ADCSR.DBLANS[4:0] bits.
(2) When A/D conversion is completed, the A/D conversion result is stored into the corresponding A/D data register (ADDRy).
(3) The ADCSR.ADST bit is automatically cleared to 0 and the 12-bit A/D converter enters a wait state. Here, a scan end interrupt request is not generated irrespective of the ADCSR.ADIE bit setting (interrupt generation upon scanning completion enabled).
(4) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by the second trigger input, A/D conversion is started on the single channel selected by the ADCSR.DBLANS[4:0] bits.
(5) When A/D conversion is completed, the A/D conversion result is stored into the A/D data duplication register (ADDBLDR), which is exclusively used in double trigger mode.
(6) If the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled), a scan end interrupt request is generated.
(7) The ADCSR.ADST bit remains 1 (A/D conversion start) during A/D conversion, and is automatically cleared to 0 when A/D conversion is completed. Then the 12-bit A/D converter enters a wait state.

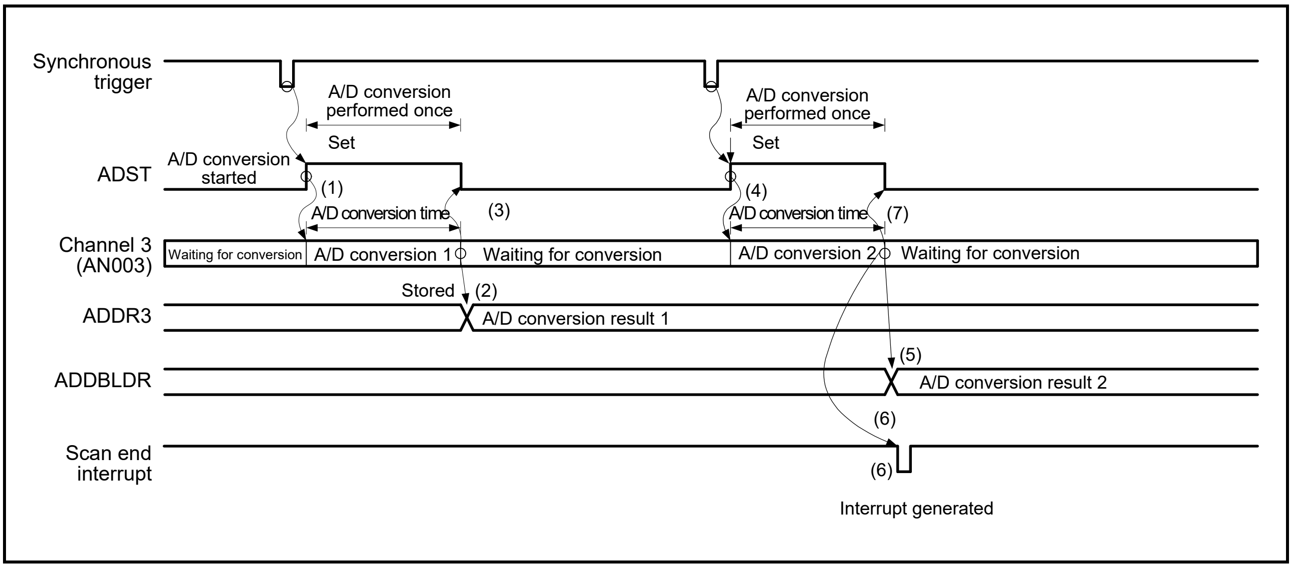


Figure 56.13 Example of Operation in Single Scan Mode (Double Trigger Mode Selected; AN003 Duplicated)

56.3.2.9 A/D Conversion in Extended Double Trigger Mode

When the double-trigger mode is selected in single-scanning mode, with TRG4AN or TRG4BN, TRG7AN or TRG7BN, ELCTRG00N or ELCTRG01N, or ELCTRG10N or ELCTRG11N selected in the TRSA[5:0] bits of the A/D conversion start trigger select register (ADSTRGR), proceed with single scanning twice as follows.
Deselect self-diagnosis and set the temperature sensor A/D conversion select bit (ADEXICR.TSSA, ADEXICR.TSSB) and internal reference voltage A/D conversion select bit (ADEXICR.OCSA, ADEXICR.OCSB) to 0.
Duplication of A/D conversion data is enabled by setting the channel numbers to be duplicated to the ADCSR.DBLANS[4:0] bits and setting the ADCSR.DBLE bit to 1. When the ADCSR.DBLE bit is set to 1, channel selection using the ADANSA0 and ADANSA1 registers is invalid. In extended double trigger mode, the ADCSR.EXTRG bit should be set to 0, and the ADCSR.TRGE bit should be set to 1. Software trigger should not be used.

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by TRG4AN input, A/D conversion is started on the single channel selected by the ADCSR.DBLANS[4:0] bits.
(2) When A/D conversion is completed, the A/D conversion result is stored into the corresponding A/D data register (ADDRy) and A/D data-duplication register A (ADDBLDRA).
(3) The ADCSR.ADST bit is automatically cleared to 0 and the 12-bit A/D converter enters a wait state. Here, a scan end interrupt request is not generated irrespective of the ADCSR.ADIE bit setting (interrupt generation upon scanning completion enabled).
(4) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by TRG4BN input, A/D conversion is started on the single channel selected by the ADCSR.DBLANS[4:0] bits.
(5) When A/D conversion is completed, the A/D conversion result is stored into A/D data duplication register (ADDBLDR) and A/D data duplication register B (ADDBLDRB).
(6) If the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled), a scan end interrupt request is generated.
(7) The ADCSR.ADST bit remains 1 (A/D conversion start) during A/D conversion, and is automatically cleared to 0 when A/D conversion is completed. Then the 12-bit A/D converter enters a wait state.

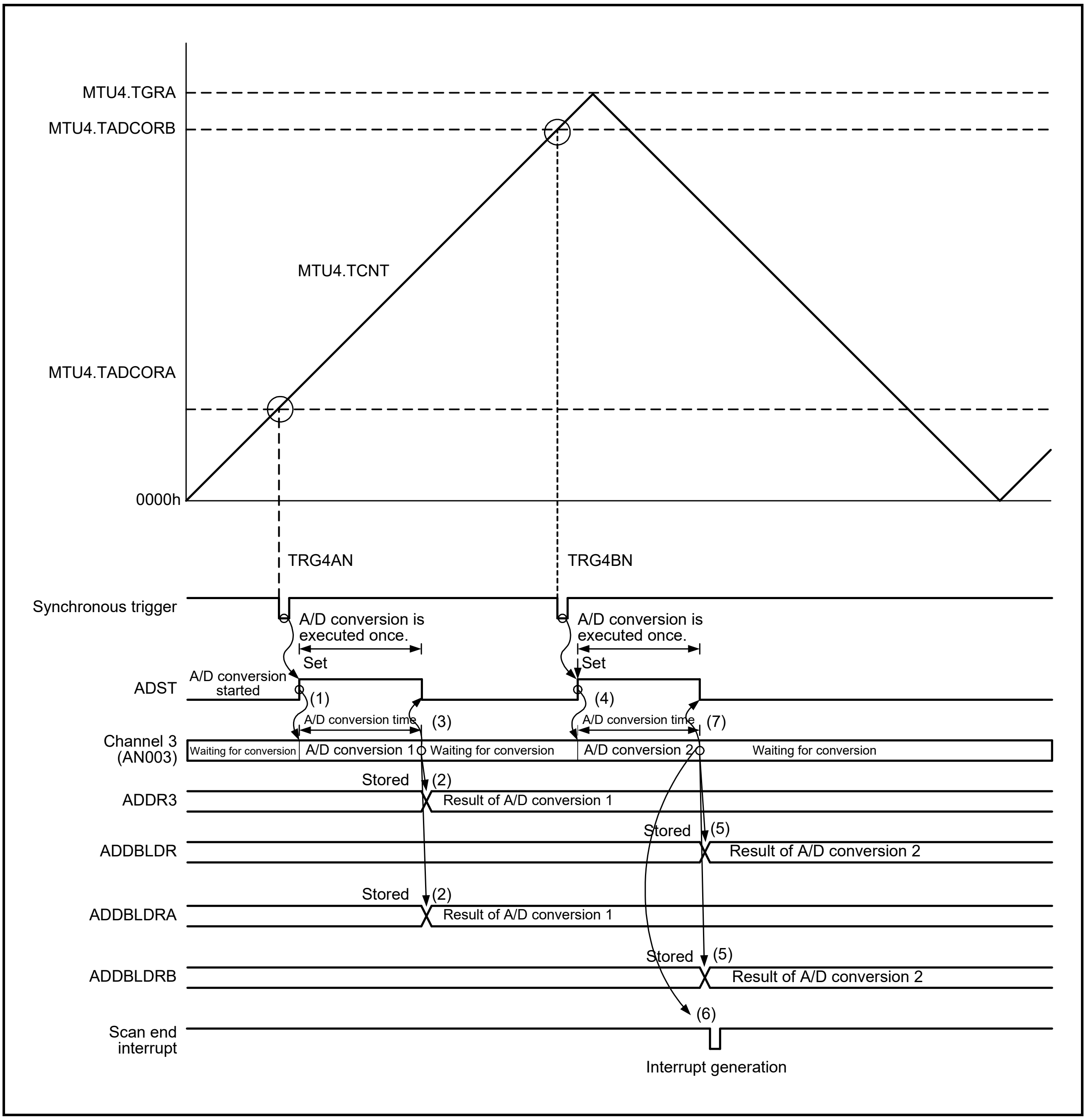


Figure 56.14 Example of Extended Operation in Double Trigger Mode (1) (Duplication Selected for AN003, TRG4AN or TRG4BN Selected, First Trigger is TRG4AN)

56.3.3 Continuous Scan Mode

56.3.3.1 Basic Operation (Without Channel-Dedicated Sample and-Hold Circuits)

In basic operation of continuous scan mode, A/D conversion is performed repeatedly on the analog input of the specified channels as below.

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, A/D conversion is performed for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.

(2) Each time A/D conversion of a single channel is completed, the A/D conversion result is stored into the corresponding A/D data register (ADDRy).

(3) When A/D conversion of all the selected channels is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
The 12-bit A/D converter sequentially starts A/D conversion for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.

(4) The ADCSR.ADST bit is not automatically cleared to 0 and steps 2 and 3 are repeated as long as the bit remains 1 (A/D conversion start). When the ADCSR.ADST bit is set to 0 (A/D conversion stop), A/D conversion stops and the 12-bit A/D converter enters a wait state.

(5) When the ADST bit is later set to 1 (A/D conversion start), A/D conversion is started again for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.

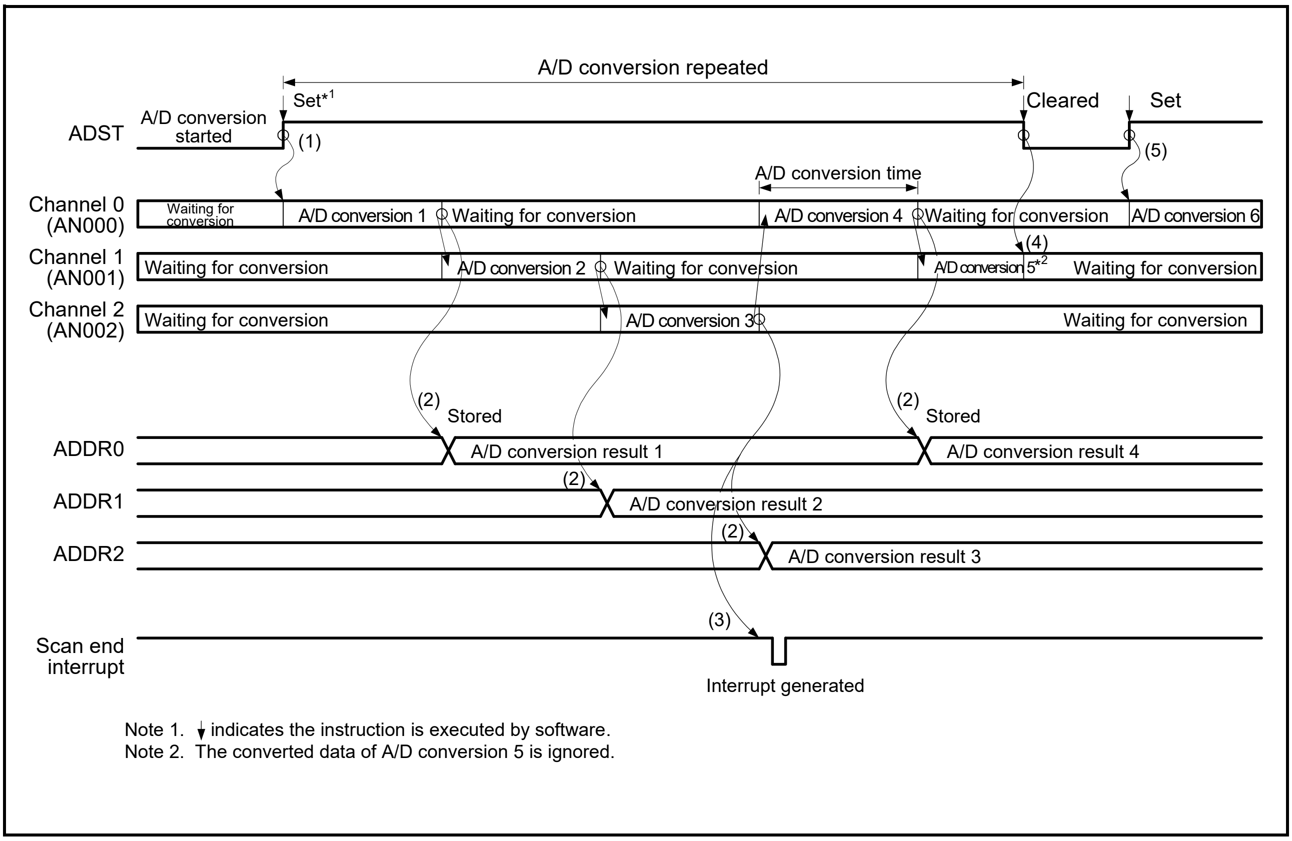


Figure 56.15 Example of Operation in Continuous Scan Mode (Basic Operation: AN000 to AN002 Selected)

56.3.3.2 Basic Operation (With Channel-Dedicated Sample-and-Hold Circuits, With Constant Sampling Disabled)

When a channel-dedicated sample-and-hold circuit is used, sample-and-hold operations are performed first, after which the analog inputs on all selected channels are A/D converted as below. The channels for which the channel-dedicated sample-and-hold circuits are to be used can be selected by the ADSHCR.SHANS[2:0] bits.

(1) Analog input sampling of all channels for which the channel-dedicated sample-and-hold circuits are to be used is started when the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, synchronous trigger input, or asynchronous trigger input.
(2) After sample-and-hold operation, A/D conversion is performed for ANn channels selected by the ADANSA0, ADANSA1 registers, starting from the channel with the smallest number n.
(3) Each time A/D conversion of a single channel is completed, the result of A/D conversion is stored in the corresponding A/D data register (ADDRy).
(4) When A/D conversion of all the selected channels is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled). At the same time, analog input sampling is started for all the channels for which the channel-dedicated sample-and-hold circuits are to be used.
(5) The ADCSR.ADST bit is not automatically cleared and steps 2 to 4 are repeated as long as the bit remains 1. When the ADCSR.ADST bit is set to 0 (A/D conversion stop), A/D conversion stops and the 12-bit A/D converter enters a wait state.
(6) When the ADCSR.ADST bit is then set to 1 (A/D conversion start), analog input sampling is started again for all the channels for which the channel-dedicated sample-and-hold circuits are to be used.

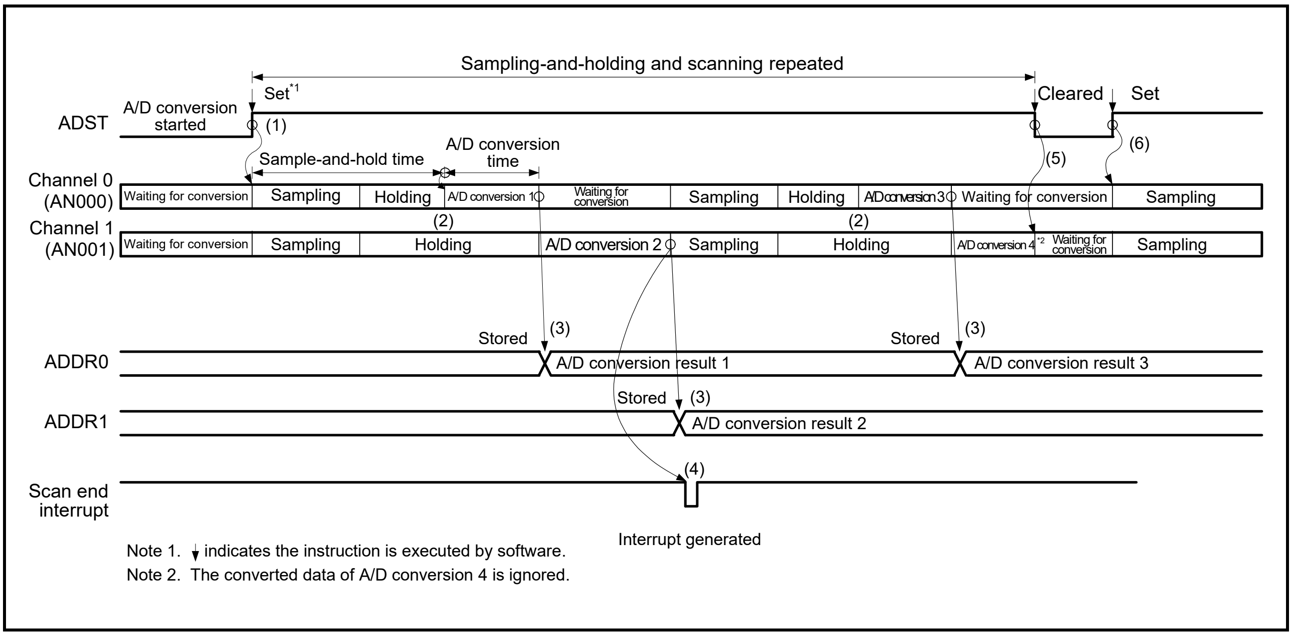


Figure 56.16 Example of Operation in Continuous Scan Mode (Channel-Dedicated Sample-and-Hold Circuits Used, AN000/AN001 Selected, With Constant Sampling Disabled)

56.3.3.3 Basic Operation (with Channel Sample-and-Hold Circuits, with Constant Sampling Enabled)

When constant sampling is enabled and a channel sample-and-hold circuit is used, sampling and holding is executed first and A/D conversion is repeated for analog input of all selected channels as described below.
The channels for which a channel-dedicated sample-and-hold circuit is used can be selected in the ADSHCR.SHANS[2:0] bits.

(1) When the ADSHMSR.SHMD bit is set to 1, the sample-and-hold circuit selected in the ADSHCR.SHANS[2:0] bits starts constant sampling.
(2) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, all channels that use the channel sample-and-hold circuit start to hold analog inputs. After setting the ADSHMSR.SHMD bit to 1 and wait at least 400 ns (when allowable impedance of the source of the enabling signal is 1 kΩ), set the ADCSR.ADST bit to 1.
(3) When the stabilizing time for the sample-and-hold circuit is elapsed, A/D conversion is performed for ANn channels selected in the ADANSA0 and ADANSA1 registers starting from the channel with the smallest number n.
(4) Each time A/D conversion of a single channels is completed, the A/D converted value is stored in the corresponding A/D data register (ADDRy), the sample-and-hold circuit resumes constant sampling.
(5) After completion of A/D conversion of all selected channels, if the ADCSR.ADIE bit is set to 1 (when interrupt generation after scanning is enabled), a scan end interrupt request is generated. In addition, all channels that use channel sample-and-hold circuit start holding of analog inputs.
(6) While the ADCSR.ADST bit holds 1 (start of A/D conversion) without being cleared, steps 3 to 5 above are repeated. When the ADCSR.ADST bit is cleared to 0 (A/D conversion stop), A/D conversion is stopped and the 12-bit A/D converter enters into a wait state.
(7) When the ADSHMSR.SHMD bit is cleared to 0, the sample-and-hold circuit stops.
(8) After that, when the ADSHMSR.SHMD bit is set to 1, the sample-and-hold circuit selected in the ADSHCR.SHANS[2:0] bits starts constant sampling.
(9) When the ADCSR.ADST bit is set to 1 (A/D conversion start), all channels that uses sample-and-hold circuit starts holding analog inputs.

Note: When only channels with a sample-and-hold circuit are selected and scanning proceeds in sequence, a period for sampling of a constant value is not secured after the second and later scans. To proceed with scanning continuously while the sample-and-hold circuit is enabled, select at least one channel from AN003 to AN007 and set the period for constant sampling of the sample-and-hold circuit to at least 400 ns (when allowable impedance of the source of the enabling signal is 1 kΩ).

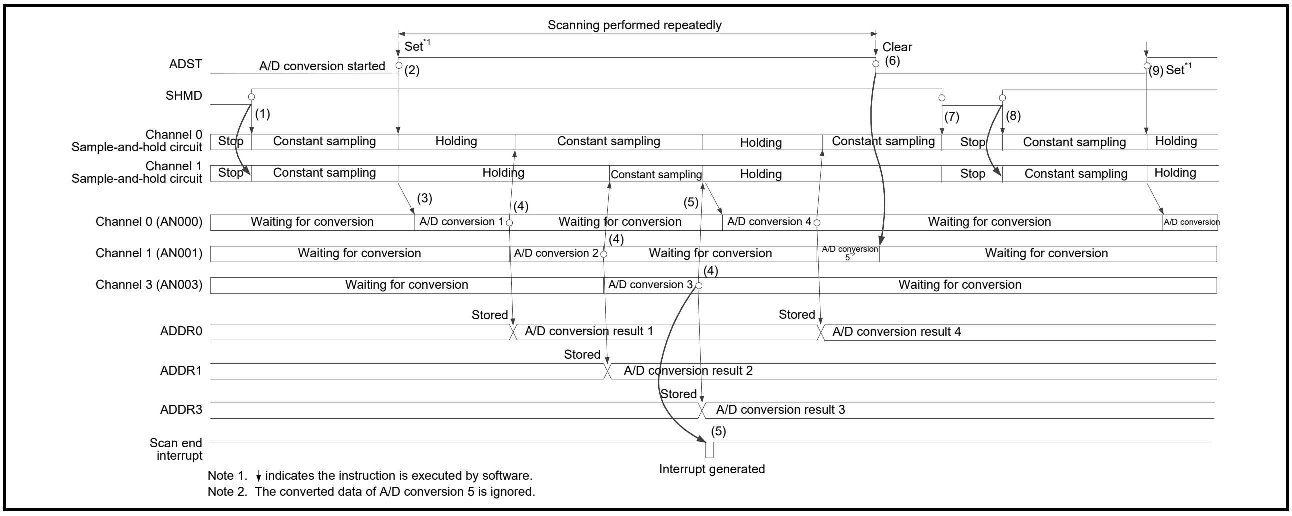


Figure 56.17 Example of Operation in Sequence Scan Mode (Channel-Dedicated Sample-and-Hold Circuits Used: AN000, AN001, and AN003 Selected, With Constant Sampling Enabled)

56.3.3.4 Channel Selection and Self-Diagnosis (Without Channel-Dedicated Sample-and-Hold Circuits)

When channels and self-diagnosis are selected at the same time, A/D conversion is first performed for the reference voltage supplied to the 12-bit A/D converter, and then A/D conversion is performed on the analog input of the selected channels, which sequence is repeated as below.

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, A/D conversion for self-diagnosis is started first.

(2) When A/D conversion for self-diagnosis is completed, the A/D conversion result is stored into the A/D self-diagnosis data register (ADRD). A/D conversion is then performed for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.

(3) Each time A/D conversion of a single channel is completed, the A/D conversion result is stored into the corresponding A/D data register (ADDRy).

(4) When A/D conversion of all the selected channels is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled). At the same time, the 12-bit A/D converter starts A/D conversion for self-diagnosis and then starts A/D conversion on ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.

(5) The ADCSR.ADST bit is not automatically cleared and steps 2 to 4 are repeated as long as the bit remains 1. When the ADST bit is set to 0 (A/D conversion stop), A/D conversion stops and the 12-bit A/D converter enters a wait state.

(6) When the ADCSR.ADST bit is later set to 1 (A/D conversion start), the A/D conversion for self-diagnosis is started again.

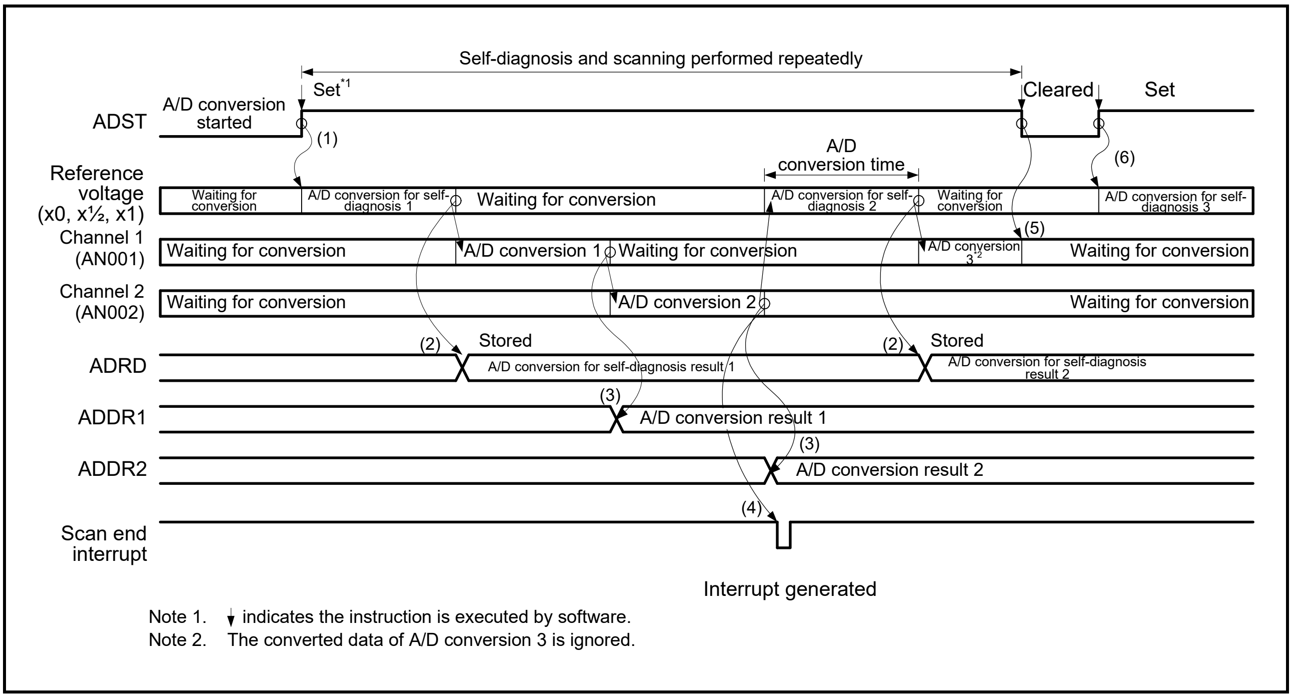


Figure 56.18 Example of Operation in Continuous Scan Mode (Basic Operation; AN001 and AN002 Selected + Self-Diagnosis)

56.3.3.5 Channel Selection and Self-Diagnosis (With Channel-Dedicated Sample-and-Hold Circuits, With Constant Sampling Disabled)

When channels and self-diagnosis are selected and a channel-dedicated sample-and-hold circuit is used, sample-and-hold operations are performed first, and A/D conversion is performed for the reference voltage supplied to the 12-bit A/D converter as below. After that, A/D conversion is performed repeatedly on the analog input of the selected channels.

(1) Analog input sampling of all channels for which the channel-dedicated sample-and-hold circuits are to be used is started when the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input.
(2) A/D conversion for self-diagnosis is started after completion of sampling and holding.
(3) When A/D conversion for self-diagnosis is completed, A/D conversion result is stored into the A/D self-diagnosis data register (ADRD), and A/D conversion is performed for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.
(4) Each time A/D conversion of a single channel is completed, the A/D conversion result is stored into the corresponding A/D data register (ADDRy).
(5) When A/D conversion of all the selected channels is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt upon scanning completion enabled). At the same time, analog input sampling is started for all the channels for which the channel-dedicated sample-and-hold circuits are to be used.
(6) The ADCSR.ADST bit is not automatically cleared and steps 2 to 5 are repeated as long as the ADCSR.ADST bit remains 1. When the ADST bit is set to 0 (A/D conversion stop), A/D conversion stops and the 12-bit A/D converter enters a wait state.
(7) When the ADST bit is later set to 1 (A/D conversion start), the A/D conversion for self-diagnosis is started again.

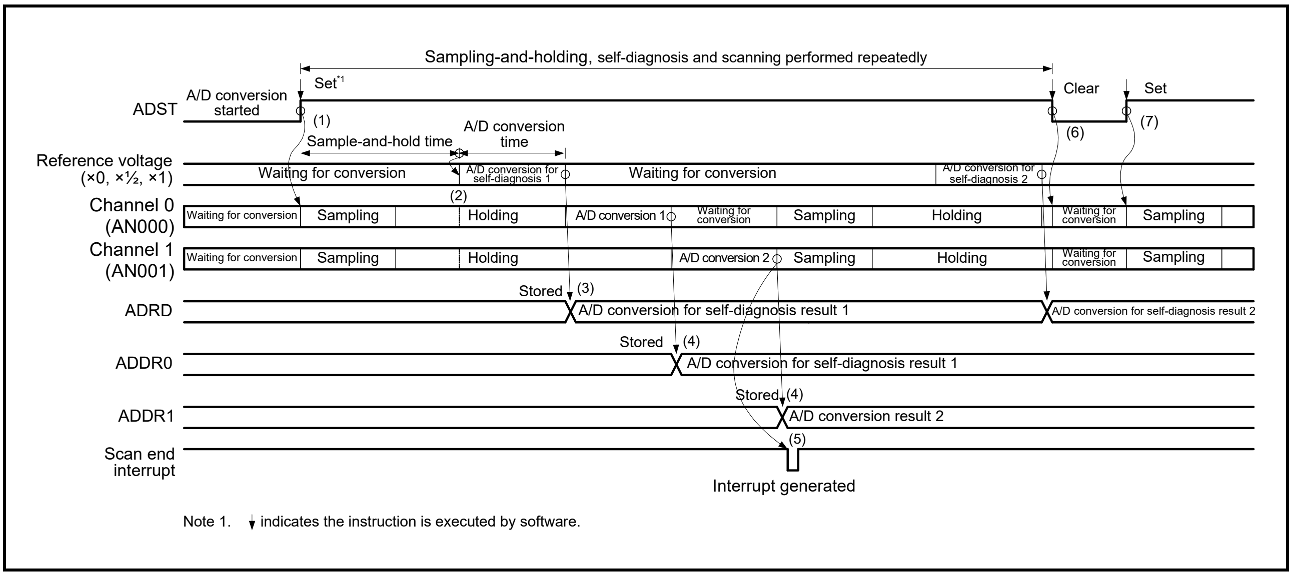


Figure 56.19 Example of Operation in Continuous Scan Mode (Channel-Dedicated Sample-and-Hold Circuits Used: AN000 and AN001 Selected + Self-Diagnosis)

56.3.3.6 Channel Selection and Self-Diagnosis (with Channel-Dedicated Sample-and-Hold Circuits, with Constant Sampling Enabled)

When self-diagnosis is selected with channels and a channel-dedicated sample-and-hold circuit is used with constant sampling enabled, sample-and-hold operations are performed first and A/D conversion is performed for the reference voltage to be supplied to the 12-bit A/D converter as described below. After that, A/D conversion is performed repeatedly on the analog input of the selected channels.

(1) The ADSHMSR.SHMD bit is set to 1, the sample-and-hold circuits selected in the ADSHCR.SHANS[2:0] start constant sampling.
(2) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, all channels that use the channel sample-and-hold circuit start to hold analog inputs. After setting the ADSHMSR.SHMD bit to 1 and wait at least 400 ns (when allowable impedance of the source of the enabling signal is 1 kΩ), set the ADCSR.ADST bit to 1.
(3) A/D conversion for self-diagnosis is started after stabilization period of the sample-and-hold circuit has been elapsed.
(4) When A/D conversion by self-diagnosis is completed, A/D converted value is stored to the A/D self-diagnosis data register (ADRD), and A/D conversion is performed for ANn channels selected by the ADANSA0 or ADANSA1 register starting from the channel with the smallest number n.
(5) Each time A/D conversion of a single channels is completed, the A/D converted value is stored in the corresponding A/D data register (ADDRy), the sample-and-hold circuit resumes constant sampling.
(6) After completion of A/D conversion of all selected channels, if the ADCSR.ADIE bit is set to 1 (when interrupt generation after scanning is enabled), a scan end interrupt request is generated. In addition, all channels that use channel sample-and-hold circuit start holding of analog inputs.
(7) While the ADCSR.ADST bit holds 1 (A/D conversion start) without being automatically cleared, steps 3 to 6 above are repeated. When the ADCSR.ADST bit is cleared to 0 (A/D conversion stop), A/D conversion is stopped and the 12-bit A/D converter enters into a wait state.
(8) When the ADSHMSR.SHMD bit is cleared to 0, the sample-and-hold circuit stops.
(9) After that, when the ADSHMSR.SHMD bit is set to 1, the sample-and-hold circuit selected in the ADSHCR.SHANS[2:0] bits starts constant sampling.
(10) When the ADCSR.ADST bit is set to 1 (A/D conversion start), all channels that uses sample-and-hold circuit again starts holding analog inputs.

Note: When only channels with a sample-and-hold circuit are selected and scanning proceeds in sequence, a period for sampling of a constant value is not secured after the second and later scans. To proceed with scanning continuously while the sample-and-hold circuit is enabled, select at least one channel from AN003 to AN007 and set the period for constant sampling of the sample-and-hold circuit to at least 400 ns (when allowable impedance of the source of the enabling signal is 1 kΩ).

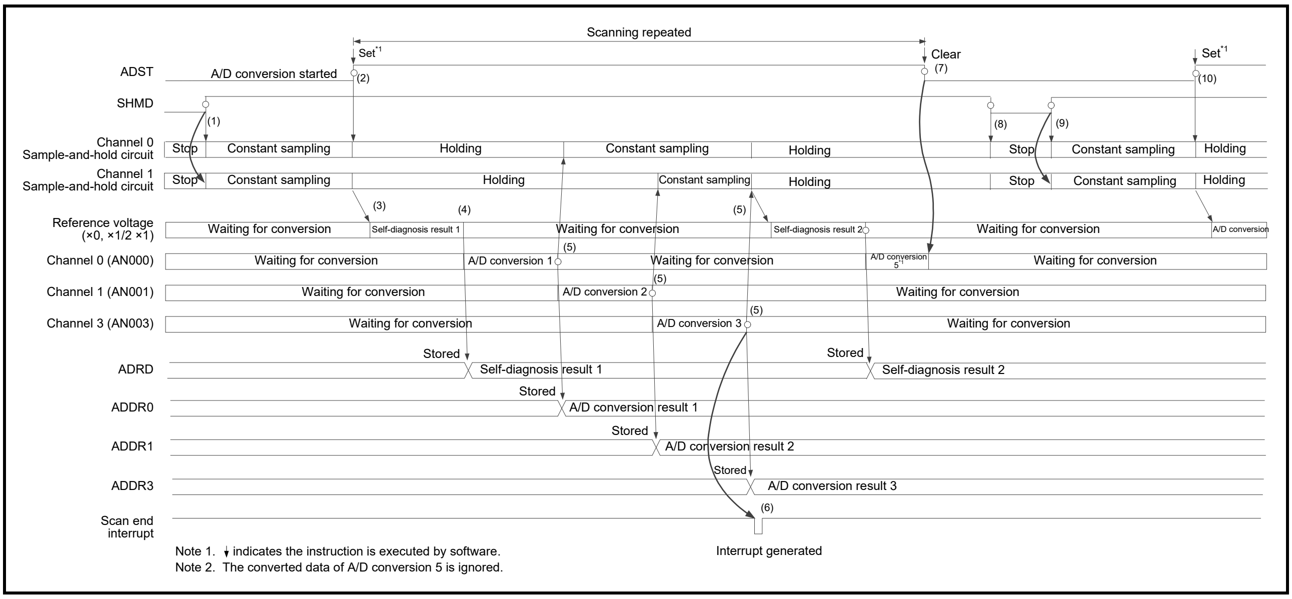


Figure 56.20 Example of Operation in Sequence Scan Mode (Channel-Dedicated Sample-and-Hold Circuits Used: AN000, AN001, and AN003 Selected + Self-Diagnosis, With Constant Sampling Enabled)

56.3.3.7 A/D Conversion With Temperature Sensor Output/Internal Reference Voltage Selected

When temperature sensor output or internal reference voltage are selected with channels, A/D conversion of analog input of the channels is performed, after that A/D conversion is performed for the temperature sensor output or internal reference voltage as described below. After that, A/D conversion is performed repeatedly on the analog input of the selected channels. When temperature sensor output and internal reference voltage are both selected, A/D conversion is performed temperature sensor output first and then internal reference voltage.
Deselecting channels and selecting only one of temperature sensor output and internal reference voltage is possible.

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, and A/D conversion is performed for ANn channels selected by the ADANSA0 or ADANSA1 register, starting from the channel with the smallest number n.
(2) Each time A/D conversion of a single channel is completed, the A/D converted value is stored in the corresponding A/D data register (ADDRy), and then A/D conversion for temperature sensor output is started.
(3) When A/D conversion of the temperature sensor output is completed, the A/D converted value is stored in the corresponding A/D temperature sensor data register (ADTSDR), and then A/D conversion for internal reference voltage is started.
(4) After completion of A/D conversion of the internal reference voltage, the A/D converted value is stored in the corresponding A/D internal reference voltage data register (ADOCDR), and if the ADCSR.ADIE bit is set to 1 (when interrupt generation after scanning is enabled), a scan end interrupt request is generated. The 12-bit A/D converter sequentially starts A/D conversion for ANn channels selected by the ADANSA0 and ADANSA1 registers, starting from the channel with the smallest number n.
(5) While the ADCSR.ADST bit holds 1 without being automatically cleared, steps 2 to 4 above are repeated. When the ADCSR.ADST bit is cleared to 0 (A/D conversion stop), A/D conversion is stopped and the 12-bit A/D converter enters into a wait state.
(6) After that, when the ADSHMSR.SHMD bit is set to 1 (A/D conversion start), A/D conversion is performed for ANn channels selected in the ADANSA0 and ADANSA1 registers starting from the channel with the smallest number n.

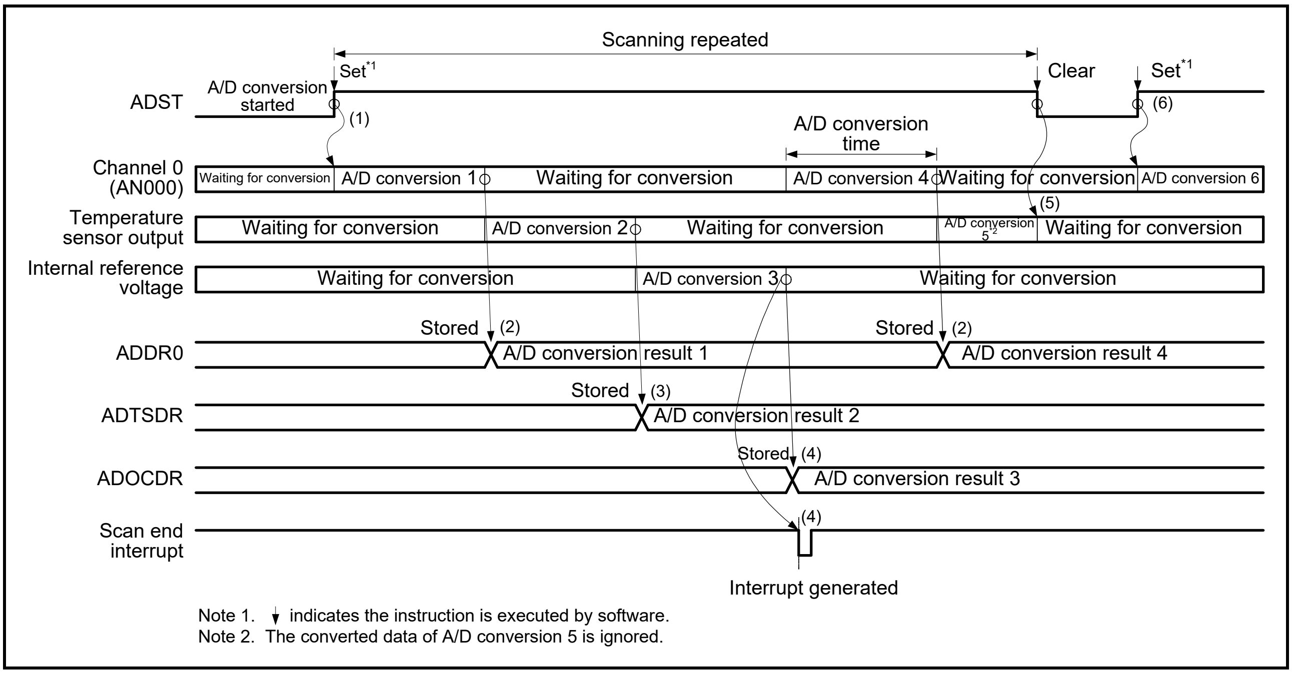


Figure 56.21 Example of Operation in Sequence Scan Mode (Basic Operation: AN000, Temperature Sensor Output and Internal Reference Voltage Selected)

56.3.4 Group Scan Mode

56.3.4.1 Basic Operation

Either two (groups A and B) or three (groups A, B, and C) can be selected as the number of the groups to be used in group scan mode.
In basic operation of group scan mode, A/D conversion is performed once on the analog inputs of all the specified channels in groups A and B, or groups A, B, and C after scan is started by a synchronous trigger as below. Scan operation of each group is similar to the scan operation in single scan mode.
The synchronous triggers of groups A and B, or groups A, B, and C can be selected using the TRSA[5:0], TRSB[5:0], and TRSC[5:0] bits in ADSTRGR, respectively. Different triggers should be used for each group A, B, and C so that scanning of groups A, B, and C does not occur simultaneously. Software trigger should not be used.
The channels to be scanned are selected using registers ADANSA0 and ADANSA1 and bits TSSA and OCSA in register ADEXICR for group A, registers ADANSB0 and ADANSB1 and bits TSSB and OCSB in register ADEXICR for group B, and registers ADANSC0 and ADANSC1 and bits TSSC and OCSC in register ADGCEXCR for group C.
When self-diagnosis is selected in group scan mode, self-diagnosis is separately executed for groups A and B, or groups A, B, and C.
The following describes operation in group scan mode using a trigger from the MTU. The TRG4AN, TRG4BN, and TRG4ABN triggers from the MTU are assumed to be used to start conversion of groups A, B and C, respectively.

(1) Scanning of group A is started by the TRG4AN trigger from the MTU.
(2) When group A scanning is completed, a scan end interrupt is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
(3) Scanning of group B is started by the TRG4BN trigger from the MTU.
(4) When group B scanning is completed, a group B scan end interrupt is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan completion enabled).
(5) Scanning of group C is started by the TRG4ABN trigger from the MTU.
(6) When group C scanning is completed, a group C scan end interrupt is generated if the ADGCTRGR.GCADIE bit is 1 (interrupt generation upon group C scan completion enabled).

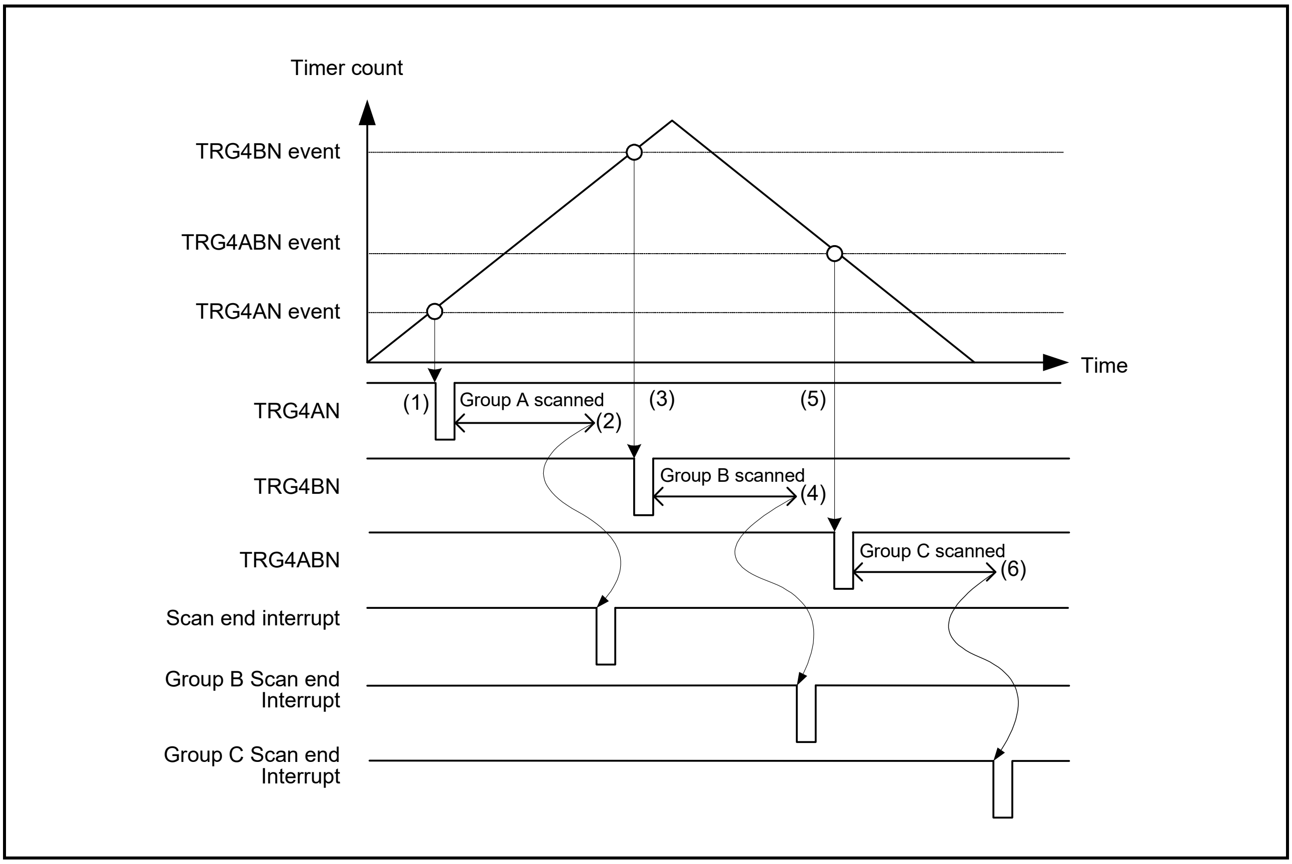


Figure 56.22 Example of Operation in Group Scan Mode (Basic Operation: Synchronous Triggers from MTU Used)

56.3.4.2 A/D Conversion in Double Trigger Mode

When double trigger mode is selected in group scan mode, two rounds of single scan operation started by a synchronous trigger are performed as a sequence for group A. For groups B and C, single scan operation started by a synchronous trigger is performed once.

In group scan mode, the synchronous triggers of groups A and B, or groups A, B, and C can be selected using the TRSA[5:0], TRSB[5:0], and TRSC[5:0] bits in ADSTRGR, respectively. Different triggers should be used for each group A, B, and C so that scanning of groups A, B, and C does not occur simultaneously. Software trigger and asynchronous trigger should not be used. When the TRG4AN or TRG4BN, TRG7AN or TRG7BN, ELCTRG00N or ELCTRG01N, or ELCTRG10N or ELCTRG11N is selected in the ADSTRGR.TRSA[5:0] bits as the synchronous trigger of the group A, operation is in the extended double trigger mode.

The channels to be scanned are selected using bits ADCSR.DBLANS[4:0] for group A, registers ADANSB0 and ADANSB1 for group B, and registers ADANSC0 and ADANSC1 for group C.

When double trigger mode is selected in group scan mode, the temperature sensor A/D conversion select bit (ADEXICR.TSSA) and internal reference voltage A/D conversion select bit (ADEXICR.OCSA) are set to 0 (not selected).

When double trigger mode is selected in group scan mode, self-diagnosis cannot be selected.

Duplication of A/D conversion data is enabled by setting the channel numbers to be duplicated to the ADCSR.DBLANS[4:0] bits and setting the ADCSR.DBLE bit to 1.

The following describes operation in group scan mode with double trigger mode using a synchronous trigger from the MTU. The TRG4ABN, TRGA0N, and TRGA1N triggers from the MTU are assumed to be used to start conversion of groups A, B, and C, respectively.

(1) Scanning of group C is started by the TRGA1N trigger from the MTU.
(2) When group C scanning is completed, a group C scan interrupt is generated if the ADGCTRGR.GCADIE bit is 1 (interrupt generation upon group C scan completion enabled).
(3) Scanning of group B is started by the TRGA0N trigger from the MTU.
(4) When group B scanning is completed, a group B scan end interrupt is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan enabled).
(5) The first scanning of group A is started by the first TRG4ABN trigger from the MTU.
(6) When the first scanning of group A is completed, the conversion result is stored into the corresponding A/D data register (ADDRy); a scan end interrupt request is not generated irrespective of the ADIE bit setting in ADCSR.
(7) The second scanning of group A is started by the second TRG4ABN trigger from the MTU.
(8) When the second scanning of group A is completed, the conversion result is stored into ADDBLDR. a scan end interrupt is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
(9) The second scanning of group B is started by the second TRGA0N trigger from the MTU.
(10) When the second scanning of group B is completed, a group B scan interrupt is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan completion enabled).
(11) The second scanning of group C is started by the second TRGA1N trigger from the MTU.
(12) When the second scanning of group C is completed, a group C scan interrupt is generated if the ADGCTRGR.GCADIE bit is 1 (interrupt generation upon group C scan completion enabled).

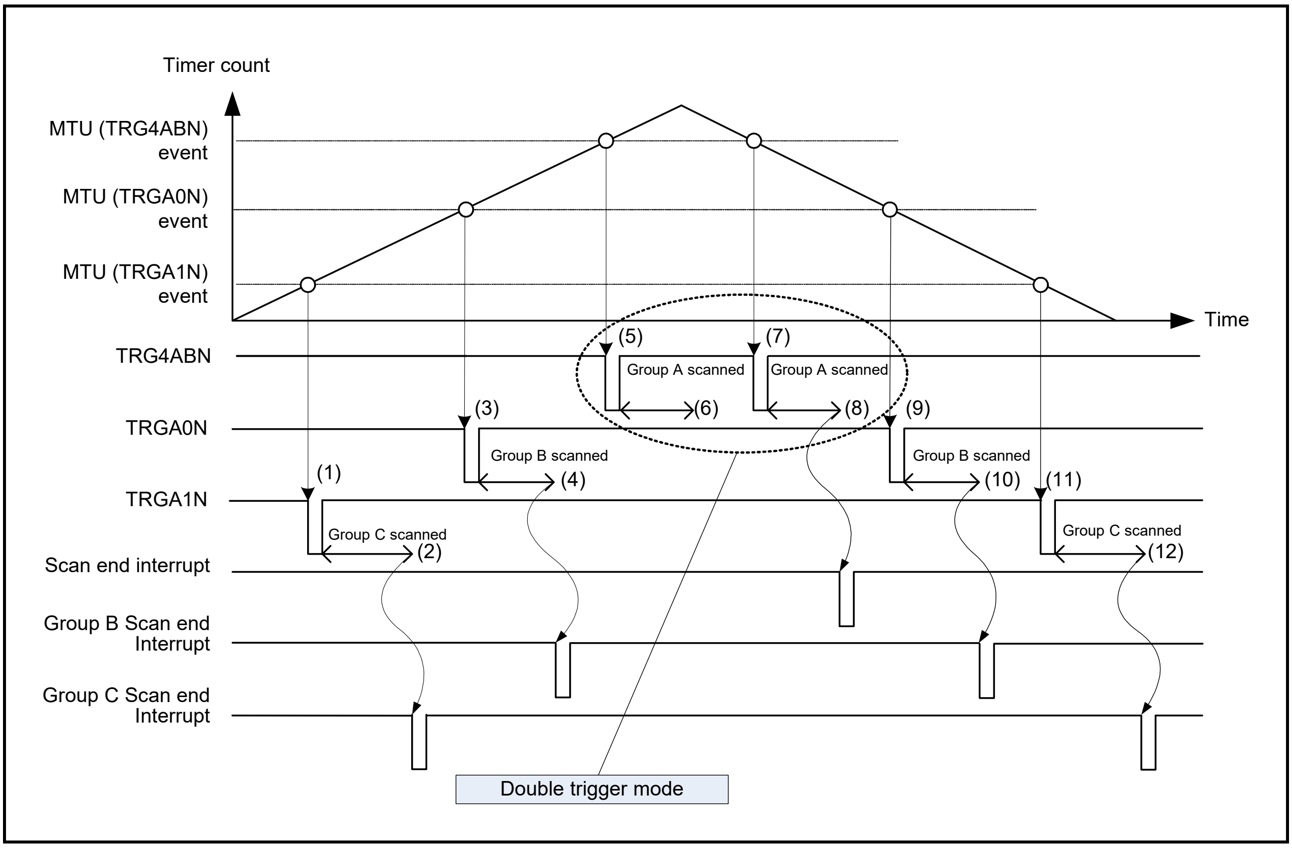


Figure 56.23 Example of Operation in Group Scan Mode with Double Trigger Mode (Basic Operation: Synchronous Triggers from MTU Used)

56.3.4.3 Operation under Group Priority Control

Setting the ADGSPCR.PGS bit to 1 in group scan mode makes operation proceed under group priority control. The group priority order is group A > group B > group C. Either two (groups A and B) or three (groups A, B, and C) can be selected as the number of the groups to be used in group scan mode. When setting the ADGSPCR.PGS bit to 1, follow the procedure described in Figure 56.24. If the procedure is not followed, proper scanning operation and data storage are not guaranteed.

In basic operation of group scan mode, if group A, B, or C is scanning, all other trigger inputs are ignored. Under group priority control, if a priority group trigger is input during A/D conversion for the low-priority group, A/D conversion for the low-priority group is discontinued and A/D conversion for the priority group proceeds.

If the ADGSPCR.GBRSCN bit is 0, the converter enters a wait state for the low-priority group on completion of the A/D conversion for the priority group.

The trigger input for the low-priority group during scanning is ignored.

If the ADGSPCR.GBRSCN bit is 1, the converter automatically restarts scanning for the low-priority group from the head of the group after A/D conversion for the priority group.

Also, the trigger input for the low-priority group generated during scanning for the priority group is enabled, and the converter automatically restarts scanning for the low-priority group after scanning for the priority group.

When the ADGSPCR.LGRRS bit is 0 while the ADGSPCR.GBRSCN bit is 1, the converter restarts scanning for the low-priority group from the head of the group. When the ADGSPCR.LGRRS bit is 1, the converter restarts scanning for the low-priority group from the channel on which scanning is discontinued.

However, when self-diagnosis is used, the converter restarts scanning from the channel on which scanning is discontinued after self-diagnosis is completed.

Table 56.15 summarizes operations in response to the input of a trigger during scanning with the settings of the ADGSPCR.GBRSCN bit.

When the ADGSPCR.GBRP bit is set to 1, scan operations in the lowest-priority group are continuously performed.

For the trigger settings in group scan mode, select a synchronous trigger for group A using the ADSTRGR.TRSA[5:0] bits, select a synchronous trigger different from that of group A for group B using the ADSTRGR.TRSB[5:0] bits, and select a synchronous trigger different from those of groups A and B for group C using the ADGCTRGR.TRSC[5:0] bits.

Set the ADSTRGR.TRSB[5:0] bits to 3Fh when setting group scan mode for two groups (the ADGCTRGR.GRCE bit to 0) and setting the ADGSPCR.GBRP bit to 1.

Set the ADGCTRGR.TRSC[5:0] bits to 3Fh when setting group scan mode for three groups (the ADGCTRGR.GRCE bit to 1) and setting the ADGSPCR.GBRP bit to 1.

Furthermore, as targets for scanning, select channels for group A by using the ADANSA0 and ADANSA1 registers, and ADEXICR.TSSA and ADEXICR.OCSA bits; select the group B channels by using ADANSB0 and ADANSB1 registers, and ADEXICR.TSSB and ADEXICR.OCSB bits; and select the group c by using ADANSC0 and ADANSC1 registers, and ADGCEXCR.TSSC and ADGCEXCR.OCSC bits respectively.

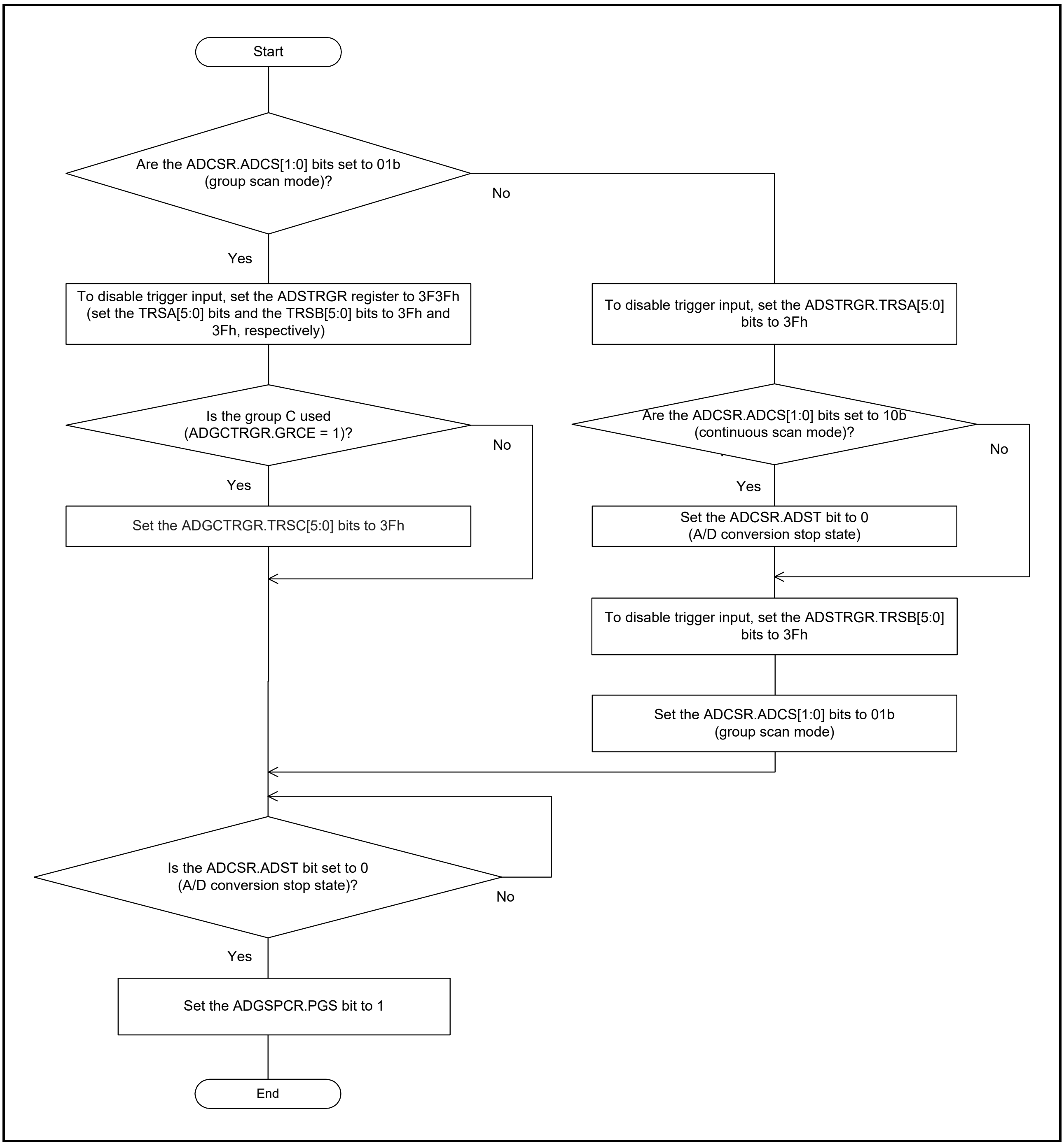


Figure 56.24 Flow of Setting the ADGSPCR.PGS Bit

Table 56.15 Control of Scanning Operations According to the Settings of the ADGSPCR.GBRSCN Bit

Scanning Operation	Trigger Input	ADGSPCR.GBRSCN = 0	ADGSPCR.GBRSCN = 1
When A/D conversion for group A is in progress	Input of trigger for group A	Trigger input is ineffective.	Trigger input is ineffective.
	Input of trigger for group B	Trigger input is ineffective.	A/D conversion is performed on group B after A/D conversion on group A is completed.
	Input of trigger for group C	Trigger input is ineffective.	A/D conversion is performed on group C after A/D conversion on group A is completed.
When A/D conversion for group B is in progress	Input of trigger for group A	A/D conversion in progress for group B discontinued and conversion for group A starts.	• A/D conversion in progress for group B is discontinued and conversion for group A starts. • A/D conversion for group B starts after conversion for group A is completed.
	Input of trigger for group B	Trigger input is ineffective.	Trigger input is ineffective.
	Input of trigger for group C	Trigger input is ineffective.	A/D conversion is performed on group C after A/D conversion on group B is completed.
When A/D conversion for group C is in progress	Input of trigger for group A	A/D conversion in progress for group C discontinued and conversion for group A starts.	• A/D conversion in progress for group C is discontinued and conversion for group A starts. • A/D conversion for group C starts after conversion for group A is completed.
	Input of trigger for group B	A/D conversion in progress for group C is discontinued and conversion for group B starts.	• A/D conversion in progress for group C is discontinued and conversion for group B starts. • A/D conversion for group C starts after conversion for group B is completed.
	Input of trigger for group C	Trigger input is ineffective.	Trigger input is ineffective.

When using group priority operation mode, refer to the following tables to select the desirable operating mode and set the registers.

Table 56.16 Group Priority Operation Setting and Operating Mode for Two Groups (ADGSPCR.PGS = 1, ADGCTRGR.GRCE = 0)

ADGSPCR			Operation
GBRSCN	LGRRS	GBRP	
0	x	0	Group priority operation for two groups (groups A and B) • When a group A trigger is input, group B scan is completed (not restarted)
1	0	0	Group priority operation for two groups (groups A and B) • After group B scan is discontinued, group B scan is restarted from the head of the channel selected with the ADANSB0 and ADANSB1 registers after group A scan is completed.
1	1	0	Group priority operation for two groups (groups A and B) • After group B scan is discontinued, group B scan is restarted from the channel on which scan was discontinued*1, among the channels selected with the ADANSB0 and ADANSB1 registers after group A scan is completed.
x	0	1	Group priority operation for two groups (groups A and B) • Single scan for group B is started continuously without start trigger input. After group B scan is discontinued, single scan is restarted from the head of the channel selected with the ADANSB0 and ADANSB1 registers after scan for group A is completed.
1	1	1	Group priority operation for two groups (groups A and B) • Single scan for group B is started continuously without start trigger input. After group B scan was discontinued, single scan is restarted from the channel on which scan was discontinued*1, among the channels selected with the ADANSB0 and ADANSB1 registers after group A scan is completed.

x = Don't care

Note 1. When self-diagnosis is used (ADCER.DIAGM = 1), A/D conversion of the discontinued channel is started after self-diagnosis.

Table 56.17 Group Priority Operation Setting and Operating Mode for Three Groups (ADGSPCR.PGS = 1, ADGCTRGR.GRCE = 1)

ADGSPCR			
GBRSCN	**LGRRS**	**GBRP**	**Operation**
0	x	0	Group priority operation for three groups (groups A, B, and C) • When a group A trigger is input, group B scan is completed (not restarted) • When a trigger for group A or B is input, group C scan is not completed (not restarted).
0	x	1	Group priority operation for three groups (groups A, B, and C) • When a group A trigger is input, group B scan is completed (not restarted) • Single scan for group C is started continuously without start trigger input. After group C scan is discontinued, scan is restarted from the head of the channel selected with the ADANSC0 and ADANSC1 registers after scan for groups A and B is completed.
1	0	0	Group priority operation for three groups (groups A, B, and C) • After group B scan is discontinued, scan is restarted from the head of the channel selected with the ADANSB0 and ADANSB1 registers after group A scan is completed. • After group C scan is discontinued, scan is restarted from the head of the channel selected with the ADANSC0 and ADANSC1 registers after scan for groups A and B is completed.
1	1	0	Group priority operation for three groups (groups A, B, and C) • After group B scan is discontinued, scan is restarted from the head of the channel selected with the ADANSB0 and ADANSB1 registers after group A scan is completed. • After group C scan is discontinued, scan is restarted from the channel on which scan was discontinued*1, among the channels selected with the ADANSC0 and ADANSC1 registers after scan for groups A and B is completed.
1	0	1	Group priority operation for three groups (groups A, B, and C) • After group B scan is discontinued, scan is restarted from the head of the channel selected with the ADANSB0 and ADANSB1 registers after group A scan is completed. • Single scan for group C is started continuously without start trigger input. After group C scan is discontinued, single scan is restarted from the head of the channel selected with the ADANSC0 and ADANSC1 registers after scan for groups A and B is completed.
1	1	1	Group priority operation for three groups (groups A, B, and C) • After scanning for group B is discontinued, once scanning for group A completes, scanning is resumed starting from the channel on which scanning was discontinued*1, among those channels selected with the ADANSB0 and ADANSB1 registers. • Single scan for group C is started continuously without start trigger input. After group C scan is discontinued, single scan is restarted from the channel on which scan was discontinued*1, among the channels selected with the ADANSC0 and ADANSC1 registers after scan for groups A and B is completed.

x = Don't care

Note 1. When self-diagnosis is used (ADCER.DIAGM = 1), A/D conversion of the discontinued channel is started after self-diagnosis.

(1) Group Priority Operation for Two Groups (ADGSPCR.PGS = 1, ADGCTRGR.GRCE = 0)

The following examples 1 to 5 show group priority operation in group scan mode (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 0) when channel 0 is selected for group A and channels 1 to 3 are selected for group B.

Operation example 1: Group A trigger input during group B scan, with rescan setting

(1) When a group B trigger input sets the ADCSR.ADST bit to 1 (A/D conversion start), scan for the ANn channels selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n.

(2) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).

(3) If a group A trigger is input during scan for group B, group B scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group A selected in the ADANSA0 and ADANSA1 registers, starts from the channel with the smallest number n.
If A/D conversion is not completed when scan was discontinued, the result is not stored in the corresponding A/D data register (ADDRy).

(4) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).

(5) A scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).

(6) If the ADGSPCR.GBRSCN bit is 1 (scan for the group is restarted after having been discontinued due to group priority operation), scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, restarts from the channel with the smallest number n while the ADCSR.ADST bit remains 1.

(7) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).

(8) A group B scan end interrupt request is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan completion enabled).

(9) The ADCSR.ADST bit is automatically cleared to 0 when scan of all selected channels is completed, and the 12-bit A/D converter enters a wait state.

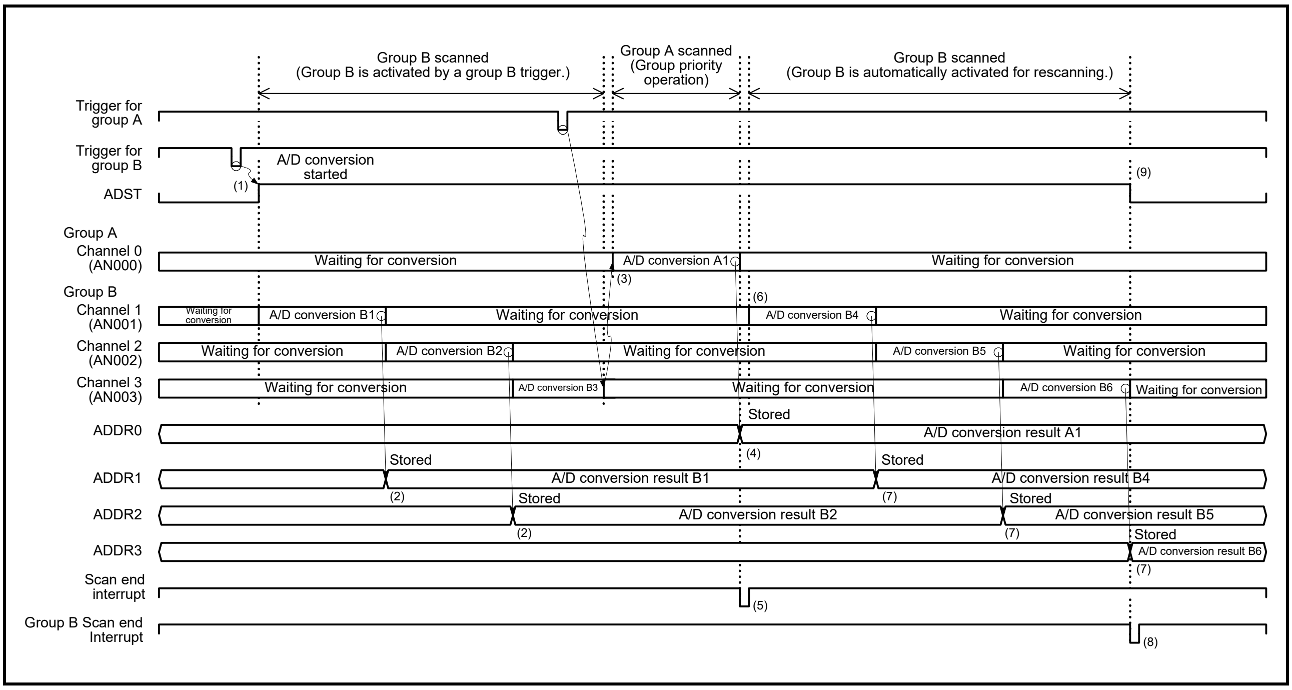


Figure 56.25 Example 1 of Group Priority Operation: Group A Trigger Input during Group B Scan, With Rescan Setting (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 0)

Operation example 2: Group A trigger input during group B rescan, with rescan setting

Figure 56.26 shows an example when a group A trigger is input during rescan operation on group B.

If a group A trigger is input, scan for group A starts even while rescan operation is in progress. Scan for group B starts after scan for group A is completed.

Operations of the ADCSR.ADST bit, storing to the A/D conversion result in the A/D data register (ADDRy), and interrupt requests are the same as example 1.

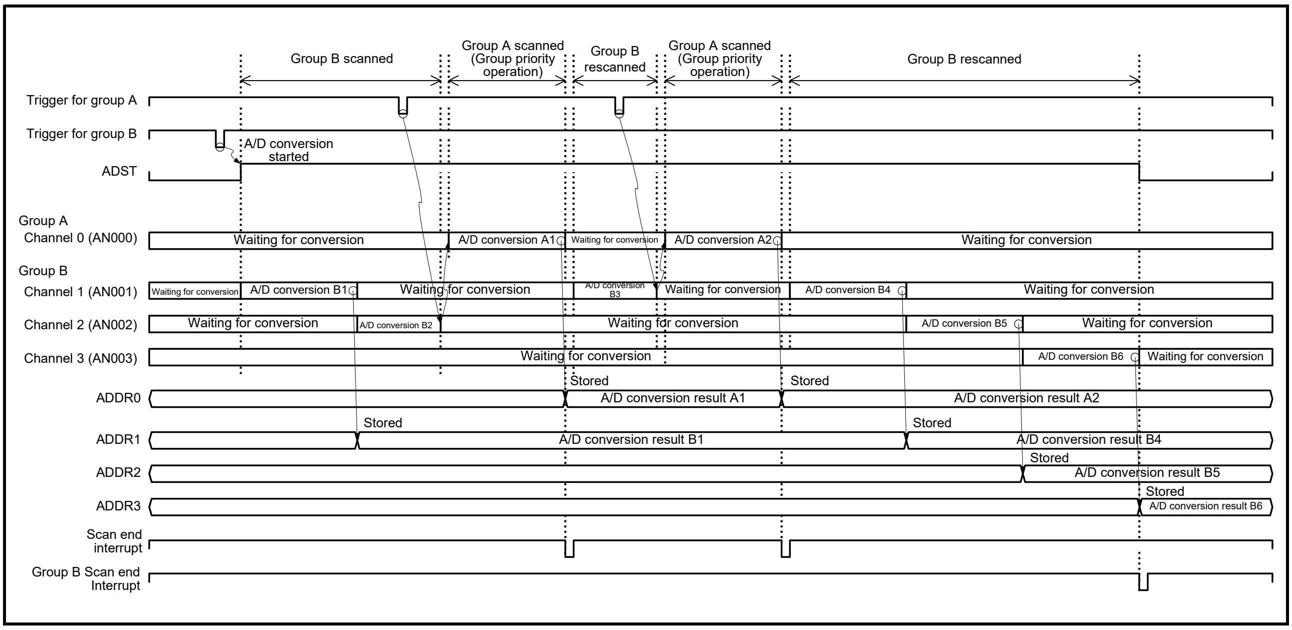


Figure 56.26 Example 2 of Group Priority Operation: Group A Trigger Input during Group B Rescan, With Rescan Setting (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 0)

Operation example 3: Group B trigger input during group A scan, with rescan setting

The following describes an example when a group B trigger is input during scan operation on group A when the ADGSPCR.GBRSCN bit is 1 (scan for the group is restarted after having been discontinued due to group priority operation).

If the ADGSPCR.GBRSCN bit is 0, all group B triggers that are input during scan operation on group A are disabled.

(1) When a group A trigger input sets the ADCSR.ADST bit to 1 (A/D conversion start), scan for the ANn channels of group A selected in the ADANSA0 and ADANSA1 registers, starts from the channel with the smallest number n.

(2) If a group B trigger is input during scan for group A, scan for group B can be started.

(3) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).

(4) After scan for group A is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).

(5) After scan for group A is completed, scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n while the ADCSR.ADST bit remains 1. When a group A trigger is input during scan for group B, group A scan starts as in example 1, and group B scan starts after group A scan is completed.

(6) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).

(7) After scan for group B is completed, a group B scan end interrupt request is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan completion enabled).

(8) The ADCSR.ADST bit is automatically cleared to 0 when scan of all selected channels is completed, and the 12-bit A/D converter enters a wait state.

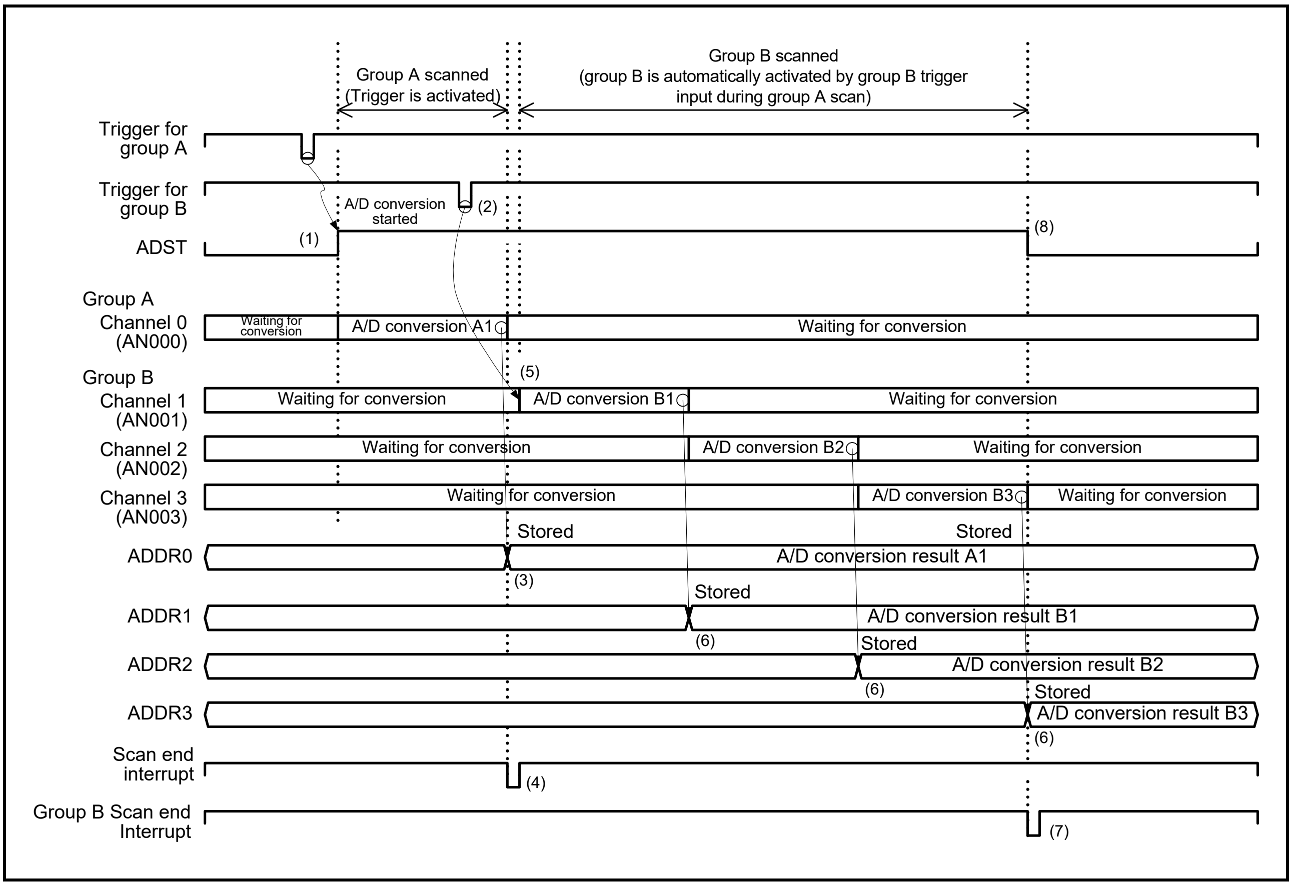


Figure 56.27 Example 3 of Group Priority Operation: Group B Trigger Input during Group A Scan, With Rescan Setting (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 0)

Operation example 4 shows group priority operation in group scan mode (ADGSPCR.GBRSCN = 0, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 0) when channel 0 is selected for group A and channels 1 to 3 are selected for group B.

Operation example 4: Group A trigger input during group B scan, without rescan setting

(1) When a group B trigger input sets the ADCSR.ADST bit to 1 (A/D conversion start), scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n.

(2) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).

(3) If a group A trigger is input during scan for group B, group B scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group A selected in the ADANSA0 and ADANSA1 registers, starts from the channel with the smallest number n.
If A/D conversion is not completed when scan was discontinued, the result is not stored in the corresponding A/D data register (ADDRy).

(4) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).

(5) After scan for group A is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).

(6) The ADST bit is automatically cleared to 0 when scan for group A is completed, and the 12-bit A/D converter enters a wait state. Scan for group B is not started until the next group B trigger is input.

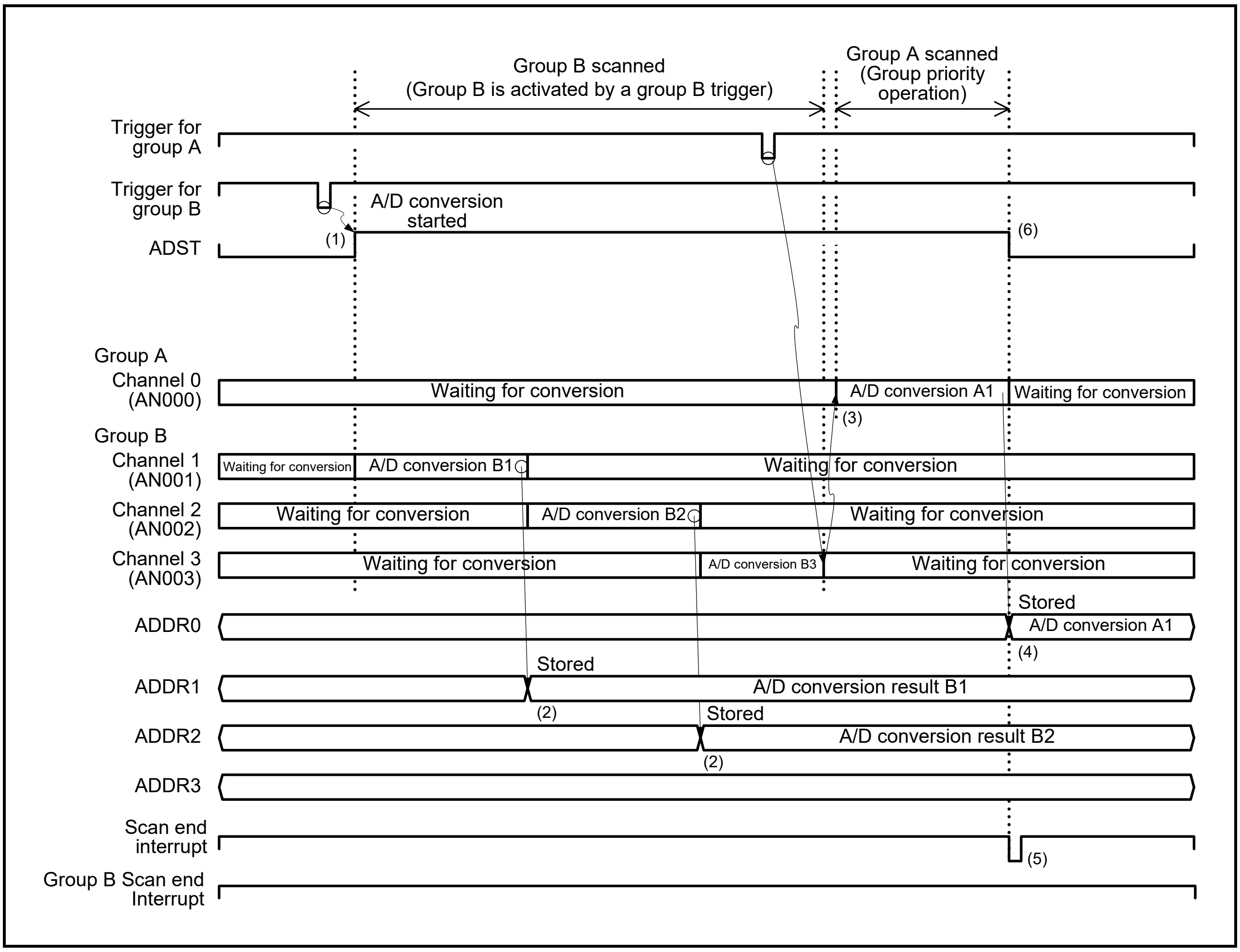


Figure 56.28 Example 4 of Group Priority Operation: Group A Trigger Input during Group B Scan, Without Rescan Setting (ADGSPCR.GBRSCN = 0, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 0)

Operation example 5 shows group priority operation in group scan mode (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 1, ADGSPCR.LGRRS = 0) when channel 0 is selected for group A and channels 1 and 2 are selected for group B.
When the ADGCTRGR.GRCE bit is set to 1, single scan mode is continuously operated on group C and scan for group B is started by trigger input.

Operation example 5: Continuous single scan operation on group B

(1) When setting ADGSPCR.GBRP to 1 sets the ADCSR.ADST bit to 1 (A/D conversion start), scan for the ANn channels selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n.
(2) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).
(3) If a group A trigger is input during scan for group B, group B scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group A selected in the ADANSA0 and ADANSA1 registers, starts from the channel with the smallest number n. If A/D conversion is not completed when scan was discontinued, the result is not stored in the corresponding A/D data register (ADDRy).
(4) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).
(5) After scan for group A is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
(6) If the ADGSPCR.GBRP bit is set to 1 (single scan is continuously operated), scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, restarts from the channel with the smallest number n while the ADCSR.ADST bit remains 1.
(7) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy).
(8) A group B scan end interrupt request is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan completion enabled).
(9) If the ADGSPCR.GBRP bit is set to 1 (single scan is continuously operated), scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, restarts from the channel with the smallest number n while the ADCSR.ADST bit remains 1.

To continuously operate single scan for group B, disable trigger input for group B.
Steps 6 to 9 are repeated as long as the ADGSPCR.GBRP bit remains 1. Clearing of the ADCSR.ADST bit to 0 is prohibited while the ADGSPCR.GBRP bit is set to 1. To forcibly stop scanning when ADGSPCR.GBRP = 1, follow the procedures for clear operation by software through the ADCSR.ADST bit shown in section 56.6.2, Notes on Stopping A/D Conversion.

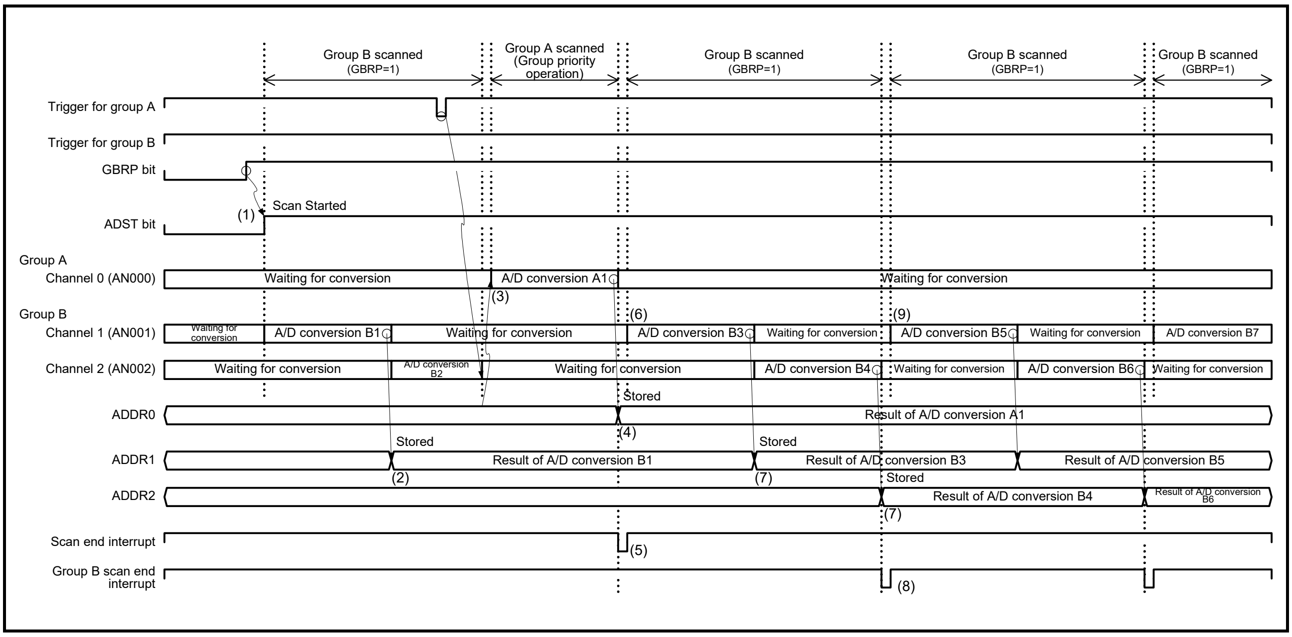


Figure 56.29 Example 5 for Group Priority Operation: Continuous Single Scan Operation on Group B (ADGSPCR.GBRP = 1, ADGSPCR.LGRRS = 1, ADGCTRGR.GRCE = 0)

(2) Group Priority Operation for Three Groups (ADGSPCR.PGS = 1, ADGCTRGR.GRCE = 1)

The following examples 1 to 5 show group priority operation in group scan mode (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 1) when channel 0 is selected for group A, channels 1 and 2 are selected for group B, and channels 3 and 4 are selected for group C.
The priority groups mean groups A and B for group C and group A for group B.

Operation example 1: Priority group trigger input during low-priority group scan, with rescan setting

(1) When a group C trigger input sets the ADCSR.ADST bit to 1 (A/D conversion start), scan for the ANn channels selected in the ADANSC0 and ADANSC1 registers, starts from the channel with the smallest number n.
(2) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(3) If a group B trigger is input during scan for group C, group C scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n. On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(4) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(5) If a group A trigger is input during scan for group B, group B scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group A selected in the ADANSA0 and ADANSA1 registers, starts from the channel with the smallest number n. If A/D conversion is not completed when scan was discontinued, the result is not stored in the corresponding A/D data register (ADDRy).
(6) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(7) A scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
(8) If the ADGSPCR.GBRSCN bit is 1 (scan for the group is restarted after having been discontinued due to group priority operation), scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, restarts from the channel with the smallest number n while the ADCSR.ADST bit remains 1. If the ADGSPCR.LGRRS bit is set to 1 at this time, scan for group B starts from the channel on which A/D conversion was discontinued.
(9) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(10) A group B scan end interrupt request is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan completion enabled).
(11) If the ADGSPCR.GBRSCN bit is 1, scan for the ANn channels of group C selected in the ADANSC0 and ADANSC1 registers, restarts from the channel with the smallest number n while the ADCSR.ADST bit remains 1. If the ADGSPCR.LGRRS bit is set to 1 at this time, scan for group C starts from the channel on which A/D conversion was discontinued.
(12) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(13) A group C scan end interrupt is generated if the ADGCTRGR.GCADIE bit is 1 (interrupt generation upon group C scan completion enabled).
(14) The ADST bit is automatically cleared to 0 when scan for group A is completed, and the 12-bit A/D converter enters a wait state.

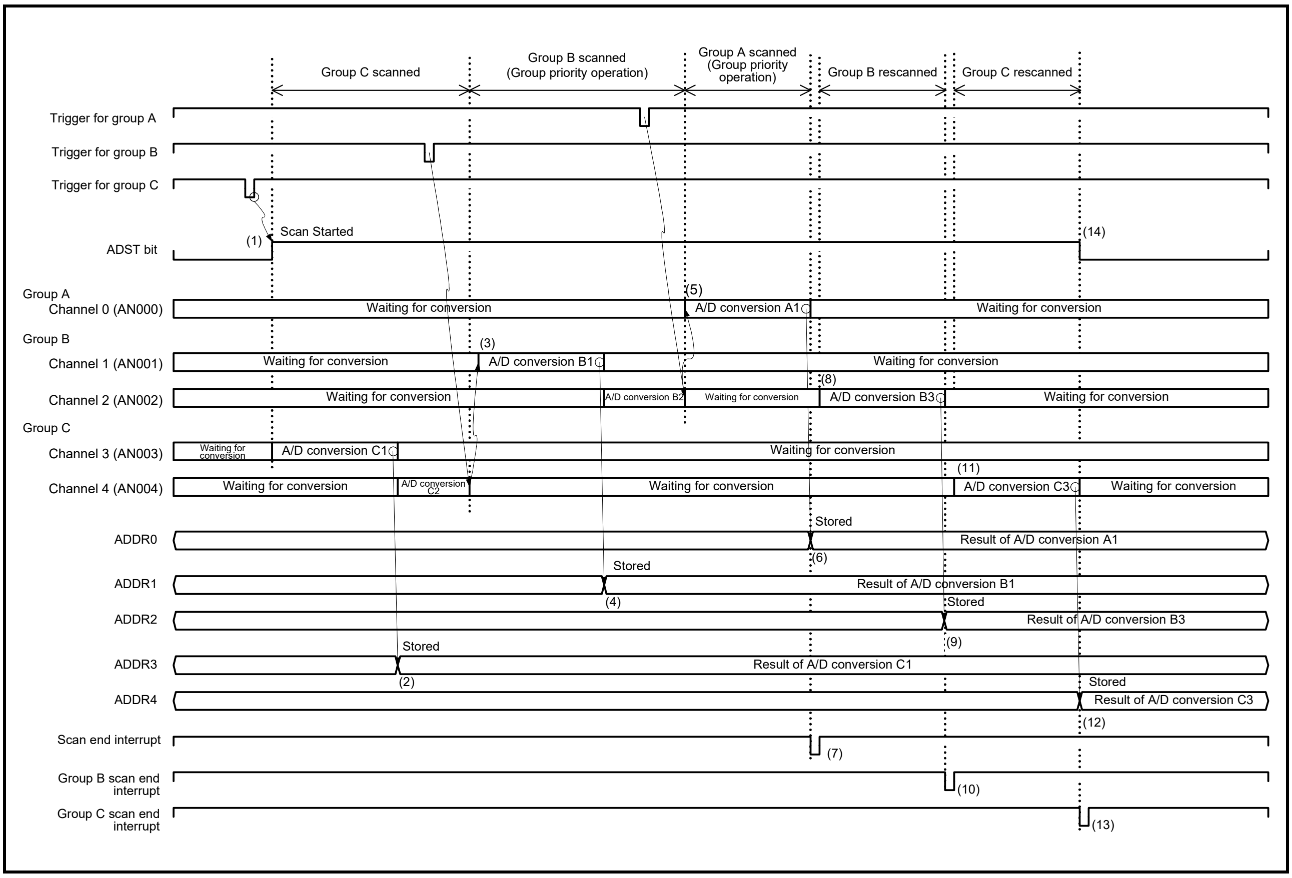


Figure 56.30 Example 1 of Group Priority Operation: Priority Group Trigger Input during Low-Priority Group Scan, With Rescan Setting (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 1)

Operation example 2: Priority group trigger input during low-priority group rescan, with rescan setting

Figure 56.31 shows an example when a group A trigger is input during rescan operation on group B.

If a trigger for the priority groups (groups A and B for group C and group A for group B) is input, scan for the priority group starts even while rescan operation on the low-priority group is in progress. After scan for the priority group is completed, scan for the low-priority group is restarted after having been discontinued.

Operations of the ADCSR.ADST bit, storing to the A/D conversion result in the A/D data register (ADDRy), and interrupt requests are the same as example 1.

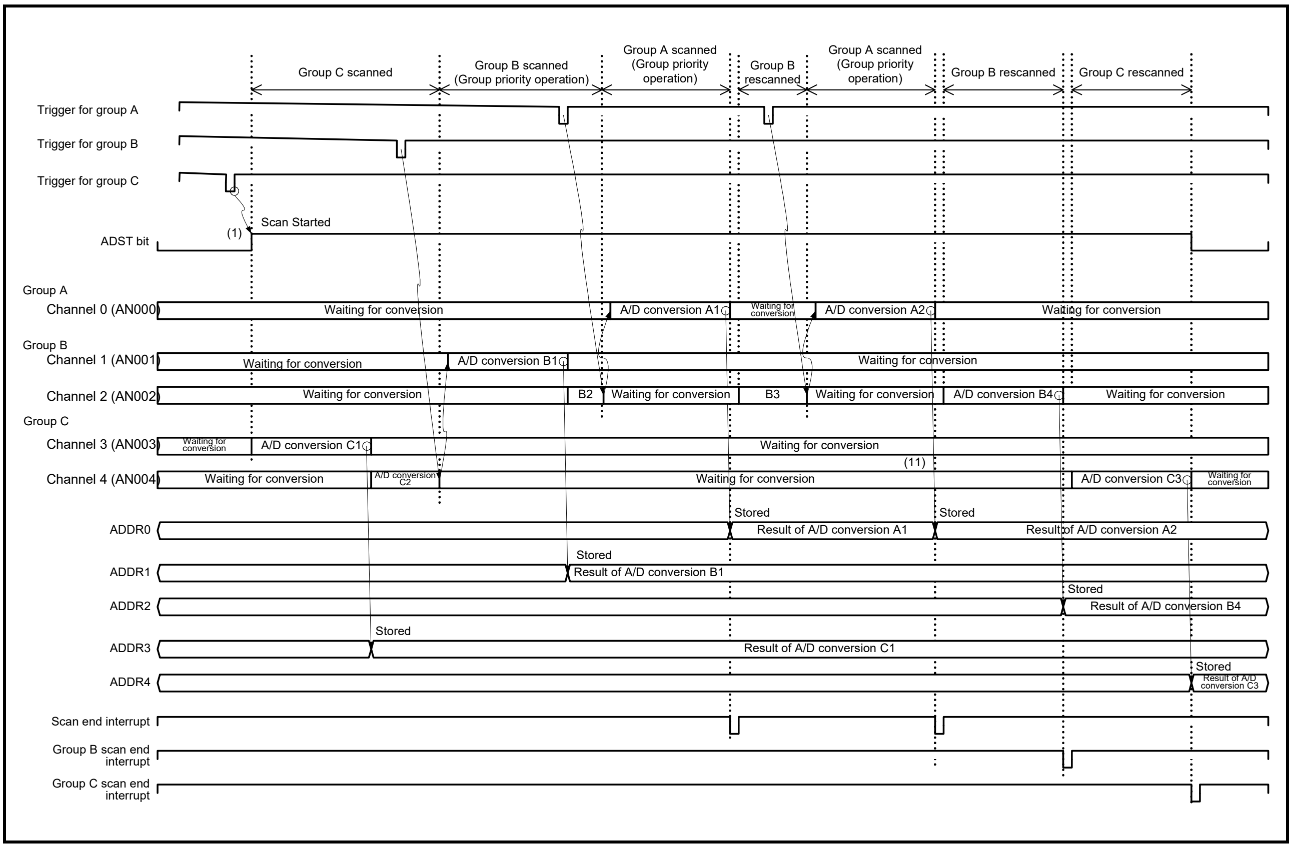


Figure 56.31 Example 2 of Group Priority Operation: Priority Group Trigger Input during Low-Priority Group Rescan, With Rescan Setting (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 1)

Operation example 3: Low-priority group trigger input during priority group scan, with rescan setting

The following describes an example when a trigger for the low-priority group is input during scan operation on the priority group when the ADGSPCR.GBRSCN bit is 1 (scan for the group is restarted after having been discontinued due to group priority operation). If the ADGSPCR.GBRSCN bit is 0, all triggers for the low-priority group that are input during scan operation on the priority group are disabled.

(1) When a group A trigger input sets the ADCSR.ADST bit to 1 (A/D conversion start), scan for the ANn channels of group A selected in the ADANSA0 and ADANSA1 registers, starts from the channel with the smallest number n.
(2) If a group B trigger is input during scan for group A, scan for group B can be started.
(3) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(4) After scan for group A is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
(5) After scan for group A is completed, scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n while the ADCSR.ADST bit remains 1. If the ADGSPCR.LGRRS bit is set to 1 at this time, scan for group B starts from the channel on which A/D conversion was discontinued.
When a group A trigger is input during scan for group B, group A scan starts as in example 1, and group B scan starts after group A scan is completed.
(6) If a group C trigger is input during scan for group B, scan for group C can be started.
(7) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(8) After scan for group B is completed, a group B scan end interrupt request is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan completion enabled).
(9) After scan for group B is completed, scan for the ANn channels of group C selected in the ADANSC0 and ADANSC1 registers, starts from the channel with the smallest number n while the ADCSR.ADST bit remains 1. If the ADGSPCR.LGRRS bit is set to 1 at this time, scan for group C starts from the channel on which A/D conversion was discontinued.
When a group A or B trigger is input during scan for group C, group A or B scan starts as in example 1, and group C scan starts after group A or B scan is completed.
(10) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(11) After scan for group C is completed, a group C scan end interrupt request is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group C scan completion enabled).
(12) The ADCSR.ADST bit is automatically cleared to 0 when scan of all selected channels is completed, and the 12-bit A/D converter enters a wait state.

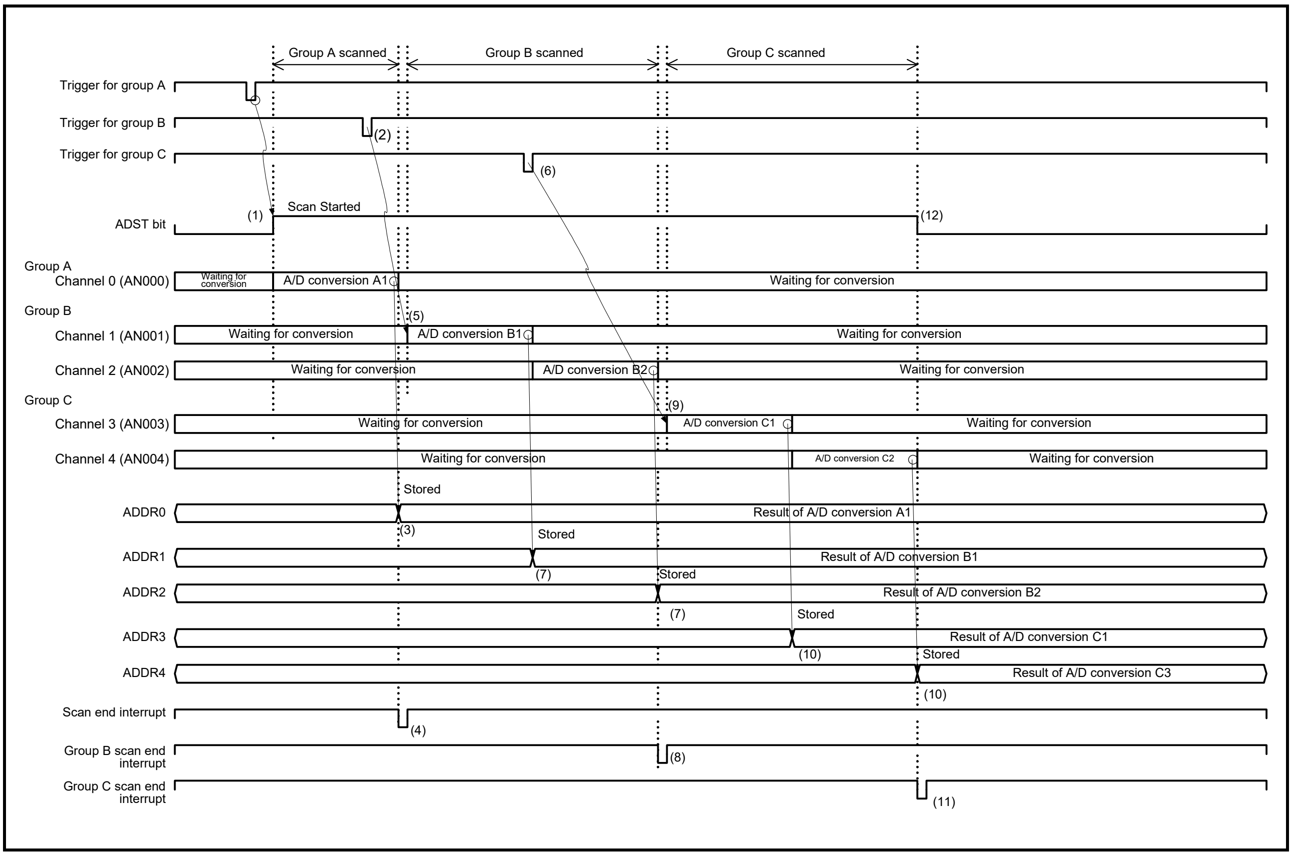


Figure 56.32 Example 3 of Group Priority Operation: Low-Priority Group Trigger Input during Priority Group Scan, With Rescan Setting (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 1, ADGCTRGR.GRCE = 1)

Operation example 4 shows group priority operation in group scan mode (ADGSPCR.GBRSCN = 0, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 0) when channel 0 is selected for group A, channels 1 and 2 are selected for group B, and channels 3 and 4 are selected for group C.

Operation example 4: Priority group trigger input during low-priority group scan, without rescan setting

(1) When a group C trigger input sets the ADCSR.ADST bit to 1 (A/D conversion start), scan for the ANn channels selected in the ADANSC0 and ADANSC1 registers, starts from the channel with the smallest number n.

(2) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)

(3) If a group B trigger is input during scan for group C, group C scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n. If A/D conversion is not completed when scan was discontinued, the result is not stored in the corresponding A/D data register (ADDRy).

(4) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)

(5) If a group A trigger is input during scan for group B, group B scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group A selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n. If A/D conversion is not completed when scan was discontinued, the result is not stored in the corresponding A/D data register (ADDRy).

(6) After scan for group A is completed, a scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).

(7) The ADST bit is automatically cleared to 0 when scan for group A is completed, and the 12-bit A/D converter enters a wait state. Scan for groups C and B is not started until the next trigger corresponding to the group is input.

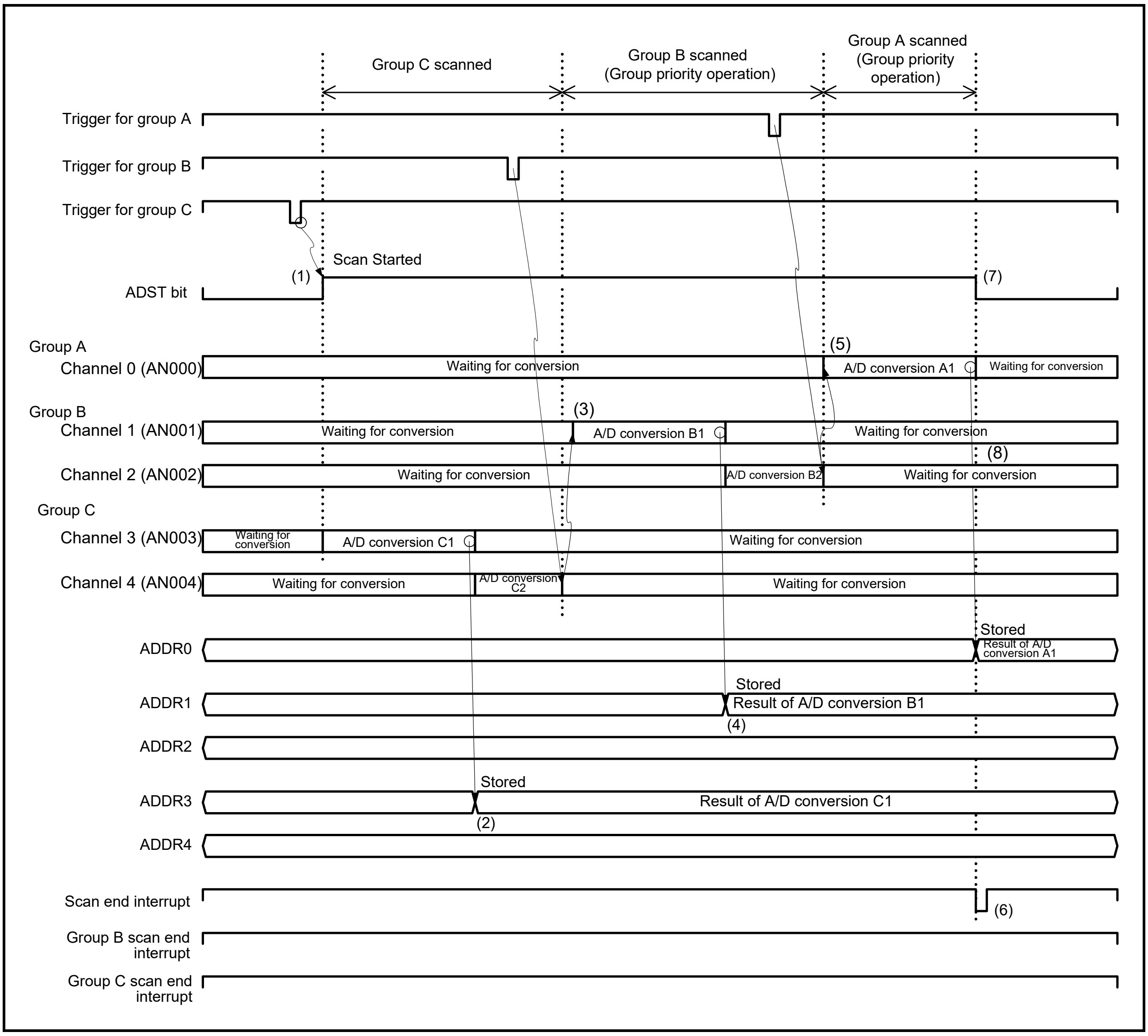


Figure 56.33 Example 4 of Group Priority Operation: Priority Group Trigger Input during Low-Priority Group Scan, Without Rescan Setting (ADGSPCR.GBRSCN = 0, ADGSPCR.GBRP = 0, ADGSPCR.LGRRS = 1)

Operation example 5 shows group priority operation in group scan mode (ADGSPCR.GBRSCN = 1, ADGSPCR.GBRP = 1, ADGSPCR.LGRRS = 1) when channel 0 is selected for group A, channel 1 is selected for group B, and channels 2 and 3 are selected for group C.
When the ADGCTRGR.GRCE bit is set to 1, single scan mode is continuously operated on group B and trigger input for group C is disabled.

Operation example 5: Continuous single scan operation on group C

(1) When setting ADGSPCR.GBRP to 1 sets the ADCSR.ADST bit to 1 (A/D conversion start), scan for the ANn channels selected in the ADANSC0 and ADANSC1 registers, starts from the channel with the smallest number n.
(2) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(3) If a group B trigger is input during scan for group C, group C scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, starts from the channel with the smallest number n. If A/D conversion is not completed when scan was discontinued, the result is not stored in the corresponding A/D data register (ADDRy).
(4) If a group A trigger is input during scan for group B, group B scan is discontinued while the ADCSR.ADST bit remains 1, and scan for the ANn channels of group A selected in the ADANSA0 and ADANSA1 registers, starts from the channel with the smallest number n. If A/D conversion is not completed when scan was discontinued, the result is not stored in the corresponding A/D data register (ADDRy).
(5) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(6) A scan end interrupt request is generated if the ADCSR.ADIE bit is 1 (interrupt generation upon scanning completion enabled).
(7) If the ADGSPCR.GBRSCN bit is 1 (scan for the group is restarted after having been discontinued due to group priority operation), scan for the ANn channels of group B selected in the ADANSB0 and ADANSB1 registers, restarts from the channel with the smallest number n while the ADCSR.ADST bit remains 1. If the ADGSPCR.LGRRS bit is set to 1 at this time, scan for group B starts from the channel on which A/D conversion was discontinued.
(8) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(9) A group B scan end interrupt request is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group B scan completion enabled).
(10) If the ADGSPCR.GBRSCN bit is 1 (scan for the group is restarted after having been discontinued due to group priority operation), scan for the ANn channels of group C selected in the ADANSC0 and ADANSC1 registers, restarts from the channel with the smallest number n while the ADCSR.ADST bit remains 1. If the ADGSPCR.LGRRS bit is set to 1 at this time, scan for group C starts from the channel on which A/D conversion was discontinued.
(11) On completion of A/D conversion on a single channel, the result is stored in the corresponding A/D data register (ADDRy)
(12) A group C scan end interrupt request is generated if the ADCSR.GBADIE bit is 1 (interrupt generation upon group C scan completion enabled).
(13) If the ADGSPCR.GBRP bit is set to 1 (single scan is continuously operated), scan for the ANn channels of group C selected in the ADANSC0 and ADANSC1 registers, restarts from the channel with the smallest number n while the ADCSR.ADST bit remains 1.

To continuously operate single scan for group C, disable trigger input for group B.
Steps 13, 11, 12, and then 13 are repeated as long as the ADGSPCR.GBRP bit remains 1.
Clearing of the ADCSR.ADST bit to 0 is prohibited while the ADGSPCR.GBRP bit is set to 1.
To forcibly stop scanning when ADGSPCR.GBRP = 1, follow the procedures for clear operation by software through the

ADCSR.ADST bit shown in section 56.6.2, Notes on Stopping A/D Conversion.

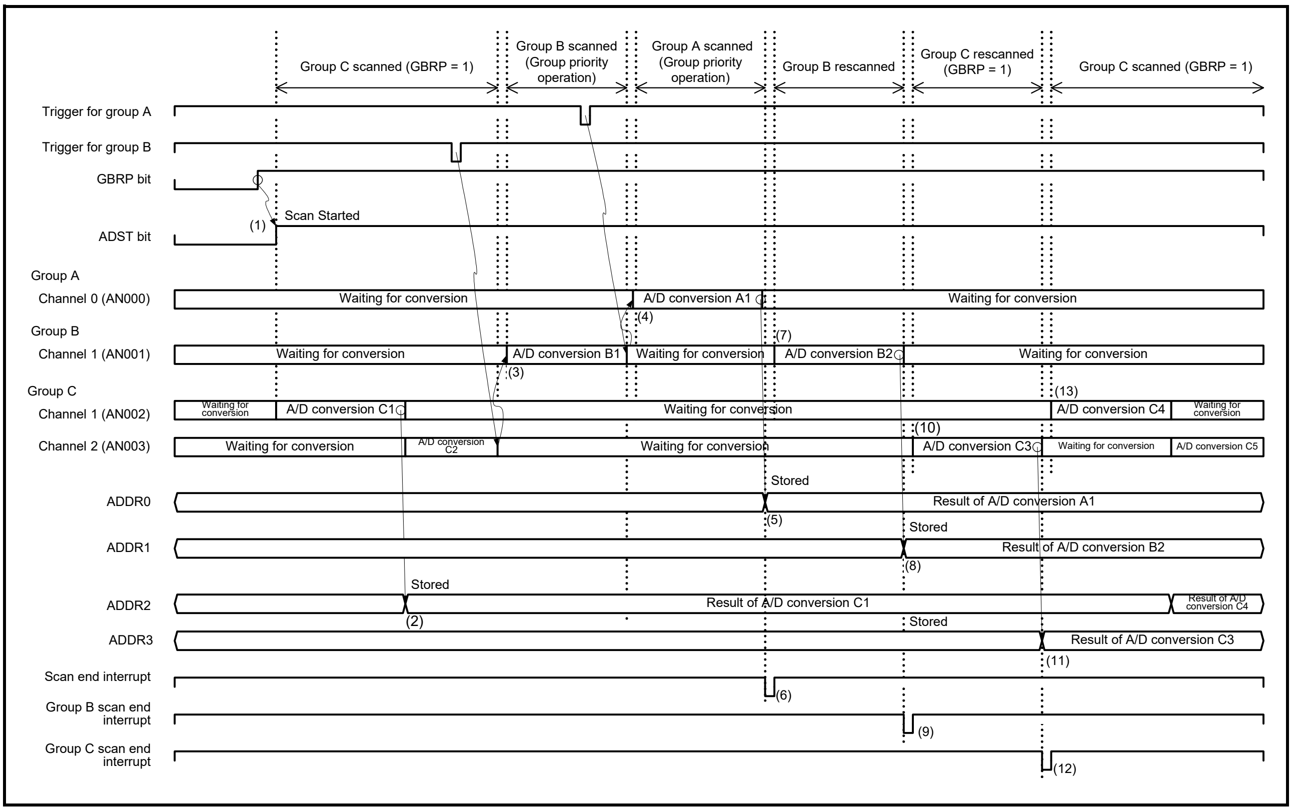


Figure 56.34 Example 5 for Group Priority Operation: Continuous Single Scan Operation on Group C (ADGSPCR.GBRP = 1, ADGSPCR.LGRRS = 1)

56.3.5 Extended Analog Input

Extended Analog Input (ANEX1) is used when an external operational amplifier is connected to the MCU and multiple analog values are A/D converted. When the extended analog input is selected, only AN100 to AN107 are selectable. Do not select AN108 to AN120, temperature sensor output, or internal reference voltage. Also, when the extended analog input is selected, self-diagnosis or disconnection detection assist function cannot be used.

56.3.5.1 Usage of ANEX1

To perform A/D conversion of multiple analog values through an operational amplifier, analog values are input by using analog input channels of the MCU (AN100 to AN107), the time-divided analog values are retrieved, and an operational amplifier is connected between ANEX0 and ANEX1.
When ANEX1 is connected, set the ADEXICR.EXSEL[1:0] bits to 01b (ANEX1 selected) and ADEXICR.EXOEN bit to 1 (ANEX0: output enabled) and then select single scan mode or sequence scan mode.
Do not set in group scan mode. Figure 56.35 shows an example of structure of the extended analog input when ANEX1 is in use, and Figure 56.36 shows operation when three channels (AN100, AN101, AN102) are selected and single scan mode is selected.

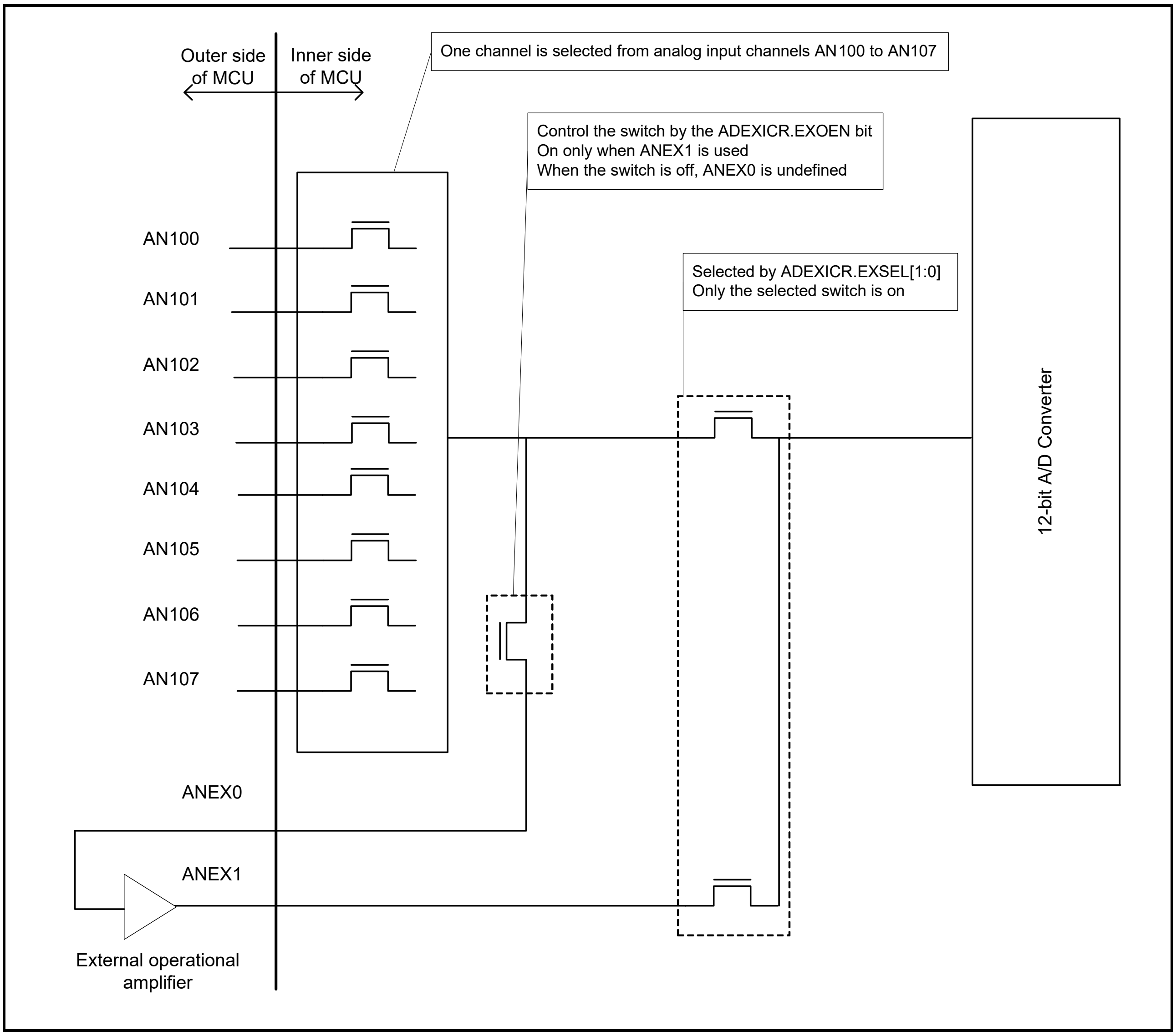


Figure 56.35 Example of Configuration of Extended Analog Input When ANEX1 is Used

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, A/D conversion is performed for the selected channels starting from the channel with the smallest number.
(2) Each time A/D conversion of a single channels is completed, the A/D converted value is stored in the corresponding A/D data register (ADDRy).
(3) After completion of A/D conversion of all selected channels, if the ADCSR.ADIE bit is set to 1 (when interrupt generation after scanning is enabled), a scan end interrupt request is generated.
(4) The ADCSR.ADST bit holds 1 during A/D conversion and is automatically cleared upon completion of A/D conversion of all selected channels, and the 12-bit A/D converter enters into a wait state.

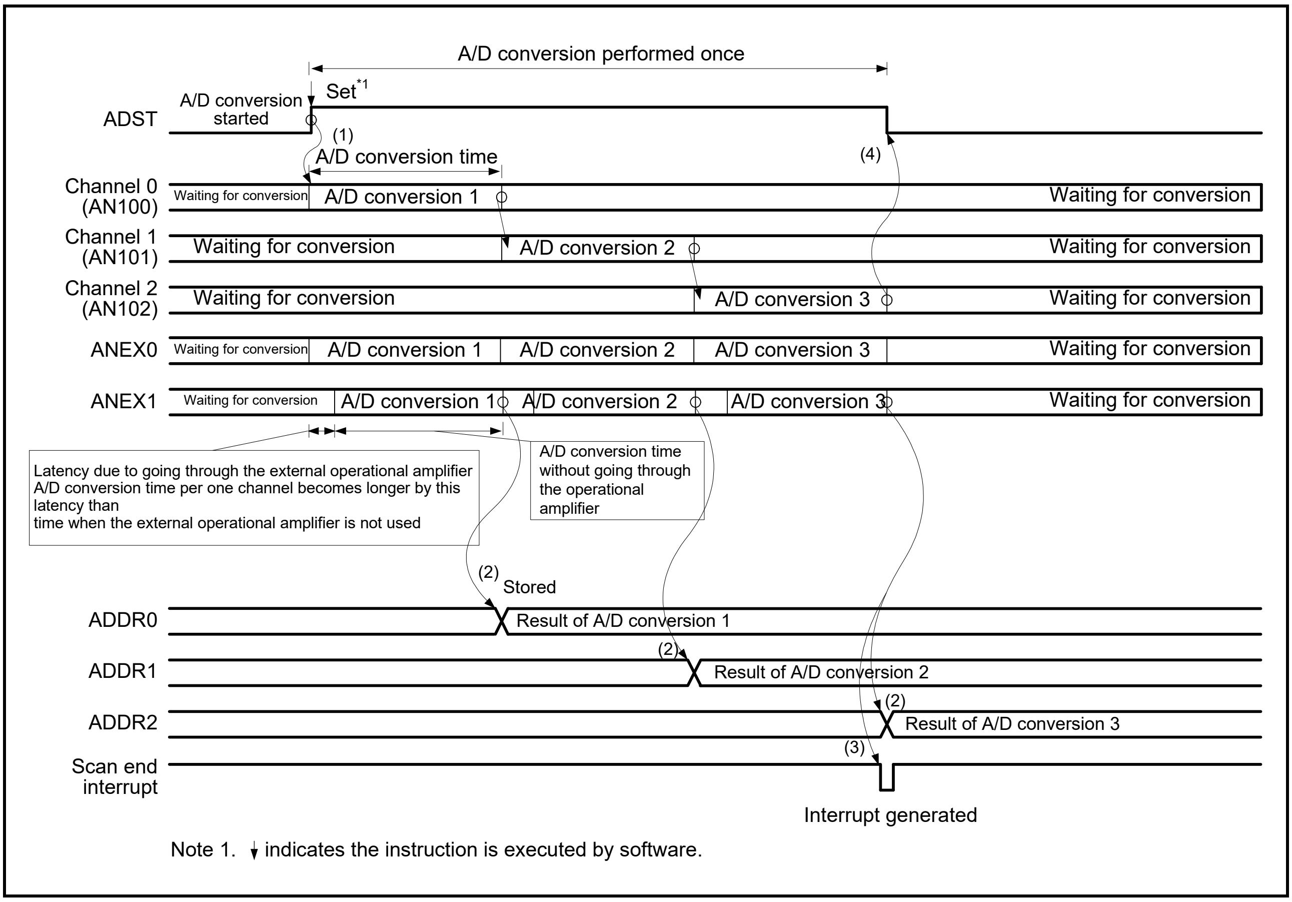


Figure 56.36 Example of Operation of ANEX1 Input (Single Scan Mode)

When the extended analog input is selected, each period for A/D conversion will be relatively longer than that of the case when analog input channels are directly A/D converted due to latency of the operational amplifier.

56.3.6 Comparison Function (Windows A/B)

56.3.6.1 Comparison Function Windows A/B

The comparison function compares the reference values set in registers ADCMPDR0, ADCMPDR1, ADWINLLB, and ADWINULB with the A/D conversion results. When this function is used, self-diagnosis or double trigger mode cannot be used. A window comparison function can also be used (when ADCMPCR.WCMPE = 1), enabling comparison of two values. Two sets of voltage level ranges can be set in the window comparison function, one for window A and one for window B.

Operation when window comparison is enabled (ADCMPCR.WCMPE = 1) in combination with continuous scan mode is described below.

(1) When the ADCSR.ADST bit is set to 1 (A/D conversion start) by software, or synchronous or asynchronous trigger input, A/D conversion is performed for selected channels, temperature sensor output, and internal reference voltage in this order.

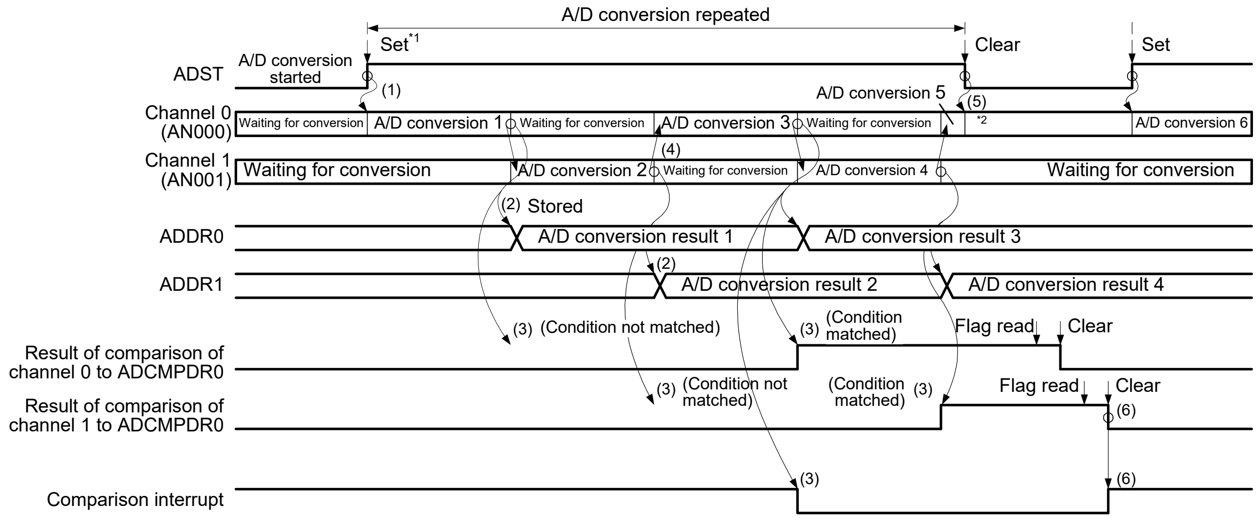
(2) Each time A/D conversion of a single channels is completed, the A/D converted value is stored in the corresponding A/D data register (ADDRy, ADTSDR, or ADOCDR). When ADCMPCR.CMPAE = 1 and window A is selected in the ADCMPANSRy and ADCMPANSER registers, the A/D converted value is compared with the setting values of the ADCMPDR0 and ADCMPDR1 registers.
When ADCMPCR.CMPBE = 1 and window B is selected in the ADCMPBNSR register, the A/D converted value is compare with the setting values of the ADWINULB and ADWINLLB registers.

(3) Upon a comparison match with the conditions specified in the ADCMPLR0, ADCMPLR1, and ADCMPLER registers, the flags of the comparison window A (ADCMPSR0.CMPSTCHA0n, ADCMPSR1.CMPSTCHA1n, ADCMPSER.CMPFTS, and ADCMPSER.CMPFOC) are set to 1 for window A.
In this case, if the ADCMPCR.CMPAIE bit is set to 1, an interrupt request S12CMPAI is generated.
Similarly, for window B, upon a match with the conditions set in the ADCMPBNSR.CMPLB register, the comparison window B flag (ADCMPBSR.CMPSTB) is set to 1. In this case, if the ADCMPCR.CMPBIE bit is set to 1, an interrupt request S12CMPBI is generated.

(4) When all selected A/D conversion and comparison are completed, scanning is performed again.

(5) Set the ADCSR.ADST to 0 (A/D conversion stop) and execute the processing for the channels in which the comparison flag is 1.

(6) After the processing above, clear all comparison flags. When comparison is re-executed, start A/D conversion again.



Note 1. ↓ indicates the instruction is executed by software.
Note 2. Data on conversion is ignored.

Figure 56.37 Example of Operation of Comparison (AN000 and AN001 for Targets for Comparison)

56.3.6.2 Restrictions on Comparison Function

The Comparison function has the following restrictions.

1. Use of self-diagnosis and double trigger mode is prohibited.
   (ADRD and ADDBLDR are not targeted for the comparison function)
2. When temperature sensor or internal reference voltage is selected for window A, operation of window B is prohibited.
3. When temperature sensor or internal reference voltage is selected for window B, operation of window A is prohibited.
4. The same channel cannot be set for window A and window B.
5. Satisfy the condition of [reference value (high side) ≥ reference value (low side)].

56.3.7 Analog Input Sampling Time and Scan Conversion Time

Figure 56.38 shows the scan conversion timing in single scan mode, in which scan conversion is activated by software or a synchronous trigger. Figure 56.39 shows the scan conversion timing in single scan mode, in which scan conversion is activated by an asynchronous trigger. The scan conversion time (t_{SCAN}) includes the start-of-scanning-delay time (t_D), channel-dedicated sample-and-hold circuit processing time (t_{SPLSH})*1, disconnection detection assistance processing time (t_{DIS})*2, self-diagnosis A/D conversion processing time (t_{DIAG})*3, A/D conversion processing time (t_{CONV}), channel-dedicated sample-and-hold circuit end time (t_{SHED})*4, and end-of-scanning-delay time (t_{ED}).
The A/D conversion processing time (t_{CONV}) consists of sampling time (t_{SPL}) and time for conversion by successive approximation (t_{SAM}). The sampling time (t_{SPL}) is used to charge sample-and-hold circuits in the A/D converter. If there is not sufficient sampling time due to the high impedance of an analog input signal source, or if the A/D conversion clock (ADCLK) is slow, sampling time can be adjusted using the ADSSTRn register.
The time for conversion by successive approximation (t_{SAM}) for unit 0 is at 13 states (ADCLK) with 12-bit resolution selected, 11 states (ADCLK) with 10-bit resolution selected, and 9 states (ADCLK) with 8-bit resolution selected; and for unit 1, 15 states (ADCLK) with 12-bit resolution selected, 13 states (ADCLK) with 10-bit resolution selected, and 11 states (ADCLK) with 8-bit resolution selected. Table 56.18 shows the scan conversion time.

The scan conversion time (t_{SCAN}) in single scan mode for which the number of selected channels is n can be determined as follows:

$$t_{SCAN} = t_D + t_{SPLSH} + (t_{DIS} \times n) + t_{DIAG} + (t_{CONV} \times n)^{*5} + t_{ED}$$

The scan conversion time for the first cycle in continuous scan mode is t_{SCAN} for single scan minus t_{ED} plus t_{SHED}.
The scan conversion time for the second and subsequent cycles in continuous scan mode is fixed to $t_{SPLSH} + (t_{DIS} \times n) + t_{DIAG} + t_{DSD} + (t_{CONV} \times n)^{*5} + t_{SHED}$.

Note 1. When no channel-dedicated sample-and-hold circuits are used, $t_{SPLSH} = 0$.
Note 2. When disconnection detection assistance is not selected, $t_{DIS} = 0$.
Note 3. When the self-diagnosis function is not used, $t_{DIAG} = 0$, $t_{DSD} = 0$.
Note 4. When no channel-dedicated sample-and-hold circuits are used, $t_{SHED} = 0$. Here, continuous scan mode is assumed. In single scan mode and group scan mode, t_{SHED} is included in the end-of-scanning-delay time (t_{ED}).
Note 5. $t_{CONV} \times n$ when the sampling time (t_{SPL}) of selected channels is the same, but it is the total of the sampling time of each channel and time for conversion by successive approximation (t_{SAM}).

Table 56.18 Times for Conversion during Scanning (in Numbers of Cycles of ADCLK and PCLK)

Item			Symbol		Type/Conditions: Synchronous Trigger (MTU)	Type/Conditions: Synchronous Trigger (TMR, TPU, ELC)	Type/Conditions: Asynchronous Trigger	Type/Conditions: Software Trigger	Unit
Scan start processing time*1, *2	A/D conversion on group under group priority control.	The low-priority group is to be stopped. (The priority group is activated after low-priority group B is stopped due to an A/D conversion source of the priority group.)	t_D		1 PCLKA + 4 PCLKB + 6 ADCLK	2PCLKB + 6 ADCLK	—	—	Cycle
		The low-priority group is not to be stopped. (Activation by an A/D conversion source of the priority group.)			1 PCLKA + 3 PCLKB + 4 ADCLK	2 PCLKB + 4 ADCLK	—	—	
	A/D conversion when self-diagnosis is enabled	A/D conversion for self-diagnosis is to be started.			1 PCLKA + 3 PCLKB + 6 ADCLK	2 PCLKB + 6 ADCLK	4 PCLKB+ 6 ADCLK	6 ADCLK	
	Other than above				1 PCLKA + 3 PCLKB + 4 ADCLK	2 PCLKB + 4 ADCLK	2 PCLKB + 4 ADCLK	4 ADCLK	
Channel-dedicated sample-and-hold processing time*1	Sampling time		t_{SPLSH}	t_{SH}	When constant sampling is disabled: Setting values of the ADSHCR.SSTSH[7:0] bits (initial value:18h) × ADCLK When constant sampling is enabled: 0				
	Wait time between sampling and A/D conversion			t_W	12 ADCLK				
Disconnection detection assistance processing time			t_{DIS}		The setting of ADNDIS[3:0] (initial value = 00h) × ADCLK				
Self-diagnosis conversion processing time*1	Sampling time		t_{DIAG}	t_{SPL}	The setting of ADSSTR0 (initial value = 0Bh) × ADCLK				
	Time for conversion by successive approximation			t_{SAM}	Unit 0		Unit 1		
					13 ADCLK (12-bit conversion resolution) 11 ADCLK (10-bit conversion resolution) 9 ADCLK (8-bit conversion resolution)		15 ADCLK (12-bit conversion resolution) 13 ADCLK (10-bit conversion resolution) 11 ADCLK (8-bit conversion resolution)		
	Normal A/D conversion is to be started after completion of self-diagnosis conversion.		t_{DED}		2 ADCLK				
	A/D conversion for self-diagnosis is to be started after completion of conversion for continuous scan on the last channel specified.		t_{DSD}		2 ADCLK				
A/D conversion processing time*1	Sampling time		t_{CONV}	t_{SPL}	The setting of ADSSTRn (n = 0 to 15, L, T, O) (initial value = 0Bh) × ADCLK				
	Time for conversion by successive approximation			t_{SAM}	Unit 0		Unit 1		
					13 ADCLK (12-bit conversion resolution) 11 ADCLK (10-bit conversion resolution) 9 ADCLK (8-bit conversion resolution)		15 ADCLK (12-bit conversion resolution) 13 ADCLK (10-bit conversion resolution) 11 ADCLK (8-bit conversion resolution)		
Channel-dedicated sample-and-hold end processing time			t_{SHED}		2 ADCLK				
Scan end processing time*1			t_{ED}		1 PCLKB + 3 ADCLK				

Note 1. For t_D, t_{SPLSH}, t_{DIAG}, t_{CONV}, and t_{ED}, see Figure 56.38 and Figure 56.39.
Note 2. This is the maximum time required from software writing or trigger input to A/D conversion start.

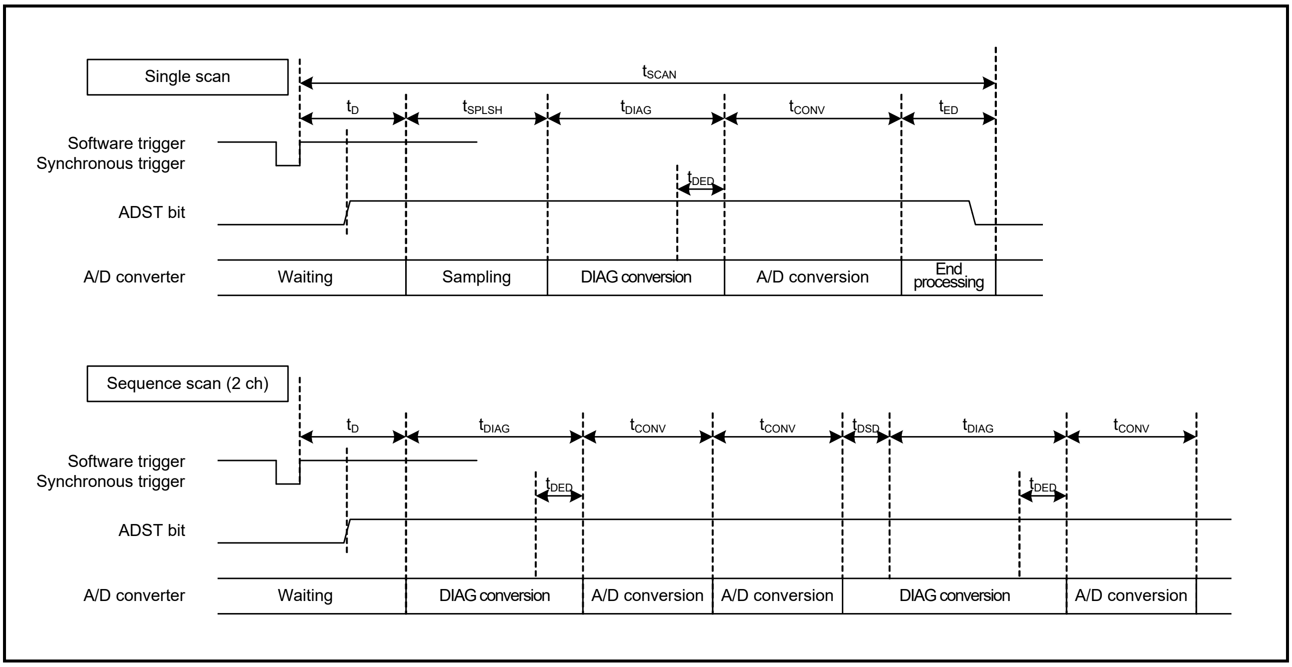


Figure 56.38 Scan Conversion Timing (Activated by Software or Synchronous Trigger)

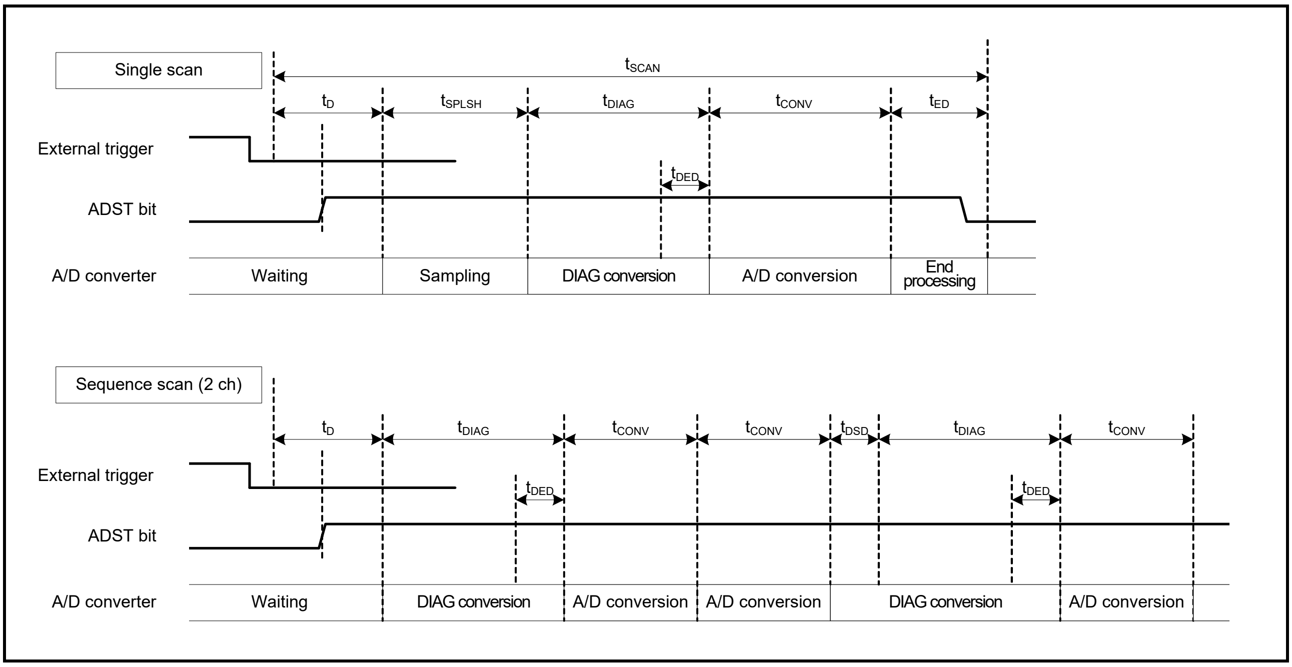


Figure 56.39 Scan Conversion Timing (Activated by Asynchronous Trigger)

56.3.7.1 Timing of Suspension and Starting of Scanning in Operation under Group Priority Control

The timings for suspension and starting of scanning in operation under group priority control that must considered are listed below.

1. The timing for suspending a scan of a group with a lower-priority and the timing for starting a scan of a group with a higher-priority.
2. The time at which scanning by the group with a lower-priority is resumed on completion of scanning by the higher-priority group when the trigger for scanning by the lower-priority group is accepted during scanning by the higher-priority group.
3. The timing for performing sequential single scans by a lower-priority group.

Figure 56.40 shows the timing diagram of each of the above cases.
The times at which scanning by group A is completed and scanning by group C resumes or scanning by group B is completed and scanning by group C resumes are the same as those for group A and group B in Figure 56.40.
The timing for consecutive single scans is the same for group B and group C.

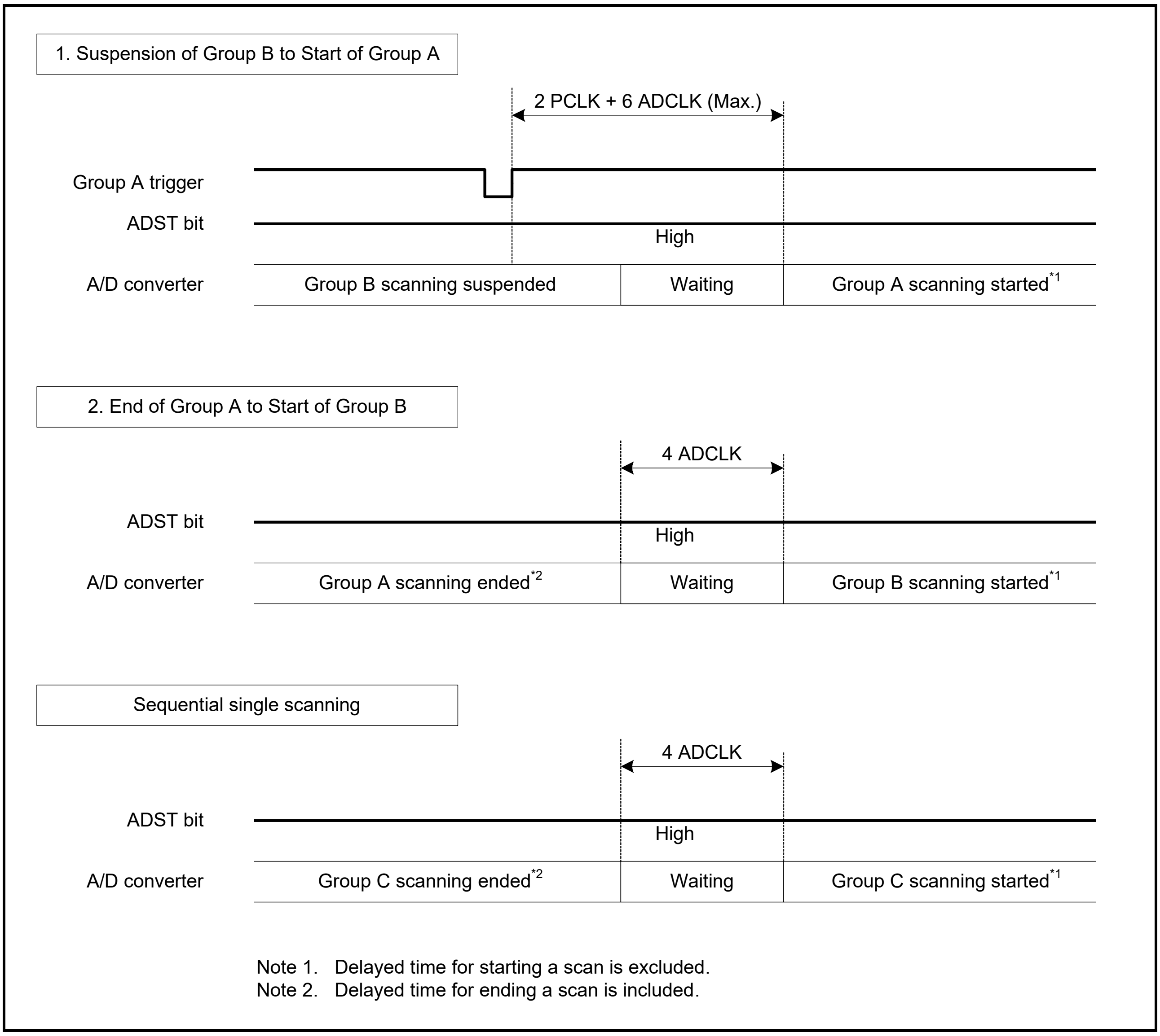


Figure 56.40 Timing of Stop/Start of Scanning in Group Priority Mode

56.3.8 Usage Example of A/D Data Register Automatic Clearing Function

Setting the ADCER.ACE bit to 1 automatically clears the A/D data registers (ADDRy, ADRD, ADTSDR, ADOCDR, ADDBLDR, ADDBLDRA, ADDBLDRB) to 0000h when the A/D data registers (ADDRy, ADRD, ADTSDR, ADOCDR, ADDBLDR, ADDBLDRA, ADDBLDRB) are read by the CPU, DTC, or DMAC.
This function enables detection of update failures of the A/D data registers (ADDRy, ADRD, ADTSDR, ADOCDR, ADDBLDR, ADDBLDRA, ADDBLDRB). The following describes the examples in which the function to automatically clear the ADDRy register is enabled and disabled.
In a case where the ADCER.ACE bit is 0 (automatic clearing disabled), if the A/D conversion result (0222h) is not written to the ADDRy register for some reason, the old data (0111h) will be the ADDRy value. Furthermore, if this ADDRy value is read into a general register using an A/D scan end interrupt, the old data (0111h) can be saved in the general register. When checking whether there is an update failure, it is necessary to frequently save the old data in the RAM or a general register.
In a case where the ADCER.ACE bit is 1 (automatic clearing enabled), when ADDRy = 0111h is read by the CPU, DTC, or DMAC, ADDRy is automatically cleared to 0000h. After that, if the A/D conversion result 0222h cannot be transferred to ADDRy for some reason, the cleared data (0000h) remains as the ADDRy value. If this ADDRy value is read into a general register using an A/D scan end interrupt at this point, 0000h will be saved in the general register. Occurrence of an ADDRy update failure can be determined by simply checking that the read data value is 0000h.

56.3.9 A/D-Converted Value Addition/Average Mode

In A/D-converted value addition mode, the same channel is A/D-converted 2, 3, 4, or 16 consecutive times and the sum of the converted values is stored in the data register. In A/D-converted value average mode, the same channel is A/D-converted two or four consecutive times and the mean of the converted values is stored in the data register. The use of the average of these results can improve the accuracy of A/D conversion, depending on the types of noise components that are present. This function, however, cannot always guarantee an improvement in A/D conversion accuracy.
The A/D-converted value addition/average mode can be used when A/D conversion of the channel select analog input, temperature sensor output (for S12AD1 only), or internal reference voltage (for S12AD1 only) is selected.

56.3.10 Disconnection Detection Assist Function

This converter incorporates the function to fix the charge for sampling capacitance to the specified state before start of A/D conversion. This function enables disconnection detection in wiring of analog inputs.
Figure 56.41 illustrates the A/D conversion operation when the disconnection detection assist function is used. Figure 56.42 shows an example of disconnection detection when precharge is selected. Figure 56.43 shows an example of disconnection detection when discharge is selected.

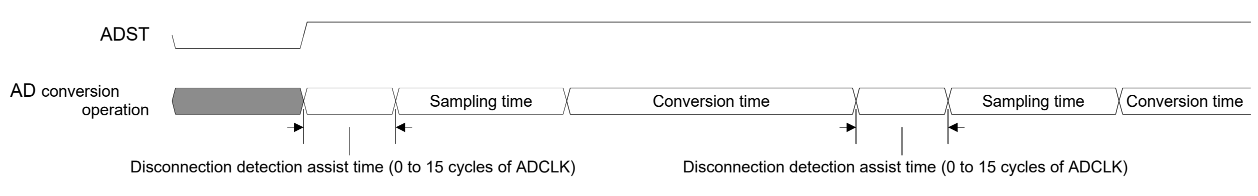
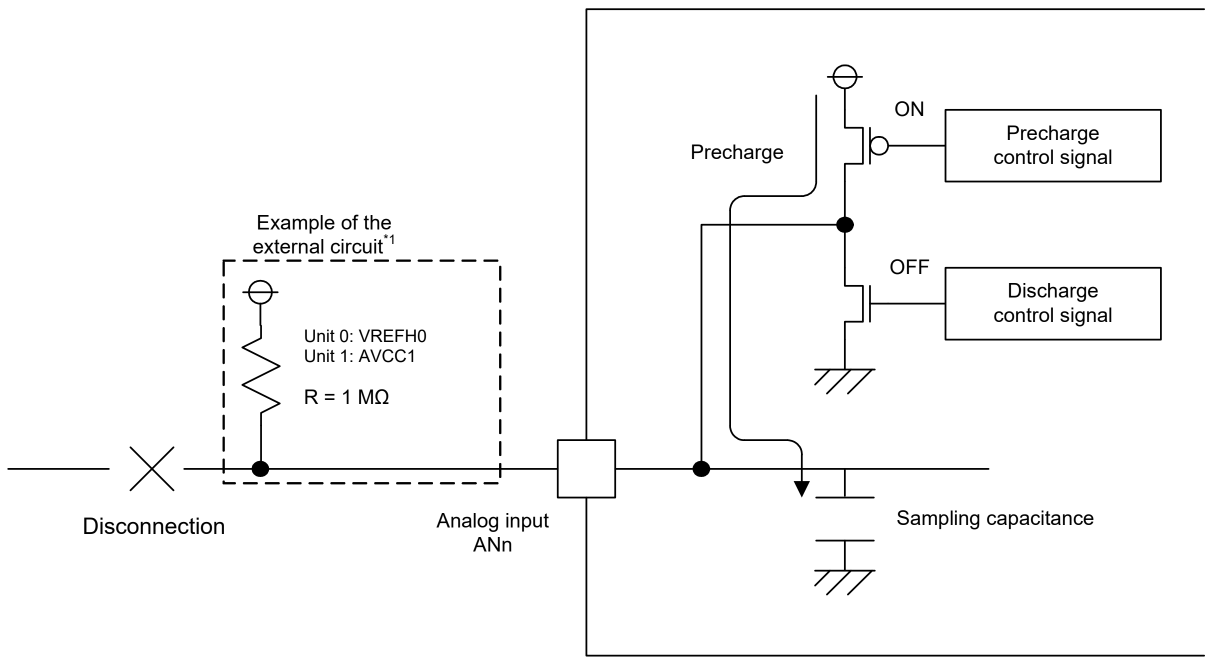
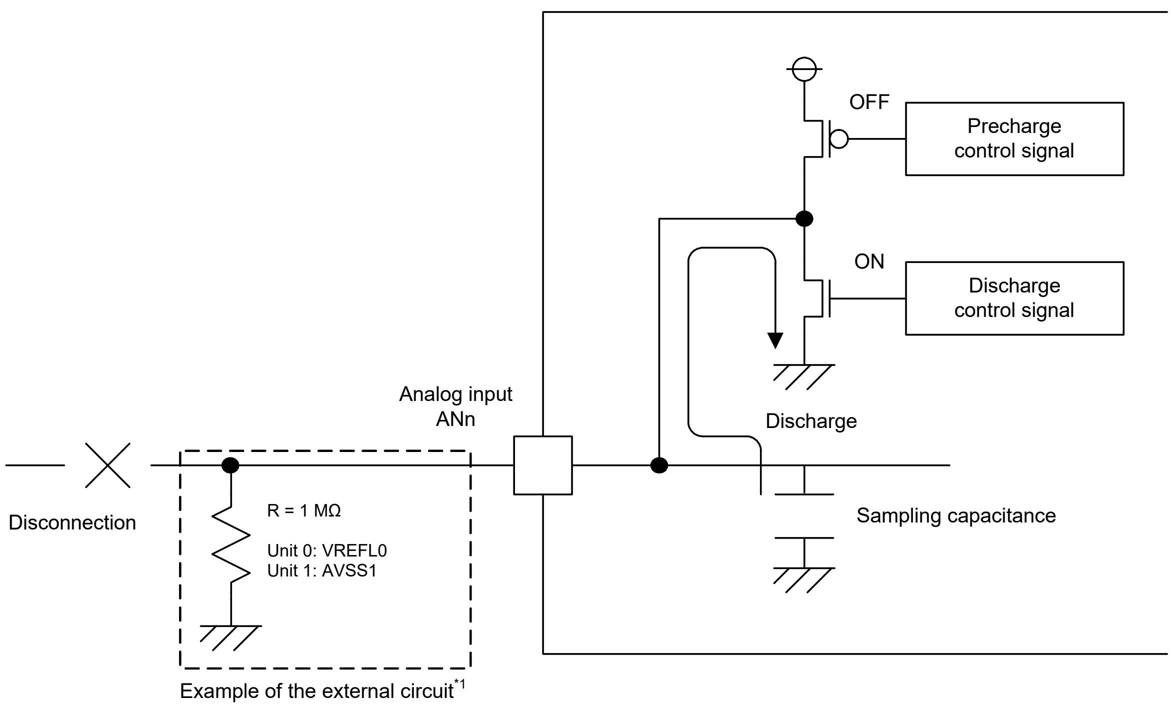


Figure 56.41 Operation of A/D Conversion When the Disconnection Detection Assist Function is Used



Note 1. The converted result should be used after fully evaluated because the result data when disconnection occurs varies depending on the external circuit.

Figure 56.42 Example of Disconnection Detection When Precharge is Selected



Note 1. The converted result should be used after fully evaluated because the result data when disconnection occurs varies depending on the external circuit.

Figure 56.43 Example of Disconnection Detection When Discharge is Selected

56.3.11 Starting A/D Conversion with Asynchronous Trigger

The A/D conversion can be started by the input of an asynchronous trigger. To start up the A/D converter by an asynchronous trigger, the A/D conversion start trigger select bits (ADSTRGR.TRSA[5:0]) should be set to 000000b, and a high-level signal should be input to the asynchronous trigger (ADTRG0# pin). Then, the ADCSR.TRGE and ADCSR.EXTRG bits should be set to 1. Figure 56.44 shows a timing of the asynchronous trigger input.
For the time from when the ADST bit is set to 1 until conversion starts, refer to section 56.6.3, A/D Conversion Restarting Timing and Termination Timing. An asynchronous trigger cannot be selected for group B or group C in group scan mode.

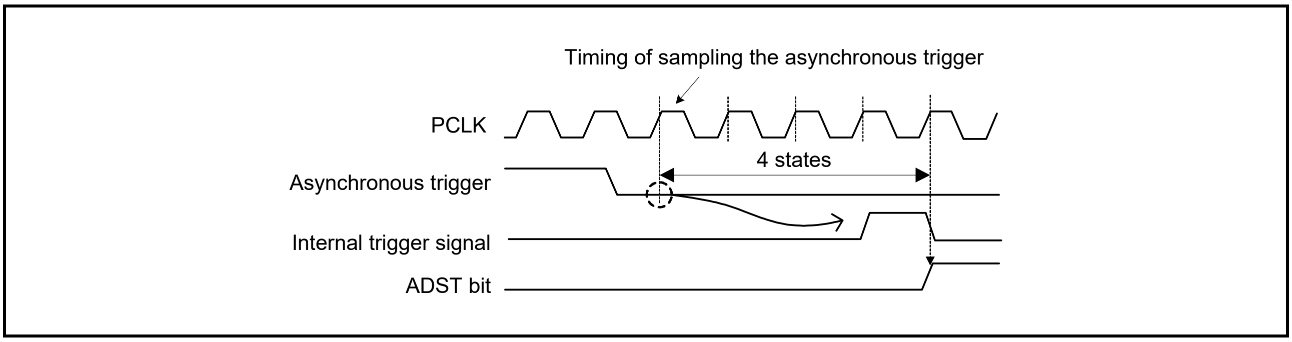


Figure 56.44 Timing of Sampling Asynchronous Trigger

56.3.12 Starting A/D Conversion with Synchronous Trigger from Peripheral Module

The A/D conversion can be started by a synchronous trigger. To start the A/D conversion by a synchronous trigger, the ADCSR.TRGE bit should be set to 1, the ADCSR.EXTRG bit should be cleared to 0, and the relevant sources should be selected by the ADSTRGR.TRSA[5:0] and ADSTRGR.TRSB[5:0] bits.

56.4 Interrupt Sources and DTC/DMAC Transfer Requests

56.4.1 Interrupt Requests

The 12-bit A/D converter can send scan end interrupt requests S12ADI/S12ADI1, S12GBADI/S12GBADI1, and S12GCADI/S12GCADI1 to the CPU.
This module can also generate interrupt requests S12CMPAI/S12CMPAI1 and S12CMPBI/S12CMPBI1 to the CPU in response to matches with a condition for comparison.
Setting the ADCSR.ADIE bit to 1 and 0 enables and disables an S12ADI/S12ADI1 interrupt, respectively; similarly, setting the ADCSR.GBADIE bit to 1 and 0 enables and disables a S12GBADI/S12GBADI1 interrupt, respectively; similarly, setting the ADGCTRGR.GCADIE bit to 1 and 0 enables and disables a S12GCADI/S12GCADI1 interrupt, respectively.
Setting the ADCMPCR.CMPAIE bit to 1 enables an S12CMPAI interrupt and setting the ADCMPCR.CMPAIE bit to 0 disables an S12CMPAI interrupt.
Setting the ADCMPCR.CMPBIE bit to 1 enables an S12CMPBI interrupt and setting the ADCMPCR.CMPBIE bit to 0 disables an S12CMPBI interrupt.
In addition, the DTC or DMAC can be activated when an S12ADI/S12ADI1, a S12GBADI/S12GBADI1, or S12GCADI/S12GCADI1 interrupt is generated. Using an S12ADI/S12ADI1, a S12GBADI/S12GBADI1, or S12GCADI/S12GCADI1 interrupt to allow the DTC or DMAC to read the converted data enables sequence conversion without burden on software.
For details on DTC and DMAC settings, see section 20, Data Transfer Controller (DTCb) and section 18, DMA Controller (DMACAa).

56.4.2 Scan Complete Event Output to ELC

The ELC can set up linked operation of a module specified in advance by using the S12ADI or S12ADI1 interrupt request signal as an event signal.
The S12GBADI or S12GBADI1 interrupt, S12GCADI or S12GCADI1 interrupt, S12CMPAI or S12CMPAI1 interrupt, and S12CMPBI or S12CMPBI1 interrupt request signals cannot be used as event signals. An event signal is output regardless of the setting of the corresponding interrupt request enable bit. The 12-bit A/D converter outputs the A/D conversion completed signals as event signals.

56.5 Allowable Impedance of Signal Source

Figure 56.45 shows an equivalent circuit of an analog input pin and an external sensor.
To perform A/D conversion accurately, the internal capacitor (Cs) must be fully charged within the sampling time. If the impedance (R0) of the signal source is high and it takes time to charge Cs, extend the sampling time with the ADSSTRn register. Conversely, if R0 is small, the sampling time can be shortened. Refer to the electrical characteristics for the permissible signal source impedance under various operating conditions.
When converting only a single pin input in single scan mode, the influence of R0 can be ignored because the input load becomes practically only the internal input resistor (Rs) by connecting an external high-capacity capacitor (C). However, because a low-pass filter is formed by R0 and C, it may not possible to follow the analog signal that changes at high speed. Insert a low-impedance buffer when converting high speed analog signals or when converting multiple pins in scan mode.

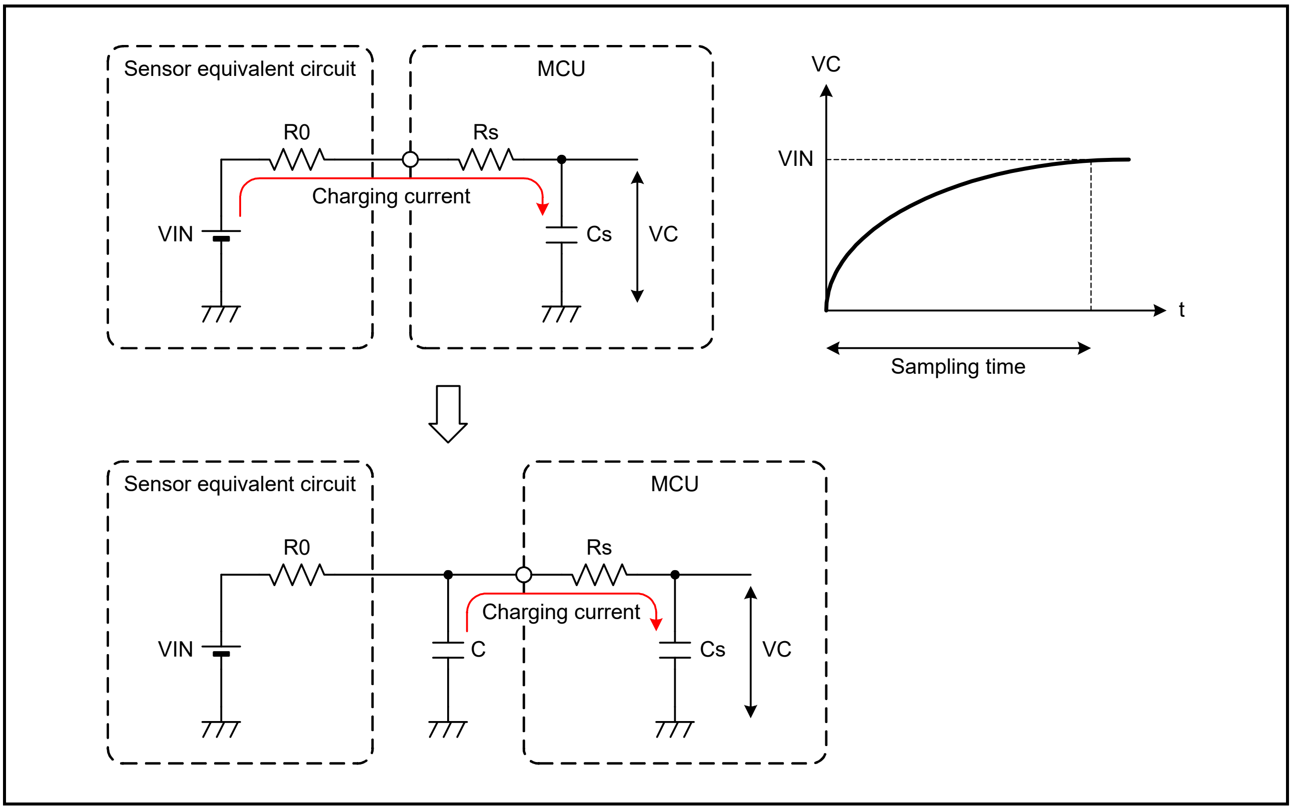


Figure 56.45 Equivalent Circuit of Analog Input Pin and External Sensor

56.6 Usage Notes

56.6.1 Notes on Reading Data Registers

The A/D data registers, A/D data duplication registers, A/D data duplication register A, A/D data duplication register B, A/D temperature sensor data register, A/D internal reference voltage data register, and A/D self-diagnosis data register should be read in 16-bit units. If a register is read twice in 8-bit units, that is, the higher-order byte and lower-order byte are separately read, the A/D-converted value having been read first may disagree with the A/D-converted value having been read for the second time. To prevent this, the data registers should never be read in 8-bit units.

56.6.2 Notes on Stopping A/D Conversion

56.6.2.1 Procedure of Stopping A/D Conversion

To stop A/D conversion when an asynchronous trigger or a synchronous trigger has been selected as the condition for starting A/D conversion, follow the procedure in Figure 56.46.

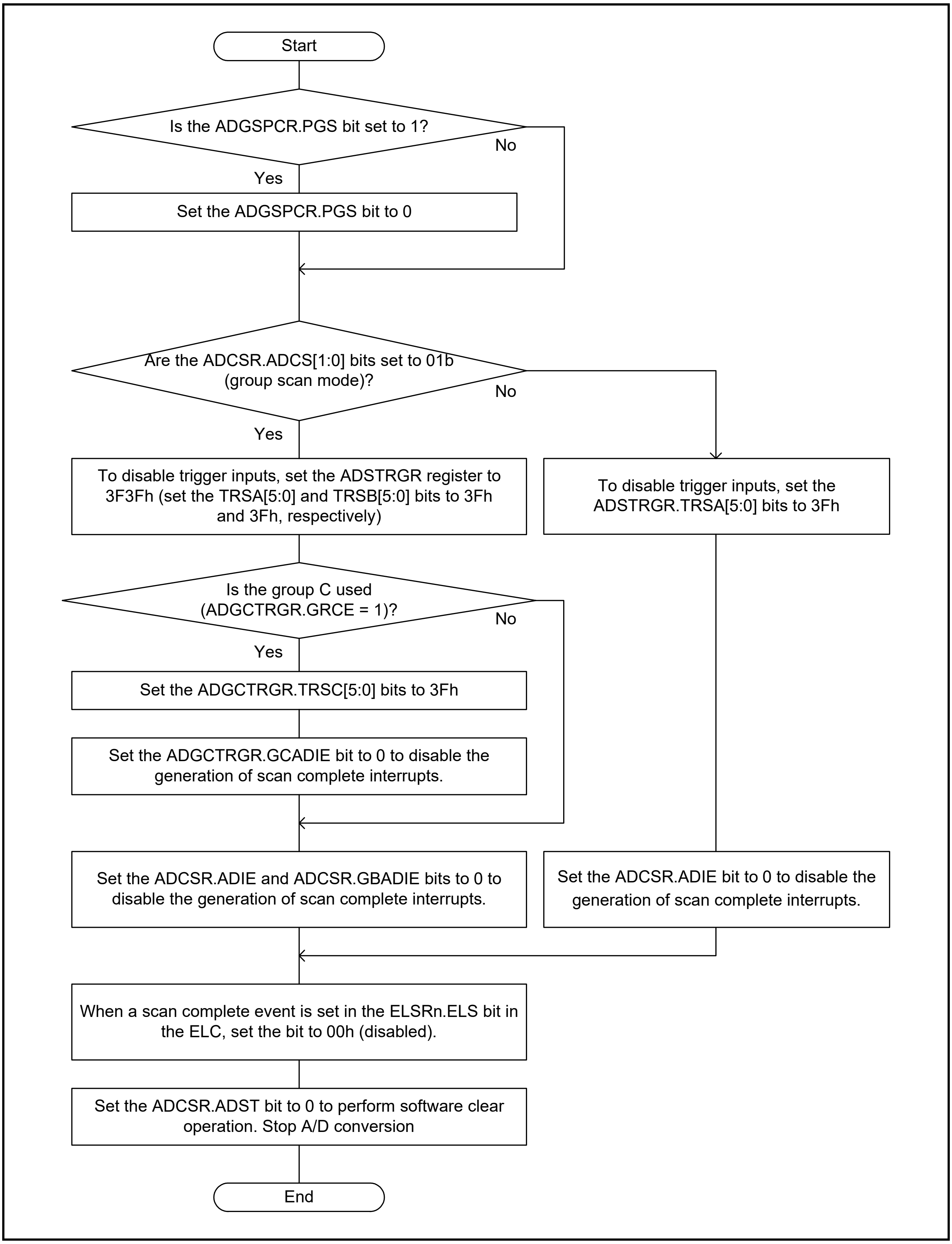


Figure 56.46 Procedure for Clear Operation by Software through the ADCSR.ADST Bit

56.6.2.2 Notes on Modes and Status Bits

The following values may be cleared or reset as described below: the voltage status used in self-diagnosis, the odd-even determination for the double-trigger mode, the bits for monitoring comparison for the comparison function.

- Re-set the settings for the voltage for use in self-diagnosis by setting the ADCER.DIAGLD bit to 1 and setting the ADCER.DIAGVAL[1:0] bits.
- Make double-trigger mode effective starting from the first scan by changing the setting of the ADCSR.DBLE bit from 0 to 1.
- Set the ADCMPCR.CMPAE and ADCMPCR.CMPBE bits to 0, which initializes the bits for monitoring comparison (MONCMPA, MONCMPB, MONCOMB).
- Initialize the constant sampling function (selected by setting the ADSHMSR.SHMD bit to 1) by setting the ADSHMSR.SHMD bit to 0. After this, if you will be using constant sampling again, wait for at least 1 cycle of ADCLK and then set the ADSHMSR.SHMD bit to 1.

56.6.3 A/D Conversion Restarting Timing and Termination Timing

It takes a maximum of six ADCLK cycles for the idle analog unit of the 12-bit A/D converter to be restarted by setting the ADCSR.ADST bit to 1. It takes a maximum of two ADCLK cycles for the operating analog unit of the 12-bit A/D converter to be terminated by setting the ADCSR.ADST bit to 0.

56.6.4 Notes on Scan End Interrupt Handling

When scanning the same analog input twice using any trigger, the first A/D-converted data is overwritten with the second A/D-converted data in the case that the CPU does not complete reading the A/D-converted data by the time the A/D conversion of the first analog input for the second scan ends after the first scan end interrupt is generated.

56.6.5 Module Stop Function Setting

Operation of the 12-bit A/D converter can be disabled or enabled by setting module stop control register. The initial setting is for operation of the 12-bit A/D converter to be halted. Register access is enabled by releasing the module stop state.

After the module stop state is released, wait for 1 µs to start A/D conversion. For details, refer to section 11, Low Power Consumption.

56.6.6 Notes on Entering Low Power Consumption States

Before entering the module stop state or software standby mode, make sure to stop A/D conversion. Here, set the ADCSR.ADST bit to 0, and secure certain period of time until the analog unit of the 12-bit A/D converter is stopped. Follow the procedure given below to secure this time.

Set the ADCSR.ADST bit to 0 by following the procedure in Figure 56.46 Procedure for Clear Operation by Software through the ADCSR.ADST Bit. Then wait for two clock cycles of ADCLK before entering the module stop mode or software standby mode.

56.6.7 Notes on Canceling Software Standby Mode

After software standby mode is canceled, wait until the crystal oscillation stabilization time or the PLL circuit stabilization time elapses, and then wait for 1 µs before starting A/D conversion. For details, refer to section 11, Low Power Consumption.

56.6.8 Pin Setting When Using the 12-bit A/D Converter

When using the 12-bit A/D converter unit 0, do not use the P40 to P47, P03, P05, and P07 pins as output pins. We also recommend not using the P00 to P02, P90 to P93, PD0 to PD7, and PE0 to PE7 pins as output pins. If any of the P00 to P02, P90 to P93, PD0 to PD7, and PE0 to PE7 pins is used for an output pin, perform A/D conversion several times, eliminate the maximum and minimum values, and obtain the average of the other results.

When using the 12-bit A/D converter unit 1, we recommend not using the P00 to P02, P90 to P93, PD0 to PD7, and PE0 to PE7 pins as output pins. If any of the P00 to P02, P90 to P93, PD0 to PD7, and PE0 to PE7 pins is used for an output pins, perform A/D conversion several times, eliminate the maximum and minimum values, and obtain the average of the other results.

56.6.9 Caution When Using an External Bus

A/D conversion at the same time as access to an external bus may produce poor results.
In this case, use a software approach, such as performing A/D conversion several times, then obtaining the average after excluding the highest and lowest values.
Deterioration in A/D conversion resolution can be reduced by changing the assignment of the external data bus. For details of the setting, refer to section 23, Multi-Function Pin Controller (MPC).

56.6.10 Error in Absolute Accuracy When Disconnection Detection Assistance is in Use

Using disconnection detection assistance leads to an error in absolute accuracy of the A/D converter. This is because an error voltage is input to the analog input pins due to the resistive voltage division between the pull-up or pull-down resistor (Rp) and the resistance of the signal source (Rs). This error in absolute accuracy is calculated from the following formula. Only use disconnection detection assistance after thorough evaluation.

Maximum error in absolute accuracy (LSB) = 4095 × Rs / Rp

56.6.11 Voltage Range of Analog Power Supply Pins

If this MCU is used with the voltages outside the following ranges, the reliability of the MCU may be affected.

- Analog input voltage range
  Voltage applied to analog input pins AN000 to AN007, AN100 to AN120: AVSSn ≤ VANn ≤ AVCCn (n = 0, 1)
- Relationship between power supply pin pairs (AVCCn–AVSSn, VCC–VSS)

  A 0.1-µF capacitor should be connected between each pair of power supply pins to create a closed loop with the shortest route possible as shown in Figure 56.47, and connection should be made so that the following condition is satisfied at the supply side: AVSSn = VSS
  When the 12-bit A/D converter is not used, the following conditions should be satisfied:
  AVCCn = VCC and AVSSn = VSS

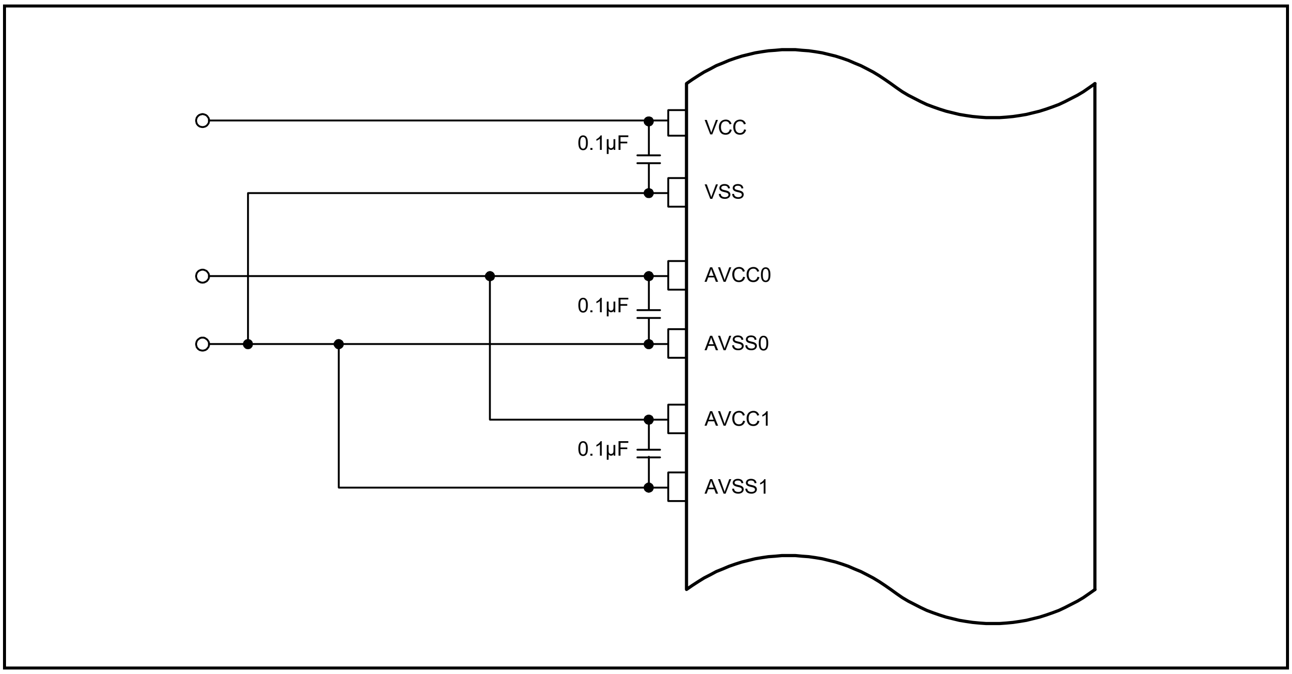


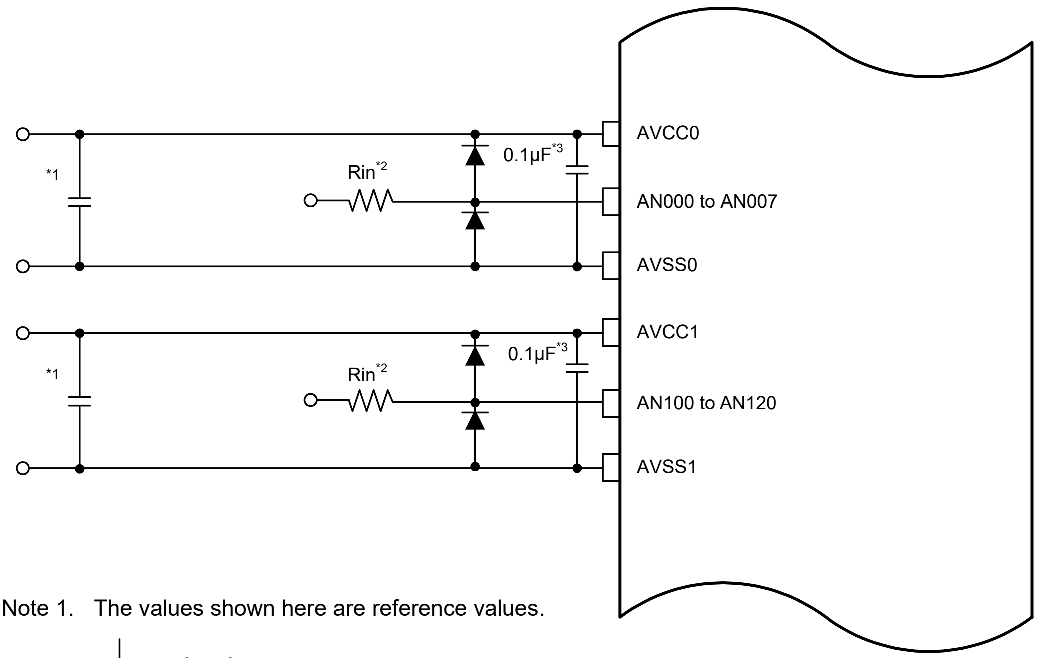
Figure 56.47 Power Supply Pin Connection Example

56.6.12 Notes on Board Design

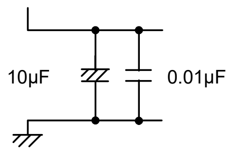
The board should be designed so that digital circuits and analog circuits are separated from each other as far as possible. In addition, digital circuit signal lines and analog circuit signal lines should not intersect or placed near each other. If these rules are not followed, noise will be produced on analog signals and A/D conversion accuracy will be affected. The analog input pins (AN000 to AN007, AN100 to AN120), and analog power supply (AVCCn) should be separated from digital circuits using the analog ground (AVSSn). The analog ground (AVSSn) should be connected to a stable digital ground (VSS) on the board (single-point ground plane connection).

56.6.13 Notes on Noise Prevention

To prevent the analog input pins (AN000 to AN007, AN100 to AN120) from being destroyed by abnormal voltage such as excessive surge, a capacitor should be inserted between AVCCn and AVSSn, and a protection circuit should be connected to protect the above analog input pins as shown Figure 56.48.



Note 1. The values shown here are reference values.



Note 2. Rin: Signal source impedance

Note 3. Place capacitors to be inserted between the power source of AVCC0, AVSS0, AVCC1, and AVSS1 as close as possible to the power supply pins to prevent deterioration in A/D conversion resolution.

Figure 56.48 Sample Protection Circuit for Analog Inputs